Master Thesis

GreenSloth: A Curated Web Resource for Validating and Comparing Peer-Reviewed Computational Models of Photosynthesis

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submitted to

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Abstract

Photosynthesis is the most vital process on Earth, sustaining life by means of food, oxygen, and other resources. This importance, however, brings forth its own challenges. It is a complex process, with many interacting components and feedback loops. This complexity has yet to deter humans from trying to understand it. Over the past centuries, photosynthesis has been a main focus of biological research. However, the scope of scientific work has since plateaued. To further explore the possibilities, more intricacies and details of the process need to be understood, and one of the most prevalent tools for doing so is mathematical modeling. These models are not new inventions, but they have grown in popularity and complexity over the years. However, the current state of the art is not without its flaws. Due to the complexity of photosynthesis, models narrow their scope to make them more manageable. This reduction, however, leads to many possible models. Over the years, the number of publications that have mentioned photosynthesis models has been ever-increasing. This increase in models, consequently, also brings a complication: the risk of being overwhelmed by the choice. There have been attempts to solve this problem, but they have only been partially successful. These solutions aim to solve problems across as many models as possible, but that fact undermines their capabilities. The models are so vastly different that many shortcuts are needed to incorporate them all into a single framework. Conversely, embracing model diversity and focusing on specific model types reduces the scope of the solution. By doing so, the light can shine on the specific type of models. More care can be taken to ensure these models are kept to a standard that follows modern guidelines like FAIR and CURE. On top of that, it allows a more direct comparison, so direct that standardized demonstrations can be used to showcase the models' capabilities. Moreover, providing a web platform to share these models with the community helps ensure they are more easily found and used. These three aspects are the core of **GreenSloth**, which comprises an entire ecosystem of tools for creating, demonstrating, and sharing ODE models of photosynthesis. The website (`https://greensloth.rwth-aachen.de/`) is the main platform for sharing the models, and the tools for creating and demonstrating them are also available as open-source GitHub repositories (`https://github.com/ElouenCorvest/GreenSloth/tree/main`). To highlight the capabilities of **GreenSloth**, five models of photosynthesis are used to showcase the ecosystem's strengths and weaknesses. These models vary in the scope of photosynthesis and in complexity, providing a good starting point as a proof of concept.

Contents

Acronyms

Cl^- chloride ion.

K^+ potassium ion.

ΔpH proton gradient between lumen and stroma.

AI Artificial Intelligence.

ATP Adenosine Triphosphate.

CBB Calvin-Benson-Bassham.

CLCE Cl^- channel.

CLI Command-line Interface.

CSS Cascading Style Sheets.

CURE Credibility, Understandability, Reproducibility, Extensibility.

Cyt b_6f Cytochrome b_6f complex.

Cytb_6f Cytochrome b_6f.

FAIR Findable, Accessible, Interoperable, Reusable.

FvCB Farquhar, von Caemmerer, and Berry.

HTML Hypertext Markup Language.

JS JavaScript.

JSON JavaScript Object Notation.

KEA3 K^+/H^+ antiporter 3.

KEGG Kyoto Encyclopedia of Genes and Genomes.

LLM Large Language Model.

MCA Metabolic Control Analysis.

MIRIAM Minimal Information Required In the Annotation of Models.

NADPH Nicotinamide Adenine Dinucleotide Phosphate.

NEON National Ecological Observatory Network.

NPQ Non-Photochemical Quenching.

ODE Ordinary Differential Equation.

PAM Pulse Amplitude Modulation.

PC_{ox} oxidised plastocyanin.

PETC Photosynthetic Electron Transport Chain.

PGA Phosphoglycerate.

PMF proton motive force.

PPFD Photosynthetic Photon Flux Density.

PQ_{ox} oxidised plastoquinone.

PSI Photosystem I.

PSII Photosystem II.

RuBisCO Ribulose-1,5-bisphosphate carboxylase-oxygenase.

RuBP Ribulose-1,5-bisphosphate.

SBML Systems Biology Markup Language.

VCCN1 voltage-gated Cl^- channel 1.

WT wild type.

List of Variables

CO_2 carbon dioxide.

H^+ proton.

O_2 oxygen.

A carbon assimilation.

B specific model quantity of interest used for calculating a control coefficient.

C_a ambient carbon dioxide (CO_2) concentration.

C_i intercellular CO_2 concentration.

$C^B_{v_i}$ control coefficient of a model quantity of interest B with respect to a specific steady-state rate v_i.

C partial pressure of CO_2 in the vicinity of RuBisCO.

F_m maximal fluorescence.

F fluorescence.

J potential rate of linear electron transport to support RuBP regeneration.

$K_{\mathrm{M|RuBisCO|CO_2}}$ michaelis-menten constant of CO_2 for RuBisCO carboxylation.

$K_{\mathrm{M|RuBisCO|O_2}}$ michaelis-menten constant of O_2 for RuBisCO oxygenation.

K_{ZSat} half-saturation constant (relative concentration of Zx) for quenching.

K_{pHSatLHC} pKa of PsbS activation, kept the same as for VDA.

K_{pHSat} half-saturation pH for de-epoxidase activity, highest activity at pH 5.8.

R_{act} RuBisCO activation state.

R_{light} respiration in the light.

T_{p} potential rate of triose phosphate utilization.

$V_{\mathrm{max|RuBisCO}}$ maximum rate of RuBisCO carboxylation.

W_{c} $v_{\mathrm{RuBisCO|carb}}$ limited by RuBisCO acivity.

W_{j} $v_{\mathrm{RuBisCO|carb}}$ limited by RuBP regeneration.

W_{p} $v_{\mathrm{RuBisCO|carb}}$ limited by triose phosphate utilization.

Γ^* CO_2 compensation point in the absence of non-photorespiratory CO_2 release.

α_{old} fraction of remaining glycolate carbon.

γ_0 Fitted quencher factor corresponding to base quenching not associated with protonation or zeaxanthin.

γ_1 Fitted quencher factor corresponding to fast quenching due to protonation.

γ_2 Fitted quencher factor corresponding to fastest possible quenching.

γ_3 Fitted quencher factor corresponding to slow quenching of Zx present despite lack of protonation.

$\mathrm{AP_{tot}}$ total adenylate stromal concentration.

DHAP dihydroxyacetone phosphate.

$\mathrm{NPQ_{max}}$ maximal extent of NPQ.

$\mathrm{PSI_{tot}}$ total PSI.

$\mathrm{PSI_{ox}}$ the fraction of PSI donors per RCII that are available for the linear electron transport.

$\mathrm{P_i}$ inorganic phosphate.

$\mathrm{Q_{active}}$ activated PSII quencher.

Q co-operative 4-state quenching mechanism.

Φ_{PSII} efficiency of photosystem II.

$\mathrm{k}_{\mathrm{cyc}}$ reaction rate constant of cyclic electron flow.

$f(\mathrm{RUBP})$ function that relatives RuBP concentration to concentration of RuBisCO active sites.

g_{s} stomatal conductance of water vapor to the atmosphere.

k_4 NPQ mechanism deactivation.

v_{ATPSynth} Adenosine Triphosphate (ATP) synthase rate.

v_{PSII} Photosystem II (PSII) rate.

v_{PSI} Photosystem I (PSI) rate.

$v_{\mathrm{RuBisCO|carb}}$ Ribulose-1,5-bisphosphate carboxylase-oxygenase (RuBisCO) carboxylation rate.

$v_{\mathrm{RuBisCO|ox}}$ RuBisCO oxygenation rate.

v_{FNR} rate of the reaction mediated by FNR.

v_{b6f} Cytochrome $\mathrm{b_6f}$ ($\mathrm{Cytb_6f}$) rate.

v_i specific steady-state rate used for calculating a control coefficient.

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1 Introduction

1.1 Preamble

Photosynthesis is a fundamental process of the very existence of life on Earth [1–3]. It provides the starting block for the food chain. It is directly responsible for the oxygen that forms the ozone layer. Additionally, it also proves to be a key player in humans' more complex lives, as, around 82 % of the world's primary energy consumption is still based on fossil fuels [4], which in turn are mostly made up of dead organic matter of photosynthetic organisms [5]. Therefore, photosynthesis has been a vital part of scientific research for centuries.

The basic principles of photosynthesis have been known since the 18th century [6], leading to advanced techniques for studying the process in greater detail. A key reason for this effort to understand photosynthesis is the potential to use it as a means to increase crop yield [7]. While breakthrough discoveries in the past introduced the Green Revolution, those methods have long reached a plateau in terms of improvement [8]. Therefore, there is a need to find new ways to change photosynthesis; however, this time in a new light. Instead of focusing on what comes out of experiments, a more directed view into the inner workings of photosynthesis is needed. To change a process, it is necessary to understand it. This is where mathematical modelling comes in.

Mathematical modelling has been a staple of the sciences for centuries [9]. From using simple geometry for calculating the distance to the sun, to simulating neutron transport in nuclear fission [10]. This method has not gone unnoticed in the world of photosynthesis research and has given rise to a large variety of models. Some focus on specific parts of the process [11, 12], some on specific ways the photosynthesis machinery reacts to the environment [13]. Some depict the process in a very simplified manner [14–16], while others try to capture the process in all its complexity [17]. There are many different ways mathematical modelling has been used to understand photosynthesis, as seen in the still-growing influx of new publications over the years (see Fig. 1).

With all these different models and ways to see photosynthesis, one cannot point to a single one as the "best" one. Each model has its own advantages and disadvantages, often tailored to answer specific questions. On top of that, many models build on each other, taking inspiration or even entire structures. It may be done to improve a model or to make it more accessible to a different audience. Sometimes, it may also be used to answer a different scientific question that was not the original model's intention. A strong example of this is the Farquhar, von Caemmerer, and Berry (FvCB) model [14], a simple model that describes the carbon assimilation (A) as a function of RuBisCO carboxylation rate ($v_{\mathrm{RuBisCO|carb}}$), RuBisCO oxygenation rate ($v_{\mathrm{RuBisCO|ox}}$), and respiration in the light (R_{light}). Even though it is simple, it has amassed a large number of citations, also in non-modelling branches. The main fields using the FvCB model are, of course, the plant sciences. However, it has also found its way into other fields, such as ecology, forestry, and more (see Fig. 2). It has become so popular that many versions have emerged. Some that take the original model as a starting point and add on more complexity [15, 18], or others that use it solely as a readout for A [17]. It has become so popular, in fact, that even misinterpretations of the original model are ingrained in photosynthesis modelling [16].

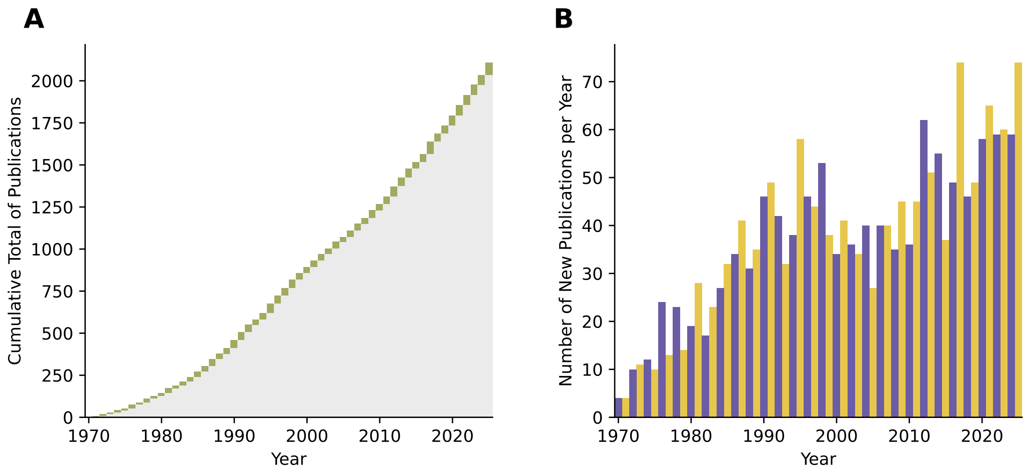

Figure 1: Results of a bibliography search for photosynthesis models. A basic bibliographic search was conducted to identify publications with the words "photosynthesis" and "model" in the title, published between 1970 and 2025. On the left, the cumulative number of publications over the years is shown, with the new publications for each year shown in green. On the right, the number of new publications per year is shown. The two colors were chosen to be easily distinguishable and to have no specific meaning. The data was obtained from the Web of Science database on the 14th of February 2026. The query used was "TI=(("photosynthesis" OR "photosynthetic") AND ("model*" OR "modelling" OR "modeling" OR "simulation" OR "representation"))", and can be found here: `https://www.webofscience.com/wos/woscc/summary/0b793bdb-ab8b-456e-a5fb-085fc470f5a2-019f07a73b/relevance/1`.

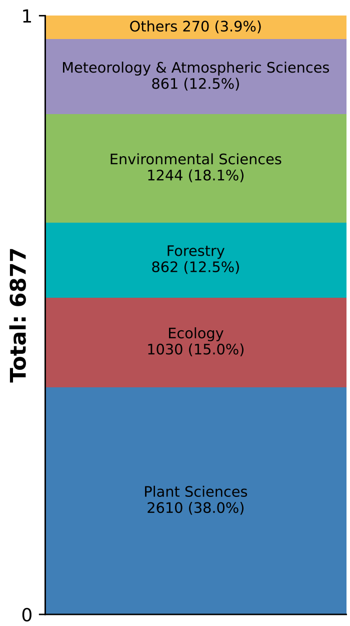

Figure 2: Number of citations of the original FvCB publication separated in categories. The citations of the original Farquhar, von Caemmerer, and Berry (FvCB) model publication [14] were obtained from the Web of Science database on the 16th of February 2026. These citations were separated into categories based on the Web of Science categories of the citing publication. The five most populated categories were taken, while the others were grouped into "Others".

While this rise in popularity of photosynthesis models brings, on the one hand, more and more tools to understand photosynthesis, it also brings a lot of confusion. Models differing in their concept, basing their work on different assumptions, and sometimes even misinterpreting starting blocks, can make it hard to see through the web of models. While it is not fair to say that the waters of photosynthesis modelling are polluted, it is fair to say there is a problem with clarity. This is not only a problem for newcomers to the field, but also experienced researchers, who may find it hard to keep up with the ever-growing number of models. This problem of clarity has already come to the attention of the scientific community, and it is not limited to photosynthesis modelling. Therefore, some proposed solutions already exist. Three key problems have been identified and solutions proposed: 1) Model Creation, 2) Model Presentation, and 3) Model Sharing. While all three are very intertwined, a clear distinction can be made between them.

1.2 Current Problems and Solutions

1.2.1 Model Creation

Model Creation can take many forms, so the transparency of the process is vital. Some may be written from scratch using Python or MATLAB. At the same time, others may be built using more specialized software for modelling, such as `mxlpy` [19], `COPASI` [20], or `Tellurium` [21]. Whatever the method used, one problem persists across all of them: the model's annotation and documentation are based on the author. This means great care must be taken to ensure the model is reproducible and understandable to others. This is not only a problem with the model description but also with the code itself. As computational methods adapt and evolve, the code may become outdated and hard to read. There have been attempts to solve this problem, one of the most notable ones being the Systems Biology Markup Language (SBML) [22].

It breaks down the concept of a model into its most basic components, such as compartments, species, reactions, and so on [22]. This creates a schematic overview of the model system, making it much easier to understand across different languages. Just as a schematic overview of photosynthesis is used in school books to explain that oxygen (O_2) is produced from water and not CO_2, SBML provides a schematic overview of the model so that different software can better understand it. While this simplification of complex systems makes it usable for a wide variety of use cases, it also makes it less sensitive to the original model.

The aim of SBML is not to be a universal language for modelling, but rather cast a big net to catch as many models as possible. However, as a net is only as good as its holes, many implementations fail to capture the full complexity. To counteract this, many extensions and upgrades have been proposed, such as the SBML Level 3 [23], which builds on the basic core described earlier and adds additional methods to support modelling and working with models. It includes packages that can render the model in a more human-readable way, capture more information needed for constraint-based modelling, and more. Overall, SBML has had a great reception in the modelling community and as support is ever-increasing; however, it cannot, and will not, solve the problem of model creation at its core.

1.2.2 Model Presentation

Reusability and reproducibility of work have always been core principles of scientific work; however, even data sharing has fallen victim to poor practices. An analysis conducted by Roche et al. [24] examined a dataset of 100 nonmolecular evolutionary or ecological articles published between 2012 and 2013 in seven leading journals and found that 56% of the articles archived incomplete data, and that 64% were not fully reproducible. This has been a widespread problem across many scientific fields, prompting the creation of the Findable, Accessible, Interoperable, Reusable (FAIR) Guiding principles [25].

These principles address the problems of data sharing and archiving and aim to create a set of rules that a scientist should follow when publishing their work. Data published under the FAIR principles not only makes it easier for humans to read and understand them, but also allows others to work upon them. This is especially helpful for modellers, as access to data is key for validating models against the real world. On top of that, these principles are also key for the presentation of models themselves, even if they cannot be directly classified as data. While work in a lab cannot be directly uploaded to a repository and downloaded elsewhere in the world, a model can. Therefore, a proposal for an extension of these FAIR principles has recently been made: the Credibility, Understandability, Reproducibility, Extensibility (CURE) Guidelines [26].

The CURE guidelines are a set of principles heavily inspired by the FAIR principles and the work done for it. They were set up with mechanistic computational models of biological systems in mind, as the machine-learning community already has a good grasp on the FAIR principles [27]. The CURE guidelines propose a checklist of baseline requirements to make it easier for an author to make their model and publication more scientifically transparent. To summarize each of the principles, a model is credible if it makes accurate and trustworthy predictions of real phenomena. It is understandable, if it is clear and comprehensible to others, through good documentation, annotation, and availability. It is reproducible, if the same results can be obtained apart from the authors themselves. It is extensible if the model can be built upon by others, both in the sense of theory and in the sense of code. These principles guide authors of models to make their work more transparent and accessible, which in turn makes it easier for others to understand and build upon it. The CURE guidelines propose nice recommendations for the presentation of models, however, they will be just that, recommendations. There are no plans on enforcing these rules to any extent, and on top of that, these guidelines have only been published as a pre-print recently, in the beginning of 2025.

1.2.3 Model Sharing

As mentioned prior, the amount of publications concerning photosynthesis models is ever-growing, which complicates the process of keeping up with the field. This issue is very persistent in many biological fields, where models are being used. To find a suitable model, one needs to go through and sift through a lot of literature, which can be very time-consuming and daunting for someone not entirely sure what they are searching for. To counteract this problem, some efforts have been made to collect a large variety of models in databases. One of the most notable ones is the BioModels database [28], which includes 1096 curated and 1668 non-curated models as of the 16^{th} of February 2026.

BioModels does not restrict what type of models can be uploaded, therefore it includes a wide assorment of models, from different fields, and with different levels of complexity. Models can range from *Homo sapiens* to *Saccharomyces cerevisiae*, from Ordinary Differential Equation (ODE) to constraint-based, or from MATLAB to Python. This wide variety of models makes it a great resource for finding a model that suits the needs of the researcher. Additionally, to counteract the problems mentioned in prior sections, models uploaded to BioModels need to adhere to the Minimal Information Required In the Annotation of Models (MIRIAM) guidelines [29]. Another set of guidelines, comparable to FAIR and CURE, that strives for reproducibility of models. To further support and encourage the use of the database, BioModels also provides an extensive curation process, where models are checked by the team behind the database to make sure they conform to the MIRIAM guidelines. After the curation process, the model is stamped with the "curated" label, pushing the credibility of the model. To access this curation process, a model has to be uploaded in an SBML format, a clever way to encourage the use of SBML and its standards. Overall, BioModels promises to make the process of sharing a model easier and more accessible, however, it still has its flaws.

A recent study done by Tiwari et al. [30] analyzed the reproducibility of 455 models taken directly from the BioModels database. The study found that only 51 % of the models were reproducible, even with the MIRIAM prerequisite in place. Out of the irreproducible models, 54 could be saved by manual empirical correction or direct author support. At the end, 37 % of the total analysed models could not be reproduced at all. The most common issues, were the missing of critical model information and annotation, wrongly described equations, or other small and easily overseen, but critical, mistakes during the authorship of the publication. To combat these problems, the authors suggest a reproducibility scoring card, that should accompany the model during the submission process. It has to be noted that this study was partly done by the team behind the BioModels database, showcasing their dedication to the problems models face in the current state of the field. Overall, BioModels is a great resource for finding and sharing models, however, it still has not reached its full potential.

1.3 GreenSloth as a Solution

It seems the problems of modelling of the modern day are known, and solutions have been proposed, however, they try to solve the issues for as many types of models as possible, which sadly makes them too broad to be truly effective. Therefore, a more specific solution may be of better help. One concentrates itself on a specific type of model, both in the sense of what it shows and how it is made. This allows the solution to be tailored to the specific needs of the field, and even enables additional ways to showcase the models. The prior mentioned solutions only focus on the building blocks of the model itself, however, they all rely on the publication of the model to showcase its capabilities. While this is the common way to look at models, having a more specialized solution allows creating a more interactive way to present the models. This is where **GreenSloth** comes in.

GreenSloth will be an entire ecosystem for the creation, presentation, and sharing of ODE models of photosynthesis. This tool, can be a solution that solves all three of the problems mentioned prior, taking a big inspiration from the prior mentioned solutions.

1.3.1 Model Creation with GreenSloth

While the actual creation of models cannot be automated and relies heavily on the author's intent, a solution cannot alleviate that part of the process. However, limiting models to Python enables the creation of tools to help the author along the way. One very big help for the author is the usage of the `mxlpy` package [19], which provides an easier way to create, simulate, and analyze ODE models. With this package, **GreenSloth** can provide a step-by-step guide to creating the model documentation, including automatic extraction of model information and equations and linking the documentation directly to the code. This way, the model-creation process is more transparent and easier for others to understand. On top of that, automatic code formatting is possible, which partitions the model into separate files and creates a custom Python package for it, making it directly useable by others.

1.3.2 Model Presentation with GreenSloth

While the base presentation of a model is documentation, the specialization of **GreenSloth** enables a more interactive presentation of different models. As only ODE models of photosynthesis are included in the **GreenSloth** ecosystem, common ways these models are analyzed and presented can be simplified to create a standardized approach for demonstrating a model's capabilities. Five popular methods have been chosen, each showcasing a proof of concept.

Plants in the wild are exposed to a wide range of conditions, and their photosynthetic machinery must be highly adaptable. One of these conditions is the light intensity and its dynamic changes during the day [31–33]. Having a model of photosynthesis simulate a day of light, using actual light data, can be a great way to show the reaction to light of the model, on top that, showing the results over the day of key photosynthetic quantities, enables the direct comparison of the model to real-world data.

As mentioned earlier, the FvCB model is widely used in photosynthesis research and is familiar to many people. It primarily describes the A of a plant by looking at the $v_{\mathrm{RuBisCO|carb}}$, $v_{\mathrm{RuBisCO|ox}}$, and R_{light} [14, 15]. However, as the $v_{\mathrm{RuBisCO|carb}}$ and $v_{\mathrm{RuBisCO|ox}}$ are directly competitive of each other, the equation can be simplified to a version, where the $v_{\mathrm{RuBisCO|ox}}$ is converted into a factor of $v_{\mathrm{RuBisCO|carb}}$, hereby including the partial pressure of CO_2 in the vicinity of RuBisCO (C) and the CO_2 compensation point in the absence of non-photorespiratory CO_2 release (Γ^*) (see Equation 1).

$$\begin{aligned} A &= v_{\mathrm{RuBisCO|carb}} - 0.5 \cdot v_{\mathrm{RuBisCO|ox}} - R_{\mathrm{light}} \\ A &= v_{\mathrm{RuBisCO|carb}} \cdot \left(1 - \frac{\Gamma^*}{C}\right) - R_{\mathrm{light}} \end{aligned} \tag{1}$$

Additionally, the $v_{\mathrm{RuBisCO|carb}}$ is represented as being limited by either the RuBisCO activity (W_{c}), the Ribulose-1,5-bisphosphate (RuBP) regeneration (W_{j}), or the triose phosphate utilization (W_{p}). The first uses C, maximum rate of RuBisCO carboxylation ($V_{\mathrm{max|RuBisCO}}$), michaelis-menten constant of CO_2 for RuBisCO carboxylation ($K_{\mathrm{M|RuBisCO|CO_2}}$), O_2, and michaelis-menten constant of O_2 for RuBisCO oxygenation ($K_{\mathrm{M|RuBisCO|O_2}}$), the second uses C, potential rate of linear electron transport to support RuBP regeneration (J), and Γ^*, while the third uses the C, Γ^*, fraction of remaining glycolate carbon (α_{old}), and potential rate of triose phosphate utilization (T_{p}). The $v_{\mathrm{RuBisCO|carb}}$ is then the minimum of these three (see Equation 2).

$$
\begin{aligned}
W_{\mathrm{c}} &= \frac{C \cdot V_{\mathrm{max|RuBisCO}}}{C + K_{\mathrm{M|RuBisCO|CO_2}} \cdot \left(1 + \frac{O_2}{K_{\mathrm{M|RuBisCO|O_2}}}\right)} \\
W_{\mathrm{j}} &= \frac{C \cdot J}{4 \cdot C + 8 \cdot \Gamma^*} \\
W_{\mathrm{p}} &= \begin{cases} \infty \text{ if } C \geq \Gamma^* \cdot (1 + 3 \cdot \alpha_{\mathrm{old}}) \\ \frac{3 \cdot C \cdot T_{\mathrm{p}}}{C - \Gamma^* \cdot (1 + 3 \cdot \alpha_{\mathrm{old}})} \end{cases} \\
v_{\mathrm{RuBisCO|carb}} &= \min\left(W_{\mathrm{c}}, W_{\mathrm{j}}, W_{\mathrm{p}}\right)
\end{aligned}
\tag{2}
$$

While the FvCB model is not made out of ODEs, it can still be implemented as a submodule inside other models, as it consists of algebraic equations. To include this in a model of photosynthesis, a bridge between the two models needs to be chosen. As many photosynthesis models include a depiction of $v_{\mathrm{RuBisCO|carb}}$ and CO_2, these two can serve as one-way connection points to the FvCB model (see Equation 1). While these two points are required, the other parameters needed in the equations can either be supplied by the photosynthesis model or be taken from the default values of the FvCB model. This way, the FvCB model can be included in a photosynthesis model, and a typical A-intercellular CO_2 concentration (C_{i}) curve can be displayed to showcase the FvCB submodule's results.

A very popular experiment to quantify plant photosynthesis is a PAM protocol [34, 35]. Here, a dark-adapted plant is exposed to several saturating light pulses over time, sometimes with varying degrees of actinic light intensity. By measuring the resulting chlorophyll fluorescence (F), the plant's Non-Photochemical Quenching (NPQ) can be calculated by using the F values at each peak of a saturating pulse, equalling the maximal fluorescence (F_{m}) (see Equation 3). As this method has been widely used in plant sciences for decades [36], it is a good way to bring a more experimental approach to a photosynthesis model. It is already a common practice among some models for validation against experimental data, reinforcing its standing as a good way to demonstrate a model's capabilities.

$$NPQ(t) = \frac{F_{\mathrm{m}}(t=0) - F_{\mathrm{m}}(t=t)}{F_{\mathrm{m}}(t=t)} \tag{3}$$

Models are often used to predict a system's behavior after a specific metabolic change. Therefore, a Metabolic Control Analysis (MCA) is a good way to showcase the effect specific parameters have on the system [37, 38]. It uses the concept of control coefficients (C), which show the relative change of a model quantity of interest (B), after a change in a steady-state rate (v_i). This can also be shown as the slope of the system's response on a log-log scale (see Equation 4).

$$C = \frac{\partial \ln(B)}{\partial \ln(v_i)} = \frac{\partial B}{\partial v_i} \cdot \frac{v_i}{B} \tag{4}$$

To simplify the calculation process, the difference of the B after a positive and negative absolute perturbation (h) of the specific steady-state rate used for calculating a control coefficient (v_i) can be used. To do that, the B is expressed as a function of the v_i by using a Taylor expansion, for both sides of the h. Due to how Taylor expansions work, from the second order and on, both equations are the same. Taking the difference of the two equations, the second order and higher terms cancel out, leaving only the first order term, which equates to the first derivative of the B in respect to the v_i. Solving for this derivative shows an approximation of the C as a function of the difference of the B after a positive and negative h of the v_i (see Equation 5).

$$\begin{aligned}
B_{\text{upper}} &= B(v_0 + h) = B(v_0) + h \cdot B'(v_0) + \frac{h^2}{2} \cdot B''(v_0) + ... \\
B_{\text{lower}} &= B(v_0 - h) = B(v_0) - h \cdot B'(v_0) + \frac{h^2}{2} \cdot B''(v_0) + ... \\
B_{\text{upper}} - B_{\text{lower}} &\approx 2 \cdot h \cdot B'(v_0) \\
\Leftrightarrow B'(v_0) &= \frac{\mathrm{d}B}{\mathrm{d}v_i} \approx \frac{B_{\text{upper}} - B_{\text{lower}}}{2 \cdot h} \\
\Rightarrow C_{v_0}^{B_0} &= \frac{\mathrm{d}B_0}{\mathrm{d}v_0} \cdot \frac{v_0}{B_0} \approx \frac{B_{\text{upper}} - B_{\text{lower}}}{2 \cdot h} \cdot \frac{v_0}{B_0}
\end{aligned} \tag{5}$$

The B can either be a specific concentration or a steady-state flux. Due to the summation theorem, the sum of all the concentration control coefficients for a specific substrate is $= 0$. The sum of all flux control coefficients for a specific flux is $= 1$ [38]. Additionally, the v_i can be observed by examining the change in a proportional parameter in that specific rate; therefore, the C is based on that parameter's change and shows the effect of that parameter on the system. While the MCA relies on many assumptions, it remains a vital tool for working with ODE models. In the case of photosynthesis models, using common substrates and fluxes as the B and showing the effects of different parameters on the system are good ways to demonstrate the differences between models directly.

To get a model to represent data extracted from the real world, fitting the model is often the solution [39, 40]. This process involves adjusting the model's parameters to minimize the difference between the model's predictions and the observed data. While doing so can be done by hand, it is often more efficient to use optimization algorithms to find the best set of parameters. These algorithms can be broken down into three steps that are performed in a loop until the best fit is found: 1) Calculating the error between the model's predictions and the observed data, 2) Adjusting the model's parameters based on the calculated error, and 3) Running the model with the new parameters to get new predictions. The first step is often done using the least-squares method, which takes all the differences between a model's predictions and the observed data, squares them, and sums them to get a single value representing the error. The second step can be performed using a variety of algorithms, one of the most popular being the Levenberg–Marquardt method [41]. This method is actually a combination of two methods, the gradient descent and the Gauss-Newton method. It uses gradient descent when the error is large and Gauss-Newton when the error is small. This way, it can efficiently find a new set of parameters to give a new try in the third step. This process is repeated until the error is minimized, and the best fit is found. While fitting is an integral part of working with models, it is a complex concept with no universally accepted method. It relies heavily on the model's complexity, the data to be fitted, and the author themselves. Even choosing which parameters to fit is not easy. There are ways to facilitate this decision, for example, an MCA, yet expertise on the model and its scientific scope is needed to make the best choice. Therefore, **GreenSloth** cannot offer a one-size-fits-all solution for fitting. However, it can provide a sample implementation of the fitting process to demonstrate a proof of concept. This way, fitting can be brought closer to the user.

1.3.3 Model Sharing with GreenSloth

To follow in the footsteps of the BioModels database, **GreenSloth** will also be a web-based platform where models can be found and downloaded. Web development and design has become an integral part of human society, as 74 % of global population are reported to be internet users [42]. Websites shape how people interact with their politics, their art, their community, and themselves [43]. Therefore, a great deal of research has been done on how to make websites more user-friendly and accessible. Good web design can be seperated intro four categories [44]: 1) Ease of navigation, 2) Site Speed, 3) Clarity and Conciseness, and 4) Visual Appeal. Mastering these four categories key not only for a good website, but also for a successful sharing platform. **GreenSloth** will be designed with these categories in mind to make it a successful platform for sharing photosynthesis models.

Having a well-thought-out design not only helps with the user experience but also enables the creation of more ways to interact with the models shared on **GreenSloth**. The concept of the website will be separated into three parts: 1) Model Search, 2) Model Summary, and 3) Model Comparisons.

Typically, to search for a model, one would type into a library of scientific publications, such as Web of Science or Google Scholar, what they wish to find and hope they have used the right keywords. Search query engineering is a skill in itself, which has been facilitated by the rise of Large Language Models (LLMs) [45] (to mixed results). However, even with the best search query, it can be hard to find the right model, as the amount of publications is increasingly overwhelming (see Fig. 1). **GreenSloth** will initially have a rudimentary search engine that lets users find models by name. However, it will be supported by a curated tag system, where custom tags added by the **GreenSloth** team to the models can be selected.

With the previously mentioned parts of the **GreenSloth** ecosystem, each model will have an entire page dedicated to it. A curated summary of the model will be shown, including its key features, assumptions, and the scientific question in mind when it was created. This way, users can quickly get an idea of what the model is about. Additionally, the entire documentation can be found on that page, from the mathematical equations to the descriptions of the meta information. Furthermore, the validation of the model reimplementation will be showcased, supported by the five previously mentioned methods to demonstrate the model's capabilities. This way, users can get a good idea of what the model is about, what it can do, and how to use it. It is vital in this process not to take ownership of the model away from the author but rather to give them a platform to showcase their work.

Finally, as the scope of **GreenSloth** is limited to specific models, all models share common ground. This opens the possibility of directly comparing models. There are many ways to compare models, but for **GreenSloth**, five key methods have been chosen to demonstrate the differences between models. The first is a direct comparison of the model's variables, showing where the two models overlap in that regard. This overlap also yields the second method: a direct comparison of simulation results. The third method is a direct comparison of the model's meta-information, reduced to its numbers. The fourth is a side-by-side comparison of the demonstrations performed with each model, as introduced beforehand. The fifth and final method is a simple showcase of each model's schematic overview, a good way to get an idea of each model's structure quickly. These comparison methods will be directly available on the **GreenSloth** platform, allowing users to easily compare different models and eventually find the one that suits their needs.

1.4 The Promise of GreenSloth

GreenSloth will be an ecosystem that aids in the creation, presentation, and sharing of ODE models of photosynthesis. It will include several tools and features to make the process of working, finding and sharing models easier and more accessible. These tools and features, while only being branches of the main tree of **GreenSloth**, are still integral parts of the ecosystem. However, as a tree continues to grow and branch out, so will **GreenSloth**. This is a big project, taking in inspiration from different fields and expertises. A project with a clear future and need in the field of photosynthesis modelling. While the shape, bark, and branches of **GreenSloth** may change and adapt over time, the core premise of this project will always stay:

Photosynthesis Models at your Pace

```
$ GreenSloth-init --help

Usage: GreenSloth-init [OPTIONS] COMMAND [ARGS]...

Options:
  --help  Show this message and exit.

Commands:
  compare-gloss-to-model    Compares glosses to model
  from-model-to-gloss       Generate temporary Glosses from model info.
  initialize                Create '<model-name>' directory.
  latex-from-model          Write LaTex from Model
  python-from-gloss         Write Python Variables from Glossaries
  update-glosses-from-main  Update glosses from main
```

Code-Block 1: The help command of the CLI of GreenSlothUtils

2 Material and Methods

2.1 GreenSlothUtils

The **GreenSlothUtils** package, written in Python, contains several utility functions to assist in packaging an already written kinetic ODE model into a more code-readable and ready for uploading to the **GreenSloth** website format. The functions included in this package vary in their complexity, from simple directory creation to rewriting model files. While **GreenSlothUtils** has been written as a package, there is no intention to upload it to package-sharing services such as PyPI. However, the package is intended to be used as a local package to assist in contributing to the **GreenSloth** website; therefore, the GitHub repository is publicly available at `https://github.com/ElouenCorvest/GreenSlothUtils.git`.

2.1.1 Create the Model Documentation

All the functions included in the package are documented in the GitHub repository; some key functions are also documented below to showcase their utility in this project. The biggest upside of this package is the ability to quickly create a new model directory with all the necessary files and format for uploading to **GreenSloth**. For ease of access, the required functions are combined into a Command-line Interface (CLI).

Easily the most important part of the CLI of **GreenSlothUtils** is the `--help` command (see Code-Block 1), which gives an overview of all the possible commands and options available in the package. Each of the other commands also includes a `--help` option to give more specific information about the command.

Using these commands following a specific order allows for a quick and easy creation of a new model directory.

1. Create the directory

First, the `initialize` command is used to create a new and complete model directory with the correct name and structure (see Code-Block 2). One required argument is the model name, which the directory should be named after. In the **GreenSloth** ecosystem, the model name is the first author's last name followed by the year of publication, e.g., `Corvest2000`. This

```
$ GreenSloth-init initialize --help

Usage: GreenSloth-init initialize [OPTIONS] <model-name>

  Create '<model-name>' directory.

Options:
  -p, --path TEXT  Path to create model directory. Defaults to path here.
  --help           Show this message and exit.
```

Code-Block 2: The initialize command of the CLI of GreenSlothUtils

command also includes one optional argument, `--path`, which allows the user to specify where the new model directory should be created. If no path is given, the directory is created in the current working directory. The directory created contains various sub-directories and files, all pre-formatted to the **GreenSloth** website standards (see Directory Tree 1).

```
Corvest2000/
├── figures/
│   ├── demonstrations.ipynb
│   └── paper_figures.ipynb
├── model/
│   ├── basic_funcs.py
│   ├── derived_quantities.py
│   ├── __init__.py
│   └── rates.py
├── model_info/
│   ├── comps.csv
│   ├── derived_comps.csv
│   ├── derived_params.csv
│   ├── params.csv
│   └── rates.csv
└── README_script.py
```

Directory Tree 1: Example directory structure created by the `initialize` command of GreenSlothUtils.

The `figures` directory consists of two Jupyter Notebooks. The `demonstrations.ipynb` includes all Model Demonstrations for the model, already written out, with each demonstration having its own cell. The model creator must replace `None` values in each string representation of the model parts needed for the demonstration. For more information on what the demonstrations entail, please refer to Section 2.3. The `paper_figures.ipynb` is a blank notebook intended to be used to recreate the figures of the model's publication. Both notebooks already include the necessary imports and helper functions to get started quickly. If the model is correctly implemented in the `model` directory, importing the model should work.

The `model` directory consists of the actual ODE model implementation in Python. For ease of use, the **GreenSloth** ecosystem sets using the `mxlpy` [19] package as a standard for writing kinetic ODE models in Python. This has been chosen due to its ease of use, speed, and continuous support from the maintainers. The most important file of the `model` directory is the `__init__.py` file, which contains the main model initialization. As a strong recommendation, the actual parts of the model, such as the rates, basic functions, and derived quantities, should be written in separate files, as shown in Directory Tree 1. This improves code readability and makes debugging easier. To help with formatting a model written using `mxlpy`, the **GreenSlothUtils** package contains a subpackage, `GreenSlothUtils.mxlpy_formatter`, which contains several functions to assist in rewriting a model into the correct format for **GreenSloth**.

The `model_info` directory contains all the necessary information about the model in a `.csv` format. This includes the model's variables, parameters, derived variables and parameters, and rates. Each of these files follows a specific format, where a short description, the mathematical depiction in the publication and in **GreenSloth**, and the variable used in Python code. Additionally, the variables and rates also include a column of Kyoto Encyclopedia of Genes and Genomes (KEGG) IDs and Glossary Indices to assist in linking the model to existing databases and the **GreenSloth** overarching glossary. The parameters, on the other hand, also include columns for units, values, and sources as given in the publication. These `.csv` files are used to automatically generate the model's glossary entries when uploading to the **GreenSloth** website; therefore, it is important that they are correctly filled out. To assist in this process, functions have been created that extract valuable information from an existing `mxlpy` model implementation and place it in the tables, already separating it into the correct columns. But more on those functions later.

Finally, the `README_script.py` file is a template script intended to write the model's README file for the **GreenSloth** website and, in general, the model's documentation. This script includes several prefilled sections, such as tables extracted from the previously mentioned `model_info` directory, a generic installation guide, and the assumptions and brief description of the model demonstrations. The user must provide specific details about the model, including a summary, a LaTeX- formatted version of the equations used, recreated publication figures, and notes for the demonstrations. Writing the README in this Python script format makes it easier to access repeating information, such as specific variables or parameters, without having to find and replace them in a Markdown file manually. Make a few changes in the script, run it, and the README file is ready to go.

2. Extract Information from the Model

Once the directory is created and the model is completely implemented, the next step is to extract necessary information from the model into the `model_info` tables. This is done using several commands of the **GreenSlothUtils** CLI.

The `from-model-to-gloss` function extracts all the necessary information from an `mxlpy` model into temporary Glossaries as a `.csv` format (see Code-Block 3). The options `--model-dir`, `--modelinfo-dir`, and `--modelgloss-dir` allow the user to specify where the model is located, where the model info tables are located, and where to store the generated Glosses, respectively. If no paths are given, the function assumes that the model directory is in the current working directory, the model info directory is in the model directory under `model_info/`, and the generated Glosses should be stored in a new directory called `model_glosses/` inside the `model_info/` directory. The option `--extract-option` allows the user to specify which parts of the model should be extracted. The possibilities are `all`, `variables`, `parameters`, `derived_variables`,

```
$ GreenSloth-init from-model-to-gloss --help

Usage: GreenSloth-init from-model-to-gloss [OPTIONS]

  Generate temporary Glosses from model info.

Options:
  -md, --model-dir TEXT         Path to model directory. Defaults to path
    here
  -mid, --modelinfo-dir TEXT    Path to model info directory. Defaults to
                                model-dir + 'model_info'
  -mgd, --modelgloss-dir TEXT   Path to where to store csvs. Defaults to
   model-
                                dir + 'model_info/model_glosses/'
  -eo, --extract-option TEXT    Parts of the model to extract.
   Possibilities:
                                   'all',
                                   'variables',
                                   'parameters',
                                   'derived_variables',
                                   'derived_parameters',
                                   'reactions',
                                   [default: 'all']
  --check / --no-check          Check for inconsistencies with
                                'compare_gloss_to_model'
  --help                        Show this message and exit.
```

Code-Block 3: The from-model-to-gloss command of the CLI of GreenSlothUtils

`derived_parameters`, and `reactions`. By default, all parts are extracted. Finally, the `--check` option allows the user to check for inconsistencies between the generated Glossaries and the existing model info tables using the `compare-gloss-to-model` command. It is recommended to point the terminal to the model's overarching directory and run the command with the default values. This is the easiest way to ensure that all paths are correctly set and still follow the custom directory structure of the **GreenSloth** Directory Tree 1. With the automatic extraction of model information, filling out the actual `model_info` tables is significantly easier. The user can then be on the safe side, knowing that no variables, parameters, or rates have been forgotten. However, only the Python variable is extracted, and the other columns must still be filled in manually by the user.

To create a bridge between the different models, overarching glossaries of variables and rates have been developed in the **GreenSloth** ecosystem. With access to these glossaries, only the ID associated with that variable or rate has to be included in the model info tables. The rest of the information is automatically filled out using the `update-glosses-from-main` command (see Code-Block 4). This command updates the existing model info tables with information from the main glossaries, such as the description, KEGG ID, and mathematical depiction in **GreenSloth**. The options `--model-dir`, `--modelinfo-dir`, and `--maingloss-dir` allow the user to specify where the model is located, where the model info tables are located, and where the main glossaries are located, respectively. If no paths are given, the function assumes that the model directory is in the current working directory, the model info directory is in the model directory

```
$ GreenSloth-init update-glosses-from-main --help

Usage: GreenSloth-init update-glosses-from-main [OPTIONS]

  Update glosses from main

Options:
  -magd, --maingloss-dir TEXT  Path to directory with main gloss.
   Defaults to
                               parent of here.
  -md, --model-dir TEXT        Path to model directory. Defaults to path
    here
  -mid, --modelinfo-dir TEXT   Path to model info directory. Defaults to
                               model-dir + 'model_info'
  --add / --no-add             Add new entries to main gloss. Defaults
   to
                               False.
  --help                       Show this message and exit.
```

Code-Block 4: The update-glosses-from-main command of the CLI of GreenSlothUtils

under `model_info/`, and the main glossaries are in the parent directory of the current working directory. The `--add` option allows the user to add new entries to the main glossary if they do not already exist. By default, this option is set to `False`. Again, it is recommended to point the terminal to the model's overarching directory and run the command with the default values. This is the easiest way to ensure that all paths are correctly set and still follow the custom directory structure of the **GreenSloth** Directory Tree 1. If the model includes variables and rates that are not in the main glossaries, this command can automatically add them, assigning a new ID. While this is a very handy feature, it is recommended to critically evaluate the new entries before running `update-glosses-from-main` with the `--add` option, to ensure that no duplicate or incorrect entries are added to the main glossaries.

Once the information has been added to the model info tables, the LaTeX depictions of each model component will serve as the primary liaison between the info tables and the subsequent model documentation, as these depictions will not change. The rest of the information may need to be updated regularly due to changes in the naming convention or other issues.

3. Correct the Model

After updating the model info tables from the main glossaries or deciding on a different name for something, it is vital to ensure the `mxlpy` model implementation still holds the same information. For that, **GreenSlothUtils** includes the `compare-gloss-to-model`, which is already implemented in the `from-model-to-gloss` command as the `--check` option (see Code-Block 3). This command compares the existing model info tables with the actual `mxlpy` model implementation, checking for inconsistencies in variable, parameter, and rate names. If any inconsistencies are found, they are printed to the terminal, allowing the user to find and correct them in the model implementation quickly. This step is crucial to ensure that the model implementation and the model info tables are in sync, as correct documentation is a key aspect of the **GreenSloth** ecosystem. It should be noted that the category of derived variables and parameters is defined by `mxlpy` logic: if something is derived from at least one variable, it is a derived variable; if it is derived only from parameters, it is a derived parameter. Therefore, if the user manually wishes to change the category of a derived quantity, they may do so in the model info tables, but must remember that the `compare-gloss-to-model` command will not check for inconsistencies in this regard.

4. Create Pointer to Python Variables

Once the model info tables and the model implementation are complete, the model documentation can be tackled. One important feature for **GreenSloth** is consistency. To ensure that all the variables, parameters, and rates mentioned in the documentation are consistent along the entire documentation file, a pointer in the `README_script.py` file can be created. To do this easily, the `python-from-gloss` command of **GreenSlothUtils** can be used (see Code-Block 5). This command creates Python variable assignments for all variables, parameters, derived quantities, and rates in the model info tables, and writes them to separate `.txt` files. On top of that, these files are also managed as a log and show the last update done with `python-from-gloss`. The options `--model-dir` and `--modelinfo-dir` allow the user to specify where the model and where the model info tables are located, respectively. If no paths are given, the function assumes that the model directory is in the current working directory and the model info directory is in the `model_info/` directory. The option `--glosstopython-dir` allows the user to specify where to store the generated Python variables. By default, this is set to the `python_written/gloss_to_python` directory in the path given to `--modelinfo-dir`. Once again, it is recommended to point the terminal to the model's overarching directory and run the command with the default values. This is the easiest way to ensure that all paths are correctly set and still follow the custom directory structure of the **GreenSloth** Directory Tree 1.

Once this command is run, the user can copy over the generated Python variable assignments into the `README_script.py` file, creating a direct pointer to the actual variables used in the model info tables. This ensures that any changes made to the variable names in the model info tables are automatically reflected in the documentation, maintaining consistency throughout. This last step of manual copying is to ensure that the user critically evaluates the changes they have made, and with the log feature, shows the last update.

5. Generate LaTeX equations

A big part of the kinetic ODE model documentation is the correct depiction of the equations used in the implementation. To assist in this process, the `latex-from-model` command of **GreenSlothUtils** can be used to automatically generate LaTeX equations from the `mxlpy` model implementation (see Code-Block 6). This command extracts the mathematical representations of variables, parameters, derived quantities, and rates from the `mxlpy` model and writes them to a `.txt` file. The options `--model-dir` and `--modelinfo-dir` allow the user to specify where the model and the model info tables are located, respectively. If no paths are given, the function assumes the model directory is in the current working directory, and the model info directory is `model_info/`. The option `-- modeltolatex - dir` allows the user to specify where to store the generated LaTeX equations. By default, this is set to the `python_written / model_to_latex` directory in the path given to `--modelinfo-dir`. Once again, it is recommended to point the terminal to the model's overarching directory and run the command with the default values. This is the easiest way to ensure that all paths are correctly

```
$ GreenSloth-init python-from-gloss --help

Usage: GreenSloth-init python-from-gloss [OPTIONS]

  Write Python Variables from Glossaries

Options:
  -md, --model-dir TEXT           Path to model directory. Defaults to path
                                  here
  -mid, --modelinfo-dir TEXT      Path to model info directory. Defaults to
                                  -md + 'model_info'
  -gpd, --glosstopython-dir TEXT  Path to gloss to python directory. Defaults
                                  to -mid + 'python_written/gloss_to_python'
  --help                          Show this message and exit.
```

Code-Block 5: The python-from-gloss command of the CLI of GreenSlothUtils

set and still follow the custom directory structure of the **GreenSloth** Directory Tree 1. To generate the LaTeX equations, **GreenSlothUtils** uses the `latexify` package [46], which is specifically designed to extract LaTeX equations from Python functions. However, sometimes Python functions are too complex for `latexify` to handle; therefore, it is recommended to critically evaluate the generated LaTeX equations before using them in the documentation. Once generated, the user can copy the LaTeX equations into the appropriate sections of the `README_script.py` file, noting that the equations already include the Python variable names used in the pointer created in the previous step.

6. Recreate the Publication figures

To validate the recreation of the `mxlpy` model implementation, it is important to recreate the figures of the original publication. Sadly, this step cannot be automated by **GreenSlothUtils** because the figures vary significantly across models. However, the `paper_figures.ipynb` notebook included in the `figures/` directory of the model structure (see Directory Tree 1) is intended to be used for this purpose. The user must enter the necessary code to recreate the figures using the implemented `mxlpy` model. It is recommended to create a `str` dictionary at the top of the notebook containing the string representations of the different parts of the model needed for the simulations. This allows easier changes to variables and parameters without having to search the entire notebook for the correct strings. Once the figures are recreated, a template in the `README_script.py` file is already included to integrate the generated figures into the documentation.

Additionally, a brief description of how this figure was created and a short analysis of the recreation process are needed. A rule of thumb for the caption of a simulation figure is to include enough details to recreate the simulation shown in the figure, without referring back to the code. This rule of thumb is often missing from scientific publications, which is why some figures may be hard, or even impossible, to recreate. If that is the case in the model at hand, it is recommended to include a brief note on what was missing to recreate the figure, to

```
$ GreenSloth-init latex-from-model --help

Usage: GreenSloth-init latex-from-model [OPTIONS]

  Write LaTex from Model

Options:
  -md, --model-dir TEXT           Path to model directory. Defaults to
   path
                                  here
  -mid, --modelinfo-dir TEXT      Path to model info directory. Defaults
   to -md
                                  + 'model_info'
  -mld, --modeltolatex-dir TEXT   Path to model to latex directory.
   Defaults to
                                  -mid + 'python_written/model_to_latex'
  --help                          Show this message and exit.
```

Code-Block 6: The latex-from-model command of the CLI of GreenSlothUtils

ensure full transparency. As this section is supposed to represent a comparison of the figures of the original publication and the `mxlpy` model implementation, showing the figures side by side would be ideal; however, due to possible copyright issues, it is best to only refer the reader to the publication for the original figures.

7. Create the Demonstrations

The last step in the documentation process is to fill out the `demonstrations.ipynb` Notebook included in the `figures/` directory of the model structure (see Directory Tree 1). This notebook already includes all the necessary code to run the model demonstrations described in Section 2.3. The user must fill in the string representations of the different parts of the model required for each demonstration. Some demonstrations may require specific pre-work, but more details can be found in Section 2.3 and in the documentation of **GreenSlothUtils**. Once the demonstrations are filled out, a brief description of each demonstration is already included in the `README_script.py` file, where the user must provide a brief analysis of each demonstration, including how it looks, whether anything did not work, and why.

8. Finishing touches

As the `README_script.py` is now filled out with all the necessary information, the last step is to run the script to generate the final `README.md` file for the model. It is recommended to critically evaluate the generated README file for any mistakes or inconsistencies, especially regarding LaTeX representations, as errors in these are very common. Once the README file is finalized, the model is ready to be included on the **GreenSloth** website.

2.2 Models

To make a good demonstration of the capabilities of **GreenSloth**, five kinetic ODE models of photosynthesis have been chosen to take part in this thesis. These five models, while all showing photosynthesis, vary in their complexity and execution. Some are based on each other, while others stand alone. In the following, a brief description of each model is given. These short summaries are the ones used on the **GreenSloth** website. Additionally, the models are validated by trying to recreate the figures of the original publication, as close as possible. Due to permissions and licensing issues, the original figures will not be present in this thesis nor **GreenSloth**. However, the publication is rightfully cited and so, the user can easily find the original figures, if they have access to the publication.

2.2.1 Bellasio2019

The Bellasio2019 [18] model is a generalized C_3 leaf-photosynthesis model, that includes simplified representations of the light and dark reactions and a stomatal behavior submodule. A lot of its implementation is based on past work by the same author and is mainly inspired by the common FvCB model. The light reactions are modified from Yin et al. (2004) and include the potential rates of ATP and NADPH production based on light intensity. This model has been created with the simple user in mind, and the author has made an effort to show its simplicity, by giving access to a Microsoft Excel Workbook containing the entire model. To showcase the model's capabilities, the author creates common steady-state carbon assimilation curves, against intercellular CO2 concentration and light intensity, and compares them to experimental data from the literature. As many models of photosynthesis rely on purely stead-state assumptions, this model is also validated in dynamic conditions, showing for example the response of the model to a fluctuation of ambient oxygen concentration.

This model was created to stay as simple as possible, while still being able to accurately represent the main features of C_3 photosynthesis. As such, it can be used as a base for more complex models, or as a starting block in larger models of plant physiology. While giving access to the entire model in an Excel Workbook format is transparent and great, the execution of said practice has been inefficient in this instance. The entire mathematical description of the model is also given in the Appendix of the publication, however there are missing or different equations between the publication and the Excel Workbook, which can lead to confusion. On top of that, the simulation protocols used for each figure are only given in small details, which leads to further confusion when trying to reproduce the results and see which equations are correct or not.

2.2.2 Fuente2024

The Fuente2024 [13] model is a kinetic model of photosynthesis that is based on Occam's razor, aiming to provide the minimal complexity to describe the core processes of this model. In this case, the model focuses on the dynamic light oscillation and its responses on the photosynthetic machinery. It focuses only on the light-dependent reactions, including simplified versions of photosystem II, photosystem I, the Plastoquinone pool, and proton and ATP concentration in the lumen and stroma. On top of that, it shows the activation of non-photochemical quenching (NPQ), the dynamics of chlorophyll fluorescence, and the rate of oxygen evolution.

The model includes the oscillating light intensity as a sinusoidal function, where the amplitude and frequency are adjustable parameters. To allow for easier comparison to other models, that often see light intensity as a constant value, the oscillation is defined around a base light intensity. However, the strength of having light with a specific frequency lies in the additional information and therefore analysis possibilities that can be performed. In this case, the model is used to create Bode plots of the response of fluorescence to light oscillations and comparing these results to experimental data from *Chlamydomonas reinhardtii*.

This simple model stays true to its name and the authors aim to provide a base model that can be easily extended, while still showing a new approach to photosynthesis modelling. Their work shows that even with a simple model, new insights can be gained by using dynamic light protocols, which may have been overlooked in traditional steady-state models. To further extend the usability of the model, the authors provide a detailed notebook written in the Wolfram language, which also shows how to recreate some of the publication's figures.

2.2.3 Li2021

The Li2021 [11] model is a kinetic model of photosynthesis that focuses more on the ion fluxes across the thylakoid membrane and their effect on the proton motive force (PMF). It was built on the Davis2017 model [47], focuses more on the photosynthetic reactions that are directly linked to the PMF, such as the water splitting at PSII, the plastoquinone oxidation at the Cytochrome b_6f complex (Cyt b_6f) complex, and more. Other photosynthetic reactions are kept as simple as possible. The Li2021 model adds two potassium ion (K^+) and two chloride ion (Cl^-) ion transport channels to the thylakoid membrane, to investigate their effects on the PMF. To validate the model, the authors compare their simulated results to experimental data from several different experiments. They show that the model can reproduce not only wild type (WT) behavior, but also the behavior of several knockout mutants. The mutants chosen, were the VCCN1, Cl^- channel (CLCE) and K^+/H^+ antiporter 3 (KEA3) knockouts and any combination thereof. After the validation, the model is then used to investigate the impact these ion channels have on the PMF and the resulting photosynthetic efficiency. Several interesting simulation protocols are used, to showcase the model's capabilities, such as a light oscillation protocol, a scan of enzyme abundance and more. Overall, this model was created to answer an already existing question in the field of photosynthesis, which is the role of ion fluxes across the thylakoid membrane.

The model itself is not well presented in the publication, but the authors do provide a link to a public GitHub repository where the model is available. It is written in Python, with many comments added to the code. The script includes many different parts of the model and simulation protocols, therefore it is not clear, what is part of the model used in the publication. The script was summarized as much as possible, to only include the parts relevant to the model, but it is not clear if this interpretation is that of the publication. Between the code and the minimal information given in the publication and its supplementary materials, there are still discrepancies, which makes it hard to fully establish the model and its parameters. However, the model shows a good example of why models of photosynthesis are important and versatile, which is why it was included in **GreenSloth**.

2.2.4 Matuszynska2016

The Matuszynska2016 [12] model, a small kinetic model, was developed to delve deeper into the effect of light memory caused by non-photochemical quenching. The systematic investigation of the Xanthophyll cycle, a combination of the pigments of violaxanthin, antheraxanthin, and zeaxanthin, sparked a series of experiments to determine whether plant light memory can be detected in a timescale of minutes to hours through pulse amplitude modulated chlorophyll fluorescence. The model was then created based on these experimental results, providing a comprehensive description of NPQ dynamics and the short-term memory of the *Arabidopsis thaliana* plant.

To keep the model as simple as possible, several processes not directly linked to NPQ have been simplified to create a dynamic ODE system consisting only of 6 different compounds. With these simplifications, the authors could fulfil an additional goal: to make a general framework that is not specific to one model organism.

To demonstrate the adaptability of their model, the authors took their calibrated *A. thaliana* model and successfully applied it to the non-model organism *Epipremnum aureum*. This adaptation allowed them to simulate realistic fluorescence measurements and replicate all the key features of chlorophyll induction, showcasing the model's versatility and potential for use in a variety of organisms.

2.2.5 Saadat2021

The Saadat2021 [48] model builds upon previous models, particularly the Matuszynska2019 model, by incorporating and modifying various reactions and aspects of photosynthesis. Overall, the model can be divided into three modules: the ascorbate-glutathione cycle, the Calvin-Benson-Bassham (CBB) cycle and thioredoxin reductase-regulated reactions, and the Photosynthetic Electron Transport Chain (PETC).

The model is primarily used to investigate the electron flows around PSI and their relevance to photosynthetic efficiency. Several different analyses have been conducted to validate the model in both steady-state and dynamic environment conditions. The most interesting is the direct comparison of a knockout mutant of the protein PGR5. This protein is known to catalyse the reduction of plastoquinone by ferredoxin. The results of this comparison align with experimental values, which are, however, not presented in the publication but are referenced. Additionally, it is noted that the results should not be interpreted as accurate quantitative data, but rather as a proof of concept for the model.

Overall, the model has one advantage over other photosynthesis models, as it also highlights the importance of other electron flows, not just the PETC. Additionally, not only are the authors open about the model's flaws, but they are also insistent on making their code and analyses available on GitHub. Therefore, this model serves as a good stepping stone for more complex models that aim to incorporate aspects of photosynthesis, which are often simplified in other models.

2.3 Model Demonstrations

2.3.1 Daylight Simulation

The Photosynthetic Photon Flux Density (PPFD) data in a 1-minute resolution used for the daylight simulation demonstration is taken from National Ecological Observatory Network (NEON) at the KONZ site, located in Kansas, USA (39.10077, -96.56307) [49], using the `neonutilities` package [50]. As only a day is shown in the simulation, only data from the 19th of June 2023 is taken into account. To limit the data to only the part of day that has sunlight, and a decent amount at that, the data is filtered to only include data that has a PPFD value higher and equal than $40\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$. This threshold is chosen as it has shown to allow most models to still simulate the photosynthetic machinery, while still being a decent representation of the actual daylight conditions. The filtered data is then used as an input for the PPFD parameter in the daylight simulation demonstration for every model. The protocol is constructed by simulating each point of PPFD data for 60 seconds. Three different outputs are shown in this demonstration: the $v_{\text{RuBisCO|carb}}$, the ATP and Nicotinamide Adenine Dinucleotide Phosphate (NADPH) ratio, and the F yield. If these outputs are not available in the model at hand, the plot is still displayed with the corresponding y-axis label, struck through.

2.3.2 FvCB Add-on

The basis of the FvCB add-on demonstration is based on the min-W variant. If the supplied model already has a A, then a generic FvCB simulation is done and compared to the results from von Caemmerer (2013) [15]. If there is no available instance of A, the two main and mandatory connection points for the add-on are the $v_{\text{RuBisCO|carb}}$ and a quantity that represents CO_2. If these two connection points are not available in the model, the simulation will not run, and only the general FvCB output will be shown.

The $v_{\text{RuBisCO|carb}}$ needs to be in a unit of $\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$. To convert from a concentration per time unit, the space that the stroma occupies in the chloroplast based on the leaf volume needs to be used. The value taken for this add-on was taken from Laisk et al. (2006) [51], where it was estimated and set to $0.0112\,\text{L}\,\text{m}^{-2}$.

As the FvCB model requires a partial pressure of CO_2, specifically the C_i, this quantity can also be given to the add-on if available in the model at hand. If it is not given, but the CO_2 is, a conversion using Henry's law is done to convert the concentration of CO_2 into a partial pressure. The default Henry's law constant used for this conversion is $3.4 \times 10^{-5}\,\text{mM}\,\mu\text{bar}^{-1}$ [52], however, the law constant can also be supplied, when it is a part of the model.

With these two quantities given in the correct format, the FvCB add-on can be added to any kinetic ODE model of photosynthesis. To finally create the representation of A, the add-on includes the Γ^* and R_light, $38.6\,\mu\text{bar}$ and $1\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$, respectively, if not given by the model. With all of these quantities included in the model, the A can be calculated (see Equation 6) and shown in the demonstration plot with a generic FvCB simulation using parameters taken from von Caemmerer (2013) [15].

$$A = v_{\text{RuBisCO|carb}} \cdot \left(1 - \frac{\Gamma^*}{C_\text{i}}\right) - R_\text{light} \tag{6}$$

2.3.3 Standard PAM Simulation

The standard PAM simulation demonstration uses a generic protocol that does not reflect actual experiments, but are in the same spirit as true experimental work. Prior to the actual protocol, the simulation of the model is done under dark conditions for 30 min. Then the actual protocol consists of 22 periods of a length of 2 min, which start with the specific light intensity of that period and ends with a saturating pulse of 3000 µmol m^{-2} s^{-1} for 0.8 s. The first two periods are in dark conditions, followed by 10 periods of actinic light of 1000 µmol m^{-2} s^{-1} and finishes with 10 periods of dark conditions again. The dark condition is set to 40 µmol m^{-2} s^{-1}, as it has been found that many models are not capable of simulating actual zero light conditions.

With this protocol, the simulation of the model is run and the F and NPQ is shown in the demonstration plot. If F is not present in the model, the corresponding plot is still displayed with the y-axis label struck through. If NPQ is not available, but F is, it is calculated by using F_m (see Equation 3).

F_m is extracted by using the simulated F values at the time points where the saturating pulses are applied. To be sure that the actual F_m is found, an interval between the two saturating pulses is taken, and the maximum value in that interval is used as F_m. Most of the time this value is correct, but for full transparency, these values are plotted as triangles on the F plot, additionally to the NPQ.

2.3.4 MCA of Photosynthesis

The MCA demonstration limits its analysis to variables and fluxes that are deemed integral parts of photosynthesis. The variables to be given should account for a representation of Phosphoglycerate (PGA), RuBP, oxidised plastoquinone ($\mathrm{PQ_{ox}}$), oxidised plastocyanin ($\mathrm{PC_{ox}}$), ATP, and NADPH. The rates should include the $v_\mathrm{RuBisCO|carb}$, PSII rate (v_PSII), PSI rate (v_PSI), Cytb$_6$f rate (v_b6f), and ATP synthase rate (v_ATPSynth). The parameters analyzed are supposed to be the control coefficients of each prior mentioned rate, and therefore the MCA simulations are run to steady-state. Each parameter is displaced by ± 0.01% and the results are put into two separate heatmaps, one for the variables and one for the fluxes. If any of the prior mentioned variables or rates are not available in the model, the corresponding row and column in the heatmaps are still displayed, but white and with less opacity.

2.3.5 Fitting of NPQ

The fitting demonstration uses experimental data taken from von Bismarck (2022) [53]. The data consists of measurements of F, F_m, and NPQ calculated (see Equation 3) under different light intensities, following a PAM protocol after a dark adaptation period of 5 min. The data was taken with Maxi Imaging-PAM (Walz, Germany) using Col-0 *A. thaliana* plants and as no details were given, the default settings of the machine are taken. That means, that each saturating pulse is 5000 µmol m^{-2} s^{-1} for 720 ms. Each period consists of a simulation of a specific light intensity and finishes with a saturating pulse, together lasting approximately 1 min. There are four different sections during this PAM protocol. Starting with one dark period with a light intensity of 0 µmol m^{-2} s^{-1}, followed by 10 high actinic light periods of 903 µmol m^{-2} s^{-1} then dropping the actinic light to 90 µmol m^{-2} s^{-1}, also for 10 periods, then again 5 dark periods to finish the protocol.

The protocol used for the simulation is based on the actual data provided. The difference of each timestamp recorded and the PPFD value at that time is taken. With this simulation protocol, the fitting procedure can be created. The parameters to be fitted are given to the fitting routine with only a bound of zero. If the default value of the parameter is lower than

$$
\begin{aligned}
\text{Standard Scale} : x_{\text{new}} \quad &= \frac{x_{\text{old}} - \mu_{\text{data}}}{\sigma_{\text{data}}} \\
\texttt{lmfit} \text{ Calculation} : \text{Residual} &= \text{weights} \cdot (x_{\text{data}} - x_{\text{model}}) \\
\text{Insert Standard Scale} : \text{Residual} &= \frac{1}{\sigma_{\text{data}}} \cdot \big((x_{\text{data|old}} - \mu_{\text{data}}) - (x_{\text{sim|old}} - \mu_{\text{data}}) \big)
\end{aligned} \tag{7}
$$

zero, then that bound will be considered the upper bound, else it is set as the lower bound. With each variation of the parameters, the simulation is run with the prior mentioned protocol. If the model includes NPQ, that output will be fitted to the experimental data. If NPQ is not available, but F is, then the F output will be used to calculate the NPQ using (see Equation 3) and then fitted to the experimental data. If neither are available, the fitting will not be done and the demonstration plot will only show the experimental data.

To actually fit the simulation output to the experimental data, the Levenberg-Marquardt method is used. This method is the most common and basic non-linear fitting algorithm, which is why it was chosen for this demonstration. On top of that, the standard scale method is implemented by default to help the fitting process. As the fitting is done using the `Model` instance from the package `lmfit` [54], the inclusion of standard scaling is done by setting the `weights` argument to the reciprocal fraction of the standard deviation of the experimental data and having both the NPQ results of the experimental data and simulation results be centered around the mean of the data (see Equation 7). As sometimes no results for the simulation occurs depending on the parameter set, a large penalty value is returned to the fitting routine to avoid these parameter sets.

After the best "possible" fit is found, according to the performed simulations, the results are plotted in the demonstration plot, showing both the experimental data points and the fitted simulation line. The F and NPQ are both shown separately for the experimental data and the best fit, and top of that a relative difference plot between the best fit or the original model parameter set is shown against the experimental data. To showcase the changes made in the fitting, a table of the changed parameters and their relative change is also displayed in the demonstration plot. The choosing of the parameters to be fitted is left to the user, as different models may have different parameters that influence NPQ more than others.

2.4 Website

2.4.1 Development

The website was written using the most basic website development tools, namely Hypertext Markup Language (HTML), Cascading Style Sheets (CSS), and JavaScript (JS). Due to a low level of expertise, the development was kept bare-bones. Therefore, no specific framework was used, except the build tool Vite, to enable an easier development and publishing process. The code of the website is available on the **GreenSloth** GitHub repository, only in a separate branch: `https://github.com/ElouenCorvest/GreenSloth/tree/website#`

Most pages and design of the website are basic and straightforward, however some aspects have been packaged in scripts to make the contribution to the site easier. Every page of a model is written purely in JS, to allow for consistency. The script takes the information directly from the model info tables and README file, from the model's respective GitHub directory, and generates the page for that model. This allows a much more streamlined contribution process, as every added model is semi-automatically added to the website, without the need for any HTML or CSS knowledge. To enable a level of security, before any model is uploaded, a small JavaScript Object Notation (JSON) file is used as a dictionary to tell the script which models are allowed to be uploaded to the website that additionally use a template HTML file with their respective name.

To compare two models, a separate page is available, which consists of two select boxes, where any of the models on the website can be selected. Here, many different aspects of the models are compared, such as the variables that are common in both models, a simple light simulation of these variables, the meta-information of the models, the two schemes, and the results of the model demonstrations side by side. All the information for this page is taken from the same sources as the pages of each model, except the light simulations. These need to be created for a model, before uploading, however, a python script that takes the included models, extracts them from the official **GreenSloth** GitHub repository, and simulates the light response curves for the different light intensities is included in the website repository.

2.4.2 Hosting

The website is hosted on an official RWTH Aachen site, which is provided by the university for free. The Vite building tool is used to build the output version of the website, which is then uploaded to the server. The website is supported by the university IT center for the backend operations, while all the content and design is maintained and belong to the Matuszyńska Lab. The latest online version of the website is available at `https://greensloth.rwth-aachen.de/`.

2.5 Additional Methods

2.5.1 Hardware

This entire Thesis and the development of the **GreenSloth** ecosystem was done on a single computer. The specifications of this computer are as follows:

- Processor: 13th Gen Intel® Core™i7-13700 @ 5.10 GHz
- Memory: 128 GB DDR4 @ 4800 MHz
- Graphics: Dedicated NVIDIA T400 (4 GB GDDR6).
- Operating System: Ubuntu 24.04.3 LTS

2.5.2 Simulations

As every model was developed using the `mxlpy` package [19], all simulations were performed using the built-in simulation functions and their respective default settings. This includes using the Assimulo integrator as the integrator for every simulation.

2.5.3 Usage of AI

To stay completely transparent during this thesis, it has to be noted that several Artificial Intelligence (AI) tools have been used to assist in the code writing, documentation, and text writing process. Several LLMs have been used, including Gemini (Google) to help with finding models, ideas and writing code snippets for visualisation. Additionally, GitHub Copilot has been used to assist in code writing, mainly to streamline the writing and debugging process. The exact parts of this thesis that have been assisted by AI tools cannot be exactly determined, as the suggestions are mainly used to inspire and speed up the writing process. However, every part of this thesis has been critically evaluated and edited by the author to ensure accuracy and quality.

3 Results

3.1 Model Validations

All five models have been successfully implemented, whereas validation using the figures from the respective publications has been successful to varying degrees. These five chosen models differ significantly in their approaches to modelling photosynthesis and complexity. While it is hard to pinpoint an exact measurement of a model's complexity, a common way is to look at the number of quantities used in the model [55]. The highest number of quantities is found in the Saadat2021 model, amassing a total of 292, which dwarfs the much smaller models, Fuente2024 and Matuszynska2016, each with a total of 57 and 73 quantities, respectively. The Bellasio2019 and Li2021 models have 134 and 96 quantities, respectively, making them both medium-sized models in respect to the models analyzed here.

Table 1: Summary of Meta-Information of All the Models.
The number of variables, parameters, reactions, and derived variables and parameters, and total sum are shown for each model. These values have been extracted directly from the model implementations. A derived quantity, is a quantity that is calculated inside the model. The separation between derived variable and parameter is based on expertise and is not based on a strict rule. However, a main aspect that was considered was with what quantity it was derived from. If the quantity was derived from even a single time-dependent quantity, it is considered a derived variable. The model used are Bellasio2019 [18], Fuente2024 [13], Li2021 [11], Matuszynska2016 [12], and Saadat2021 [48]

	Variables	Parameters	Reactions	Derived Variables	Derived Parameters	Total
Bellasio2019	13	82	17	6	16	134
Fuente2024	5	31	10	9	2	57
Li2021	13	43	20	13	7	96
Matuszynska2016	6	42	8	11	6	73
Saadat2021	30	167	46	25	24	292

*Editor's note: In the next sections, the figures of the original publications of each model have been recreated. An attempt was made to complete the recreation process as closely as possible to make the comparison as easy as possible. It is important to note that all figures in this section are our own recreations and **none** have been taken from the original publications. On top of that, these results will only show the differences between the recreations and the original figures. They will not present the actual scientific findings of these figures, as they were already reported in each publication. It is advised to look at the original publications, as these figures are not included here.*

3.1.1 Bellasio2019

All five figures of the publication [18] could be recreated, however, each to different degrees. In the recreation of the publication's third figure, all but the d subfigure show a matching recreation of the curves as the publication (see Fig. 3). The d subfigure, showing the stomatal conductance of water vapor to the atmosphere (g_s), shows a higher value for both the ambient O_2 and low O_2 curves in the lower ambient CO_2 concentration (C_a) range. In contrast, at the other end of the spectrum, both curves show a much higher similarity to the publication. It should be noted that the publication curves of the C_a scan do not align on the x-axis, a difference that has not been reproduced. This same issue persists in the fourth figure; however, its recreation was successful (see Fig. 4).

The fifth figure shows mixed results in its recreation (see Fig. 5). In all subfigures of the recreation, the first phase of the transition was placed before the zero point of the x-axis, as in the publication. This was done to make it easier to interpret whether the simulation run that came first also had problems. The results of A, $v_{ATPSynth}$, rate of the reaction mediated by FNR (v_{FNR}) at a transition of $50\,\mu mol\,m^{-2}\,s^{-1}$ to $1500\,\mu mol\,m^{-2}\,s^{-1}$ match the publication well (see Fig. 5a), while the reverse transition shows a slightly slower acclimation to the new light intensity (see Fig. 5b). This same issue persists with the function that relatives RuBP concentration to concentration of RuBisCO active sites (f(RUBP)) in the high to low acclimation (see Fig. 5d), while the f(RUBP) in the low to high transition has the same results at the start, but ends at a lower value than the publication (see Fig. 5c). Both the RuBisCO activation state (R_{act}) and the g_s were identically recreated (see Fig. 5c and d). The results of PGA end at the identical value to the publication. However, the transition is much faster in the low to high light and much slower in the high to low light (see Fig. 5e and f). The inorganic phosphate (P_i) results show the same slow acclimation during the low-to-high transition; however, the recreation recovers from the fall to a much higher value than the publication (see Fig. 5e). During the high-to-low transition, the recreation shows a small negative dip after the transition point, then rises to a stable value. At the same time, the publication has a steady rise to a peak, which falls off a small bit to the same stable value as the recreation (see Fig. 5f). The RuBP and dihydroxyacetone phosphate (DHAP) curves in the high to low transition both show a very similar pattern to the publication, except for the same acclimation problem explained prior with f(RUBP) (see Fig. 5f). On the low-to-high transition, however, the DHAP curve is deemed identical to that reported in the publication. In contrast, the RuBP curve shows a faster, but lower-peak, acclimation than the publication. On top of that, the stable value reached by the RuBP curve is also lower in the recreation$f$(RUBP) (see Fig. 5e). Lastly, both the ratios of ATP and NADPH to their respective totals show a much faster acclimation in the low to high transition, while stabilizing at a higher value than the publication (see Fig. 5g). The same acclimation issue as with f(RUBP) is seen in the high to low transition, while the stable value of the ATP curve is higher and that of the NADPH curve is lower compared to the publication (see Fig. 5h)

The sixth figure shows no similarity to the publication at all. However, many transition points, the zero-point on the axis, show a change in the slope of the continuing curves, hinting at a change in the simulation (see Fig. 6). The seventh and final figure on the other hand, was successfully recreated, with both the A and efficiency of photosystem II (Φ_{PSII}) curves showing the same pattern and values as the publication (see Fig. 7). The only difference, is that the A at the transition point of ambient to low O_2 is much smoother in the recreation than it is in the publication.

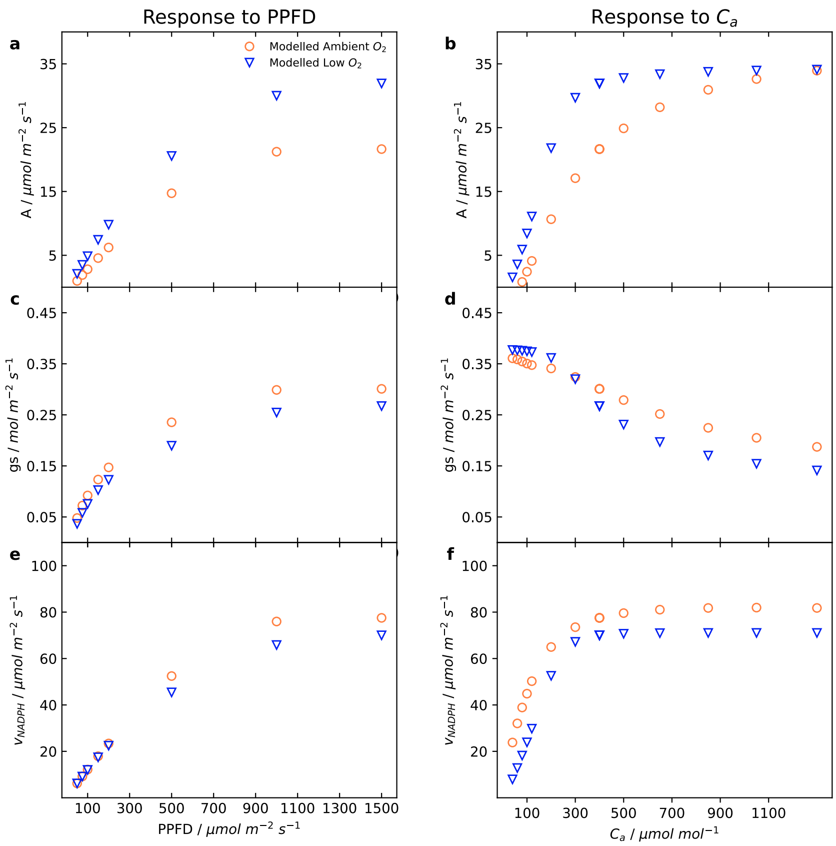

Figure 3: Simulated quasi-steady-state PPFD and C_a response curves of different rates. A quasi-steady-state scan of Photosynthetic Photon Flux Density (PPFD) and ambient CO_2 concentration (C_a) at two different oxygen (O_2) concentrations, 210 000 µbar in orange and 20 000 µbar in blue. The rates shown, from top to bottom, are carbon assimilation (A), stomatal conductance of water vapor to the atmosphere (g_s), and rate of the reaction mediated by FNR (v_{FNR}). The scan of PPFD is done at a fixed ambient CO_2 concentration (C_a) of 400 µbar, while the scan of C_a is done at a fixed PPFD of 1500 $\mu mol\ m^{-2}\ s^{-1}$. The other parameters are kept at their default values. The PPFD values used are 50, 75, 100, 150, 200, 500, 1000, and 1500. For C_a are 400, 300, 200, 120, 100, 80, 60, 40, 400, 400, 400, 400, 500, 650, 850, 1050, and 1300. As a steady-state cannot always be reached within this model, a quasi-steady-state is assumed to be at a maximum time of 1800 s. This figure is recreated from figure 3 of the original publication of the Bellasio2019 model [18]. The experimental data could not be plotted because there was no access to them. In the publication, it is stated that a scan of intercellular CO_2 concentration (C_i) was performed instead of C_a; however, for various reasons, C_a is used here, even though the x-axis values are not identical.

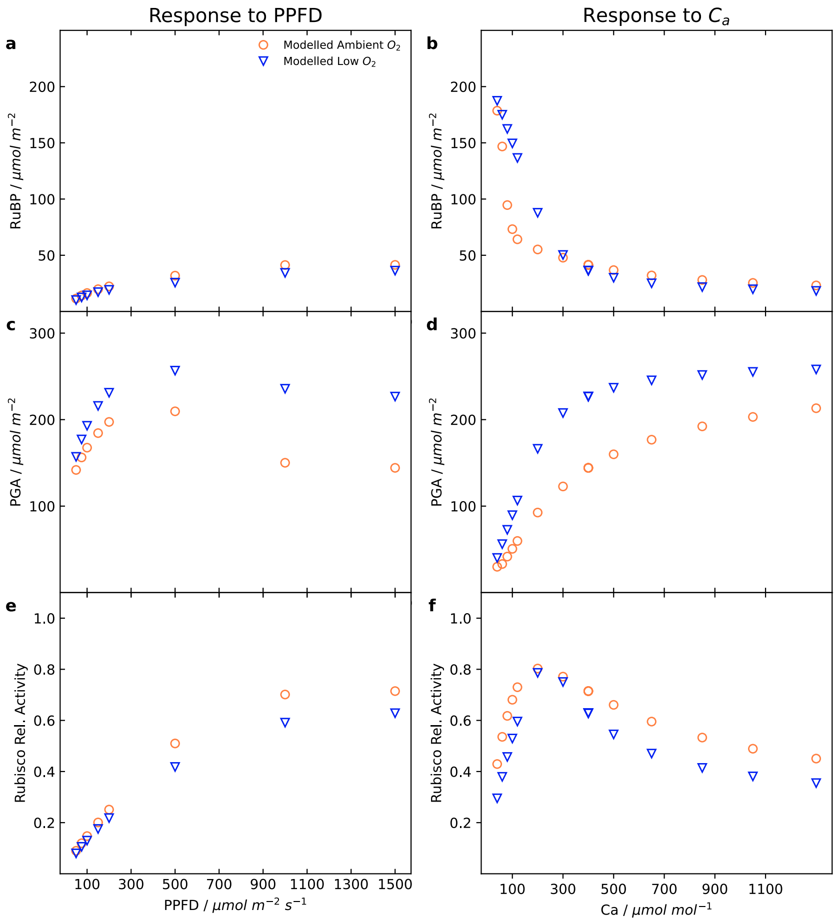

Figure 4: Simulated quasi-steady-state PPFD and C_a response curves of metabolic concentrations.
A quasi-steady-state scan of Photosynthetic Photon Flux Density (PPFD) and ambient CO_2 concentration (C_a) at two different oxygen (O_2) concentrations, 210 000 µbar in orange and 20 000 µbar in blue. The metabolic concentrations shown, from top to bottom, are Ribulose-1,5-bisphosphate (RuBP), Phosphoglycerate (PGA), and the relative Ribulose-1,5-bisphosphate carboxylase-oxygenase (RuBisCO) activity. The first two mentioned were multiplied by the mesophyll volume per m^2 of leaf (V_m) to come to a per area unit, and the last is a product of the function that relatives RuBP concentration to concentration of RuBisCO active sites (f(RUBP)) and the RuBisCO activation state (R_{act}). The scan of PPFD is done at a fixed ambient CO_2 concentration (C_a) of 400 µbar, while the scan of C_a is done at a fixed PPFD of 1500 µmol m^{-2} s^{-1}. The other parameters are kept at their default values. The PPFD values used are 50, 75, 100, 150, 200, 500, 1000, and 1500. For C_a are 400, 300, 200, 120, 100, 80, 60, 40, 400, 400, 400, 400, 500, 650, 850, 1050, and 1300. As a steady-state cannot always be reached within this model, a quasi-steady-state is assumed to be at a maximum time of 1800 s. This figure is recreated from figure 4 of the original publication of the Bellasio2019 model [18]. The experimental data could not be plotted because there was no access to them. In the publication, it is stated that a scan of intercellular CO_2 concentration (C_i) was performed instead of C_a; however, for various reasons, C_a is used here, even though the x-axis values are not identical.

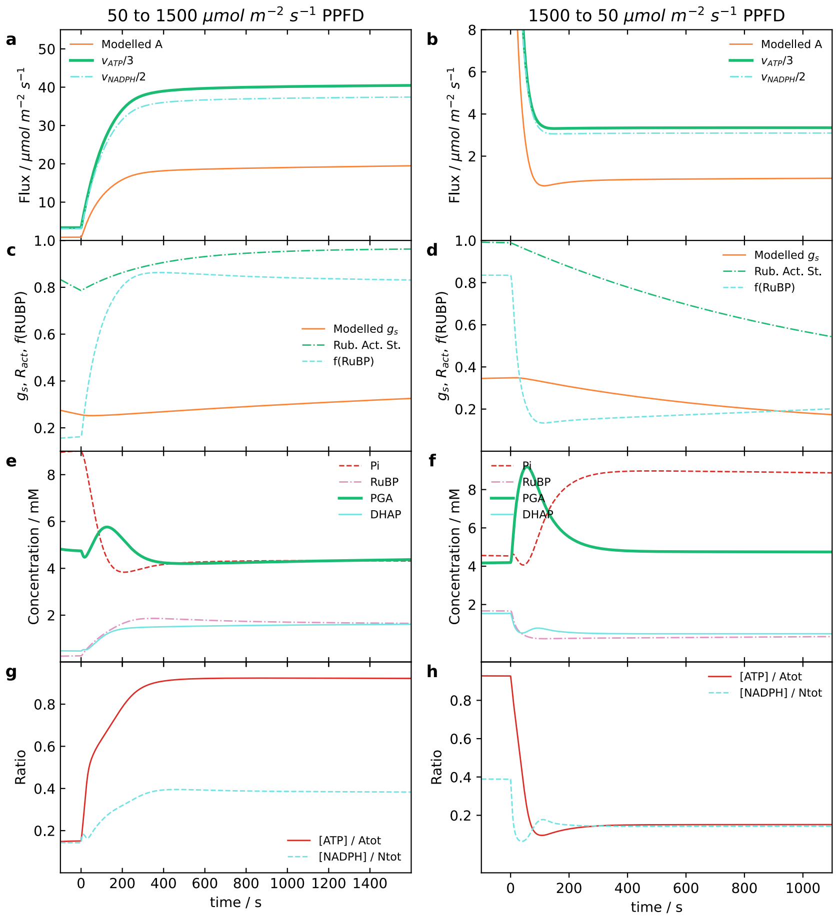

Figure 5: Simulated transition between different PPFD.
The simulations were done with an acclimation period of 400 s and a second period that reaches quasi-steady-state at 1800 s. Each period is done with a different Photosynthetic Photon Flux Density (PPFD) to show a transition. On the left, a low to high light intensity, 50 µmol m^{-2} s^{-1} and 1500 µmol m^{-2} s^{-1}, transition is done, while on the right the reverse is done. This switch is the zero point on the x-axis. The simulation is run using the default parameters, except ambient CO_2 concentration (C_a) (= 350 µbar), maximum rate of RuBisCO carboxylation ($V_{max|RuBisCO}$) (= 0.18 mmol m^{-2} s^{-1}), value of τ in the dark (τ_0) (= −0.12), time constant for increase in g_S ($K_{i|g_S}$) (= 3600 s), and time constant for decrease in g_S ($K_{d|g_S}$) (= 1200 s). These parameters were chosen to mimic the environment of the experimental work done. The top row shows the carbon assimilation (A), the ATP synthase rate ($v_{ATPSynth}$), and rate of the reaction mediated by FNR (v_{FNR}). The top-middle row shows the stomatal conductance of water vapor to the atmosphere (g_s), RuBisCO activation state (R_{act}), and function that relatives RuBP concentration to concentration of RuBisCO active sites (f(RUBP)). The bottom-middle row shows the inorganic phosphate (P_i), Ribulose-1,5-bisphosphate (RuBP), Phosphoglycerate (PGA), and dihydroxyacetone phosphate (DHAP). Lastly, the bottom row shows the ratio of Adenosine Triphosphate (ATP) and Nicotinamide Adenine Dinucleotide Phosphate (NADPH) to their respective totals. This figure is recreated from figure 5 of the original publication of the Bellasio2019 model [18]. The experimental data could not be plotted, as there was no access to it.

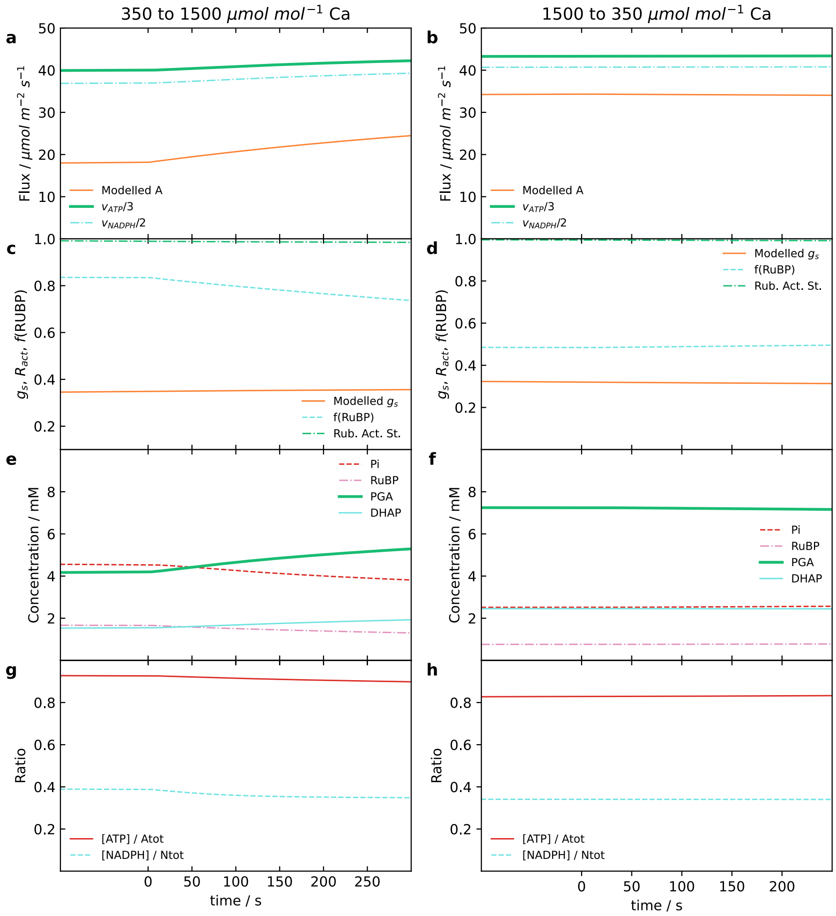

Figure 6: Simulated transition between different C_a.
The simulations were done with an acclimation period of 400 s and a second period that reaches quasi-steady-state at 500 s. Each period is done with a different ambient CO_2 concentration (C_a) to show a transition. On the left, an ambient to high carbon dioxide (CO_2) concentration, 350 µbar and 1500 µbar, transition is done, while on the right the reverse is done. This switch is the zero point on the x-axis. The simulation is run using the default parameters, except Photosynthetic Photon Flux Density (PPFD) ($= 1500\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$), maximum rate of RuBisCO carboxylation ($V_{\mathrm{max|RuBisCO}}$) ($= 0.18\,\mathrm{mmol\,m^{-2}\,s^{-1}}$), value of τ in the dark (τ_0) ($= -0.12$), time constant for increase in g_S ($K_{\mathrm{i}|g_S}$) ($= 3600\,\mathrm{s}$), and time constant for decrease in g_S ($K_{\mathrm{d}|g_S}$) ($= 1200\,\mathrm{s}$). These parameters were chosen to mimic the environment of the experimental work done. The top row shows the carbon assimilation (A), the ATP synthase rate (v_{ATPSynth}), and rate of the reaction mediated by FNR (v_{FNR}). The top-middle row shows the stomatal conductance of water vapor to the atmosphere (g_s), RuBisCO activation state (R_{act}), and function that relatives RuBP concentration to concentration of RuBisCO active sites (f(RUBP)). The bottom-middle row shows the inorganic phosphate (P_i), Ribulose-1,5-bisphosphate (RuBP), Phosphoglycerate (PGA), and dihydroxyacetone phosphate (DHAP). Lastly, the bottom row shows the ratio of Adenosine Triphosphate (ATP) and Nicotinamide Adenine Dinucleotide Phosphate (NADPH) to their respective totals. This figure is recreated from figure 6 of the original publication of the Bellasio2019 model [18].

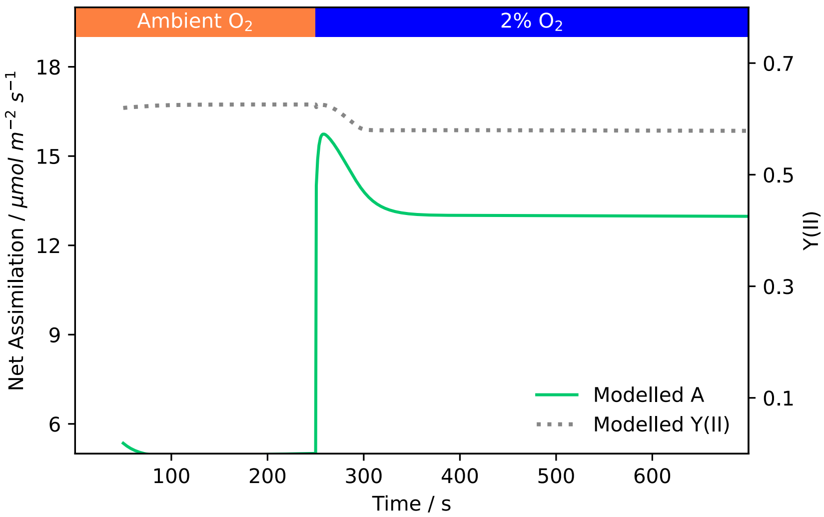

Figure 7: Simulated transition from ambient to low O_2 concentration. Response of the model to a transition from ambient oxygen (O_2) concentration (210 000 µbar), until steady-state is reached, to low oxygen (O_2) concentration (20 000 µbar). The simulation is run using the default parameters, except Photosynthetic Photon Flux Density (PPFD) (= 300 µmol m^{-2} s^{-1}) and ambient CO_2 concentration (C_a) (= 200 µbar). Shown in the plot are carbon assimilation (A) (left y-axis) and efficiency of photosystem II (Φ_{PSII}) (right y-axis). This figure is recreated from figure 6 of the original publication of the Bellasio2019 model [18]. The experimental data could not be plotted, as there was no access to it.

3.1.2 Fuente2024

The recreation of the second figure of the Fuente2024 model [13] shows a good match to the publication, except for the subfigure at the top right (see Fig. 8). The PC_{ox} curve fits the same trend as the publication, but is shifted slightly higher. The same applies to the third figure, where all but the PQ_{ox} and the F curves are near to identical to the publication (see Fig. 9). Both inaccurate curves are shifted slightly lower in the recreation, especially noticeable in the higher light intensity range.

The fourth figure shows many similarities to the publications, yet still encompasses issues (see Fig. 10). The low oscillating light simulation (between 50 µmol m^{-2} s^{-1} and 150 µmol m^{-2} s^{-1}) shows a consistent similarity to the publication for the lumenal pH, the ratio of ATP to total adenylate stromal concentration (AP_{tot}), and the ratio of the fraction of PSI donors per RCII that are available for the linear electron transport (PSI_{ox}) to total PSI (PSI_{tot}) for all six different frequencies. The curves of the activated PSII quencher (Q_{active}) during the low oscillating light stay true to the publication in the lower frequencies ($\frac{1}{10000}$ s^{-1}, $\frac{1}{1000}$ s^{-1}, $\frac{1}{100}$ s^{-1}, the first three rows respectively), while shifting downwards in the higher frequencies (s^{-1}, $\frac{1}{0.01}$ s^{-1}, $\frac{1}{0.001}$ s^{-1}). The recreated PQ_{ox} in low oscillation show an upward shift in the lower frequencies, while in the frequencies of $\frac{1}{0.01}$ s^{-1} and $\frac{1}{0.001}$ s^{-1} the shift is ever so slightly down. The high oscillating light simulation (between 50 µmol m^{-2} s^{-1} and 950 µmol m^{-2} s^{-1}) is more consistent in its recreation, while still incorporating differences. Both the PQ_{ox} and the ratio of PSI_{ox} show an upward shift compared to the publication in all frequencies. In the lower frequencies ($\frac{1}{10000}$ s^{-1}, $\frac{1}{1000}$ s^{-1}, $\frac{1}{100}$ s^{-1}) the lumenal pH, the Q_{active}, and the ratio of ATP all are identical to the publication. However, in the other frequencies, the latter mentioned is shifted upwards, while the other two

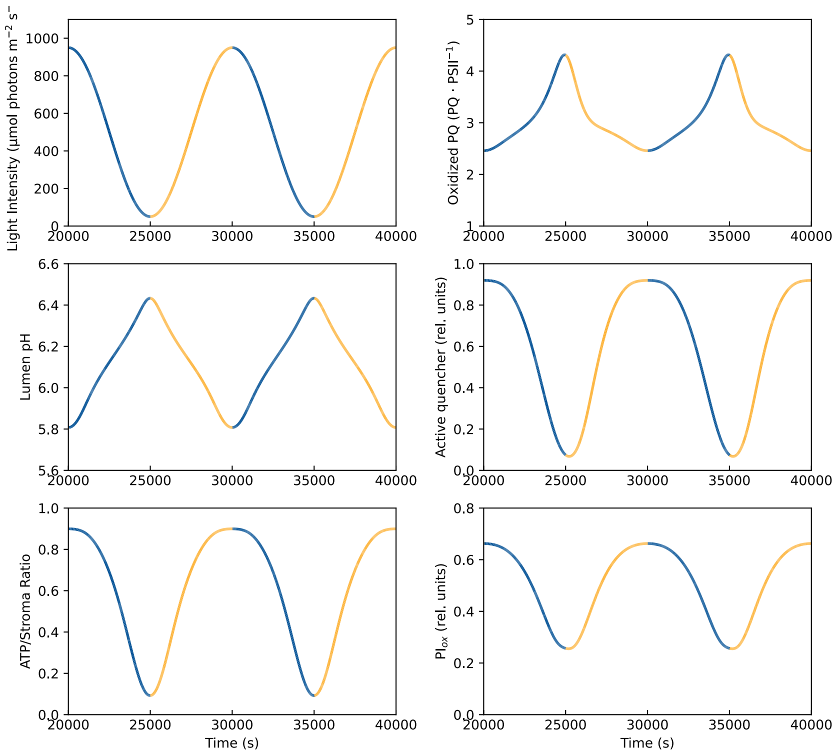

Figure 8: Results at the end of a long simulation with oscillating light.
The light intensity used in the simulation is oscillating between $50\,\mu mol\,m^{-2}\,s^{-1}$ and $950\,\mu mol\,m^{-2}\,s^{-1}$ with a frequency of $\frac{1}{10000}\,s^{-1}$, as seen in the upper left plot. The ascent of light is shown in yellow, while the descent is shown in blue. The results shown are the lumenal pH (middle left), the ratio of Adenosine Triphosphate (ATP) in the stroma to the total adenylate stromal concentration (AP_{tot}) (bottom left), the oxidised plastoquinone (PQ_{ox}) (top right), the activated PSII quencher (Q_{active}) (middle right), and the the fraction of PSI donors per RCII that are available for the linear electron transport (PSI_{ox}) (bottom right). To calculate the lumenal pH, the following equation was used: $pH_{lu} = -\log_{10}\left(\left[H^+_{lu}\right] \cdot 10^{-6}\right)$. The simulation is run using the default parameters and is simulated to a time of 40 000 s, whereas only the last 20 000 s are shown in the figure. This figure is recreated from figure 2 of the original publication of the Fuente2024 model [13].

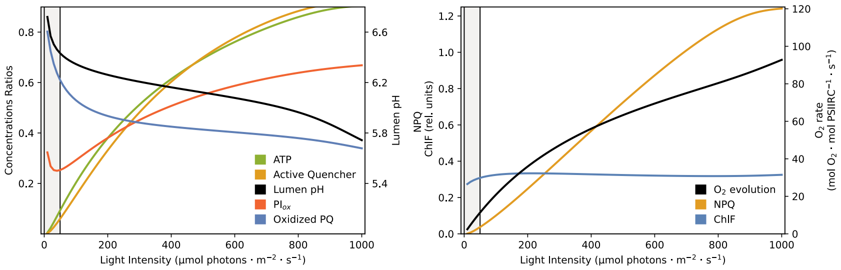

Figure 9: Steady-state scan of light intensities for several components.
A steady-state scan of light intensities, from $0\,\mu mol\,m^{-2}\,s^{-1}$ to $1000\,\mu mol\,m^{-2}\,s^{-1}$ was done. The left plot shows the ratio of Adenosine Triphosphate (ATP) in the stroma to the total adenylate stromal concentration (AP_{tot}) (green), the activated PSII quencher (Q_{active}) (yellow), the lumenal pH (black), the ratio of the fraction of PSI donors per RCII that are available for the linear electron transport (PSI_{ox}) to the total PSI (PSI_{tot}) (orange), and the ratio of oxidised plastoquinone (PQ_{ox}) to the total plastoquinone (PQ_{tot}). To calculate the lumenal pH, the following equation was used: $pH_{lu} = -\log_{10}\left(\left[H^+_{lu}\right] \cdot 10^{-6}\right)$. The right plot shows the O_2 evolution (black), the Non-Photochemical Quenching (NPQ) (yellow), and the chlorophyll fluorescence (F) (blue), all directly taken from the simulation results. Note the two different y-axis on both plots to showcase different scales. The simulations are run using default parameters, while removing the oscillating light mechanism by setting the amplitude of the oscillation to zero. The light intensity is then inputted as a constant value for each simulation. This figure is recreated from figure 3 of the original publication of the Fuente2024 model [13].

down. It has to be noted that most mentioned shifts are minimal, except for the shifts of the Q_{active} in the low frequencies of both oscillating lights and the shifts of the PQ_{ox} in the three highest frequencies.

The fifth figure also has similar issues as the last (see Fig. 11). The O_2 evolution rate is similar to the publication for all the frequencies at the low oscillation, while at the high oscillation it shows an upward shift in the four lower frequencies, while having a downward shift in the other frequencies. The NPQ shows a good match to the publication in both oscillations for the three lower frequencies, while shifting in the higher frequencies. In the low oscillation, the NPQ shifts downwards, while in the high oscillation it shifts upwards. The chlorophyll F holds a consistent trend that is comparable to the publication in the low oscillation, and only shows a downward shift in the three highest frequencies. However, the high oscillation results not only show a shift, but also do not follow the correct curvature. In the recreation, the curves are all much more flattened than in the publication, especially in the lower frequencies.

As the prior figures show, the implementation struggled significantly with the F, therefore the recreation of the last three figures of the publication, all showing various aspects of this variable, was deemed not possible.

3.1.3 Li2021

Some subfigures of the recreation of the third figure of the Li2021 model [11] show a good match to the publication (see Fig. 12). In the top row, the NPQ curve of the WT with a light intensity of $500\,\mu mol\,m^{-2}\,s^{-1}$ (first row, third plot) follows the same curve as the other light intensity, a stark rise at the start that forms a peak that slowly falls off to a stable value until the 20 min mark, and then quickly falls to zero during the dark period. While in the publication, it slowly rises to the stable value and stays there, without first forming a peak, then also quickly drops to zero in the dark. The two last plots could not be reproduced at all, as not enough information

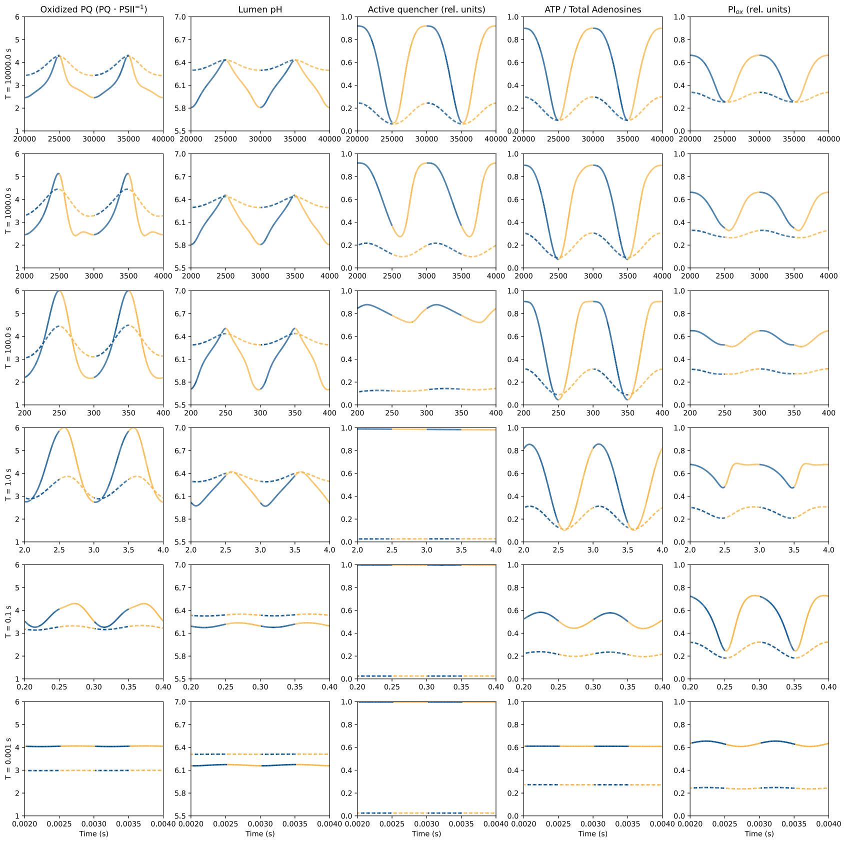

Figure 10: Results of dependent variables under differing light intensity oscillations conditions.
Different simulations were done using different settings of the light oscillation. Each row of plots correspond to a different frequency used in the oscillation, with $\frac{1}{10000}\,\mathrm{s}^{-1}$, $\frac{1}{1000}\,\mathrm{s}^{-1}$, $\frac{1}{100}\,\mathrm{s}^{-1}$, $\frac{1}{1}\,\mathrm{s}^{-1}$, $1\,\mathrm{s}^{-1}$, and $\frac{1}{0.001}\,\mathrm{s}^{-1}$, from top to bottom. Additionally, each frequency was simulated twice with differing amplitudes of oscillation. The oscillation was either between $50\,\mathrm{\mu mol\,m^{-2}\,s^{-1}}$ and $950\,\mathrm{\mu mol\,m^{-2}\,s^{-1}}$ (solid) or between $50\,\mathrm{\mu mol\,m^{-2}\,s^{-1}}$ and $150\,\mathrm{\mu mol\,m^{-2}\,s^{-1}}$ (dashed). The ascent of light is shown in yellow, while the descent is shown in blue. The results shown are the oxidised plastoquinone (PQ_{ox}), the lumenal pH, the activated PSII quencher (Q_{active}), the ratio of Adenosine Triphosphate (ATP) in the stroma, and the the fraction of PSI donors per RCII that are available for the linear electron transport (PSI_{ox}), from left to right. To calculate the lumenal pH, the following equation was used: $\mathrm{pH_{lu}} = -\log_{10}\left(\left[\mathrm{H}^+_{\mathrm{lu}}\right]\cdot 10^{-6}\right)$. The simulations are run using the default parameters, while changing the settings of the light oscillation as described. This figure is recreated from figure 4 of the original publication of the Fuente2024 model [13].

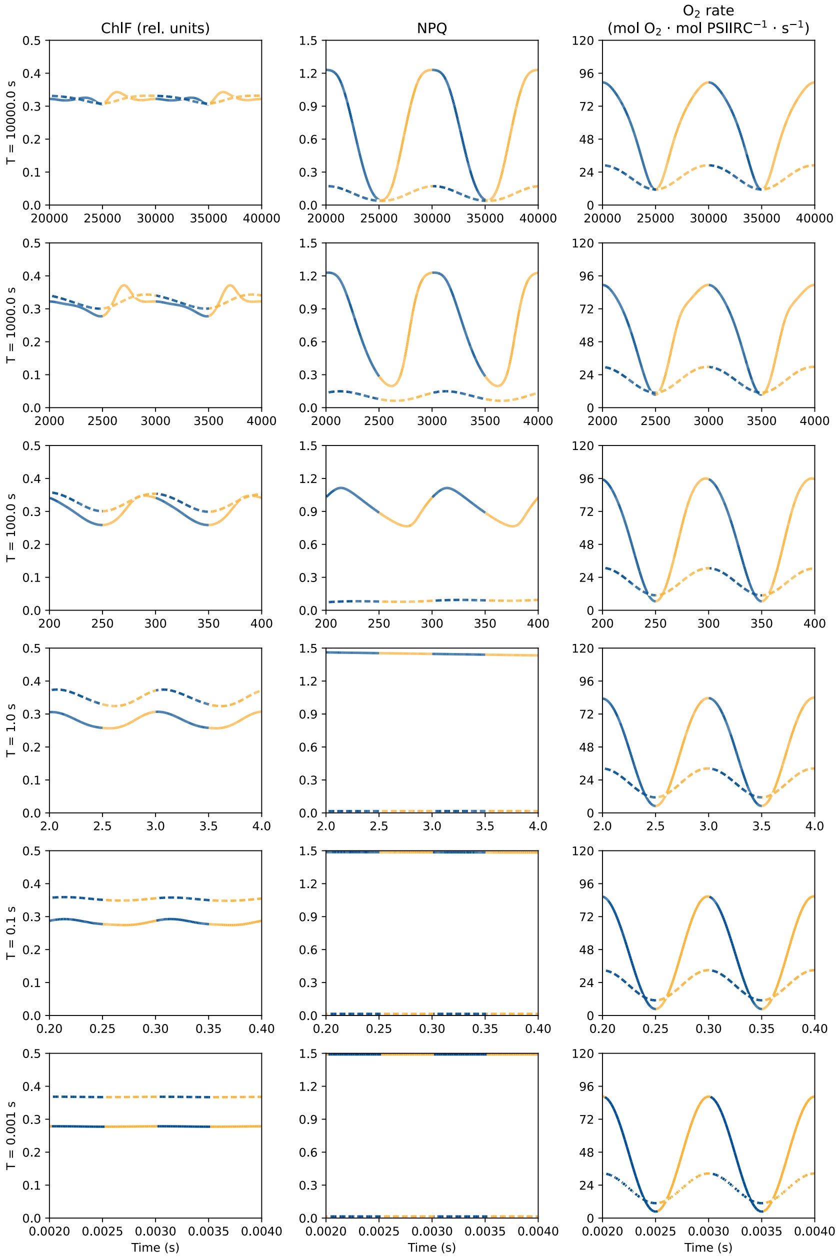

Figure 11: Results of independent variables under differing light intensity oscillations conditions.
Different simulations were done using different settings of the light oscillation. Each row of plots correspond to a different frequency used in the oscillation, with $\frac{1}{10000}\,\mathrm{s}^{-1}$, $\frac{1}{1000}\,\mathrm{s}^{-1}$, $\frac{1}{100}\,\mathrm{s}^{-1}$, $\frac{1}{1}\,\mathrm{s}^{-1}$, $1\,\mathrm{s}^{-1}$, and $\frac{1}{0.001}\,\mathrm{s}^{-1}$, from top to bottom. Additionally, each frequency was simulated twice with differing amplitudes of oscillation. The oscillation was either between $50\,\mathrm{µmol\,m^{-2}\,s^{-1}}$ and $950\,\mathrm{µmol\,m^{-2}\,s^{-1}}$ (solid) or between $50\,\mathrm{µmol\,m^{-2}\,s^{-1}}$ and $150\,\mathrm{µmol\,m^{-2}\,s^{-1}}$ (dashed). The ascent of light is shown in yellow, while the descent is shown in blue. The results shown are the chlorophyll fluorescence (F), the Non-Photochemical Quenching (NPQ), and the O_2 evolution, from left to right. The simulations are run using the default parameters, while changing the settings of the light oscillation as described. This figure is recreated from figure 5 of the original publication of the Fuente2024 model [13].

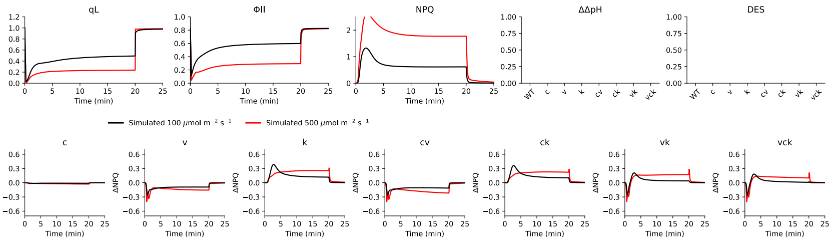

Figure 12: Simulation results of simple light protocol under differing light intensities and mutants.
A simple light protocol consisting of a light period of 20 min, with a light intensity of 100 µmol m^{-2} s^{-1} (black) or 500 µmol m^{-2} s^{-1} (red), and then a dark period of 5 min was used. This protocol was simulated for several genotypes of *Arabidopsis thaliana*, including the wild type (WT), and the knockout mutants of the Cl^- channel (CLCE) (c), voltage-gated Cl^- channel 1 (VCCN1) (v), K^+/H^+ antiporter 3 (KEA3) (k), and every variation thereof (cv, ck, vk, vck). The results shown in the top row are the (), efficiency of photosystem II (Φ_{PSII}), and the Non-Photochemical Quenching (NPQ) of the WT simulation, for each light intensity. Additionally, there are two empty plots, that are left to make the figure more comparable to the publication, but could not be reproduced. The bottom row shows the difference of NPQ between the mutant and the WT simulations, for both light intensities. The mutant depicted in the plot is shown in the title of each subplot. The simulations were run using the default parameters, while changing the Photosynthetic Photon Flux Density (PPFD) to match the light intensities. To create each mutant model, the corresponding rate constant of the rate being knockout was set to zero, for e.g. the rate constant of KEA3 (k_{KEA}). This figure is recreated from figure 4 of the original publication of the Li2021 model [11].

was given to be able to recreate them using the model. The first four plots of the bottom row were able to be recreated fully, while the last three show discrepancies with the light intensity of 500 µmol m^{-2} s^{-1}. As these plots show the difference between the mutant and the WT NPQ, the problem of the prior described plot persists in these plots, with the beginning of each curve being shifted upwards compared to the publication.

The fourth figure was recreated to a good extent, with every plot representing the 100 µmol m^{-2} s^{-1} light intensity showing a good match to the publication (see Fig. 13). However, figure the 500 µmol m^{-2} s^{-1} light intensity plots show a slight discrepancy, just like in the last figure. All curves follow the approximate trends, while some, like the proton gradient between lumen and stroma (ΔpH) and the PMF curves of the WT end at too high of a value, or the beginning of the curves show small dips and peaks, like the flux of concentration of K^+ in the lumen (${K_{lu}}^+$) in the VCCN1 mutant that rises and forms two peaks, one after another. These inconsistencies can be found throughout the recreated figure, however, only in the plots of the higher light intensity.

The recreation of the fifth figure has also some mixed results (see Fig. 14). The top row shows a good representation of the first plot, but the second plot has the same problem as the prior plots. The NPQ difference of the 500 µmol m^{-2} s^{-1} light intensity curve has a too strong rise at the beginning, while getting to the same stable value as the publication. These issues can also be seen in the last plot of the first row, as this plot shows a 2 min scan of the differences of NPQ at different light intensities. Therefore, all curves in this plot are shifted upwards in comparison to the publication. The first two plots of the middle row show a very good recreation of the publication, while the curve for VCCN1 in the last plot ends at a slightly too high value, compared to the publication. The bottom row starts out with a good match in the first column,

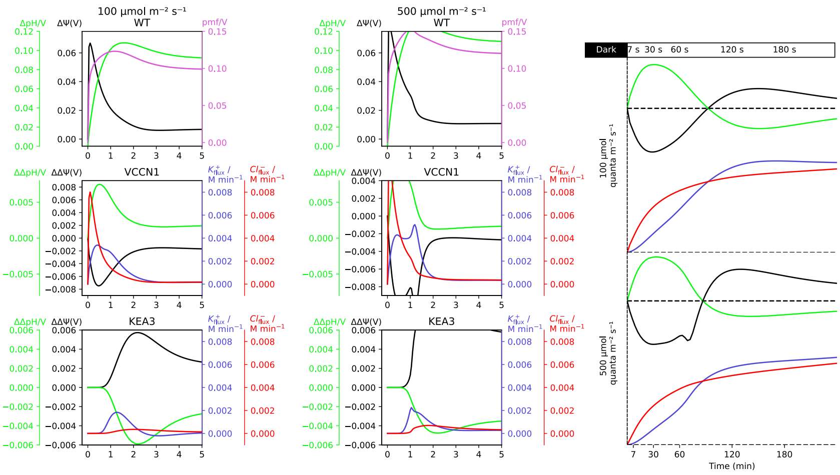

Figure 13: Simulation results of KEA3 and VCCN1 knockout mutants under differing light intensities.

The models of the wild type (WT) (top row), voltage-gated Cl^- channel 1 (VCCN1) knockout mutant (middle row), K^+/H^+ antiporter 3 (KEA3) knockout mutant (bottom row), and the combination of both with the additional Cl^- channel (CLCE) knockout (right column) were simulated under a light intensity of $100\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ (left column, and top row of right column), $500\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ (middle column and bottom row of right column). The results shown in the top row of the two left columns are the proton motive force (PMF) (pink), the proton gradient between lumen and stroma (ΔpH) (green), and the electric potential difference between lumen and stroma ($\Delta\Psi$) (black), all in Volts, for the WT. In the two last rows of the left columns all curves are differences between the WT and corresponding mutant. These results are the proton gradient between lumen and stroma (ΔpH) (green) and the electric potential difference between lumen and stroma ($\Delta\Psi$) (black) and the fluxes of the concentration of K^+ in the lumen (${K_{lu}}^+$) (blue) and concentration of Cl^- in the lumen ($Cl-_{lu}$) (red). The fluxes are taken from the model, by getting the right-hand side of each time point of the corresponding Ordinary Differential Equation (ODE) and by multiplying by 60, to convert from per second to per minute. The same results are plotted on the right column. The simulations were run using the default parameters, while changing the Photosynthetic Photon Flux Density (PPFD) to match the light intensities. To create each mutant model, the corresponding rate constant of the rate being knockout was set to zero, for e.g. the rate constant of KEA3 (k_{KEA}). This figure is recreated from figure 4 of the original publication of the Li2021 model [11].

but the middle column shows some small mismatches. In both initial plots, the points of the multiplier factor of 10 and 100 are shifted away from the zero point for the KEA3 mutant, while they shift to the zero point for the VCCN1 mutant, compared to the publication. However, the steady-state plots both are a good match to the publication, which cannot be said to the last plot of the bottom row. All the points of the higher multiplier factors are a good match, however, the points of both light intensities of the lower factors are shifted both upwards, where the 500 $\mu mol\, m^{-2}\, s^{-1}$ light intensity points are shifted much more than the 100 $\mu mol\, m^{-2}\, s^{-1}$ light intensity points.

3.1.4 Matuszynska2016

In the recreation of the fourth figure of the Matuszynska2016 model [12], all three plots show a very good match to the publication (see Fig. 15). As access to the experimental data was possible was given, the experimental values were also plotted. The only thing to note, is that these data points had to be shifted on the x-axis to fit the actual peaks of the simulation. It is assumed that, that was also done for the publication, which is why it was done here.

The fifth figure consists of two different subfigures, the top showing a striking similarity to the publication except for the lumenal pH curve of 300 $\mu mol\, m^{-2}\, s^{-1}$ and 900 $\mu mol\, m^{-2}\, s^{-1}$ light intensity, both shifted slightly downwards in the recreation (see Fig. 16). However, both show the same trend through the time series. The phase plane trajectory on the other hand, shows only a slight similarity. The highest lumenal pH reaches at higher co-operative 4-state quenching mechanism (Q) activity is that of 7, while in the publication it is near to 8. Overall, the trajectories are all shifted downwards in the recreation, except for the trajectory of 100 $\mu mol\, m^{-2}\, s^{-1}$, which shows the most similarity to the publication. On top of that, the steady-state values are also all shifted downwards and the points attributed to the light intensities, are not in the same places as in the publication.

The last figure could also be successfully recreated, with only two small discrepancies (see Fig. 17). The first is that the first value of the Φ_{PSII} curve in the recreation is much lower than in the publication and the second is that the recreation misses the last value of the Φ_{PSII} curve. Other than that, both simulation results and experimental data show a very good match to the publication. Here it should also be noted that the experimental data was shifted on the x-axis to fit the peaks of the simulation, as was done for the prior figure.

3.1.5 Saadat2021

The second, third, fifth, and sixth figure of the Saadat2021 model [48] were successfully recreated, without any discrepancies (see Fig. 18, Fig. 19, Fig. 21, and Fig. 22 respectively). The fourth figure, however, shows the correct trends, but the parameter range, where limit cycle oscillations of the reaction rate constant of cyclic electron flow (k_{cyc}) parameter were observed, could not be recreated. This issue made it therefore impossible to fully recreate the figure, which is why the lines in the recreation are much more jagged than in the publication (see Fig. 20).

The last figure of the publication includes the same parameter range as the fourth figure, therefore the same issue is observed in the recreation. However in this case, the recreation could

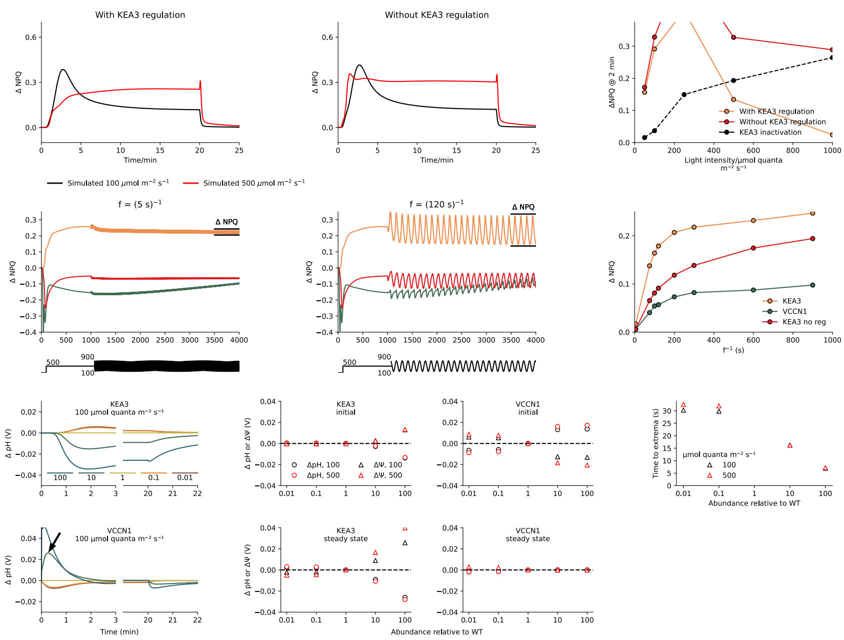

Figure 14: Simulation results of KEA3 regulation, oscillating light, and different abundances of KEA3 and VCCN1.
Three different types of simulation was performed here. The top row shows results of simulations that show the effect of K^+/H^+ antiporter 3 (KEA3) regulation. The first two plots show the results of a simulation following a simple light protocol, with a light period of 20 min and a dark period of 5 min. Each plot consists of two simulations, each showing a simulation at either a light intensity of 100 µmol m^{-2} s^{-1} (black) or 500 µmol m^{-2} s^{-1} (red). The left plot describes difference of the KEA3 knockout mutant to the WT simulation, while the right plot describes the difference of a simulation without the K^+/H^+ antiporter 3 (KEA3) regulation mechanism to the WT simulation. To remove the regulation of KEA3, the regulation of KEA3 activity by NADPH (qL_{act}) was set to a constant value of one. These plots show the Non-Photochemical Quenching (NPQ) over the time series, which is also shown on the right plot, however, as a scan of light intensities at the 2 min mark. Additionally, a difference curve is also drawn in the plot (black and dashed). The middle row shows the reaction of the model to oscillating light. To simulate this new form of light, a simple sinus curve was used, with the following equation: $\mathrm{PPFD} = \mathrm{PPFD}_{\mathrm{base}} + \mathrm{PPFD}_{\mathrm{amp}} \cdot \sin(2\pi \cdot \mathrm{f} \cdot t)$. In this equation, the $\mathrm{PPFD}_{\mathrm{base}}$ shows the value where the oscillation should happen, the $\mathrm{PPFD}_{\mathrm{amp}}$ shows the amplitude of the oscillation, and the f shows the frequency of the oscillation. The two first plots show the difference of NPQ to the WT of the KEA3, voltage-gated Cl^- channel 1 (VCCN1), and without KEA3 regulation mutants, at two different frequencies. The left at a frequency of $\frac{1}{5}$ s^{-1}, and the right at a frequency of $\frac{1}{120}$ s^{-1}. The simulations were first run to 1000 s at a light intensity of 500 µmol m^{-2} s^{-1} without any oscillation, and then until 4000 s with an oscillation between 100 µmol m^{-2} s^{-1} and 900 µmol m^{-2} s^{-1}. The last plot shows the difference between the extrema of the last wave at varying frequencies of all three mutants. The last row shows the effect of different abundances of KEA3 and VCCN1 on the proton gradient between lumen and stroma (ΔpH) and electric potential difference between lumen and stroma ($\Delta\Psi$). The abundance of either KEA3 or VCCN1 was changed by a factor of 100, 10, 1, 0.1, and 0.01, by multiplying the factor to the respective rate constant of the transporter. The simulations run follow the same light protocol as the first row, show the difference of ΔpH to the normal abundance (1) as a time series for 100 µmol m^{-2} s^{-1}. The four plots in the middle show the initial and steady-state values of ΔpH (circles) and $\Delta\Psi$ (triangles) at 100 µmol m^{-2} s^{-1} (black) and 500 µmol m^{-2} s^{-1} (red), for each abundance, also as a difference to the normal abundance. The initial values were taken as a difference of the value at 50 s to the start of the respective simulation, while the steady-state values were taken at 20 min. The last plot on the right shows the time it took to reach the extrema of the VCCN1 abundance time series. An example is marked as an arrow in the bottom left plot. The time point of the extrema of both light intensities is then plotted. All simulations named here were run using the default parameters, unless stated otherwise. To create each mutant model, the corresponding rate constant of the rate being knockout was set to zero, for e.g. the rate constant of KEA3 (k_{KEA}). This figure is recreated from figure 5 of the original publication of the Li2021 model [11].

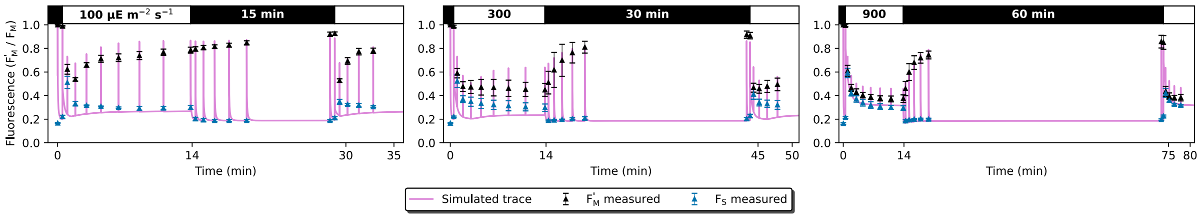

Figure 15: Experimental and simulated PAM protocol with three different light levels and durations of *Arabidopsis thaliana*.
A Pulse Amplitude Modulation (PAM) protocol was done using *Arabidopsis thaliana* plants, with three different light levels and durations. The protocols start with a saturating pulse, followed by a dark period of 30 s, then a light period of 14 min that starts with a saturating pulse and continues with 7 additional ones, all an accumulative 20 seconds apart (+30 s, +50 s, +70 s, etc. from the start of the period). Then another dark period of differing lengths, also starting with a saturating pulse and going along with 5 additional ones, also an accumulative 20 seconds apart. To end the protocol, a final light period of 5 min, with a saturating pulse to start and 4 additional ones, also an accumulative 20 seconds apart. The three different protocols only differ in the light intensities of the light periods and the duration of the second dark period. The first protocol, shown on the left, has a light intensity of 100 $\mu mol\,m^{-2}\,s^{-1}$ and a second dark period of 15 min. The second protocol, shown in the middle, has a light intensity of 300 $\mu mol\,m^{-2}\,s^{-1}$ and a second dark period of 30 min. The third protocol, shown on the right, has a light intensity of 900 $\mu mol\,m^{-2}\,s^{-1}$ and a second dark period of 60 min. The experimental values shown, are the base fluorescence (F) (blue) and the maximal fluorescence (F_m) (black). Three replicates for each measurement were done, but only the mean values and standard deviation are shown. The data was taken from the original publication [12], therefore all the other meta-information is to be read there. The simulation (pink) was done using the default parameters and changing the Photosynthetic Photon Flux Density (PPFD) to match the light intensities of the protocols and saturating pulses of 2000 $\mu mol\,m^{-2}\,s^{-1}$. Additionally, the PPFD was converted to an internal activation rate, that was calibrated to three light intensities of *A. thaliana*. This was done by following equation: $\mathrm{Light} = 0.0005833 \cdot \mathrm{PPFD}^2 + 0.2667 \cdot \mathrm{PPFD} + 187.5$. This figure is recreated from figure 4 of the original publication of the Matuszynska2016 model [12].

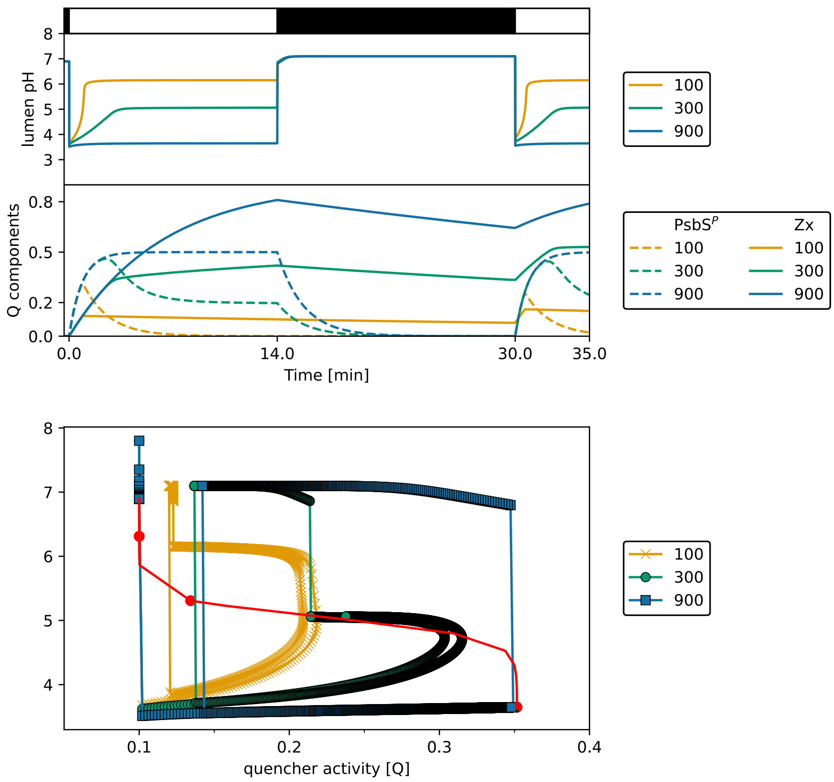

Figure 16: Visualisation of lumenal pH and quencher components in response to different light intensities.
A protocol of dark and light periods was used to simulate the model at three different light intensities. The protocol starts with a dark period of 30 s, followed by a light period of 14 min, then another dark period of 16 min, and ending with a final light period of 5 min. The three different light intensities used in the light periods were 100 µmol m^{-2} s^{-1} (yellow), 300 µmol m^{-2} s^{-1} (green), and 900 µmol m^{-2} s^{-1} (blue). The time series of each simulation is shown in the top plot, for the lumenal pH and the concentration of the quencher components protonated PsbS protein (PsbS$^{\mathrm{P}}$) and zeaxanthin (Zx). The bottom plot shows a phase plane trajectory of the co-operative 4-state quenching mechanism (Q) and the lumenal pH for each light intensity. Each simulation was done with the default parameters of the model, whereas the light intensities were inputted using the conversion of Photosynthetic Photon Flux Density (PPFD) to an internal activation rate for *Arabidopsis thaliana* by following equation: Light $= 0.0005833 \cdot \mathrm{PPFD}^2 + 0.2667 \cdot \mathrm{PPFD} + 187.5$. This figure is recreated from figure 5 of the original publication of the Matuszynska2016 model [12].

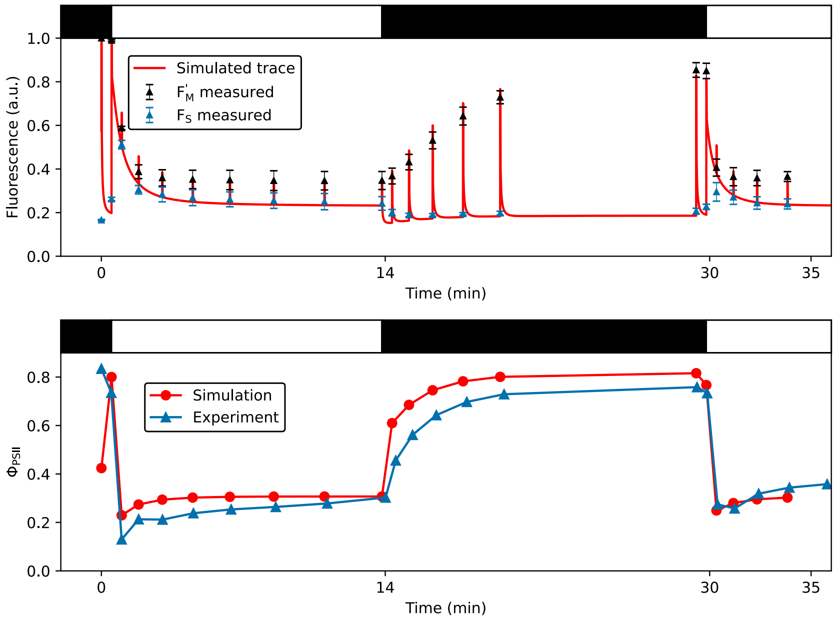

Figure 17: Experimental and simulated PAM protocol of *Epipremnum aureum*.
A Pulse Amplitude Modulation (PAM) protocol was done using *Epipremnum aureum* plants that starts with a saturating pulse, followed by a dark period of 30 s, then a light period of 14 min of a light intensity of that starts with a saturating pulse and continues with 7 additional ones, all an accumulative 20 seconds apart (+30 s, +50 s, +70 s, etc. from the start of the period). Then another dark period of 16 min, also starting with a saturating pulse and going along with 5 additional ones, also an accumulative 20 seconds apart, except for the last that occurs 30 s. To end the protocol, a final light period of 5 min, with a saturating pulse to start and 4 additional ones, also an accumulative 20 seconds apart. The light intensity used for the light periods is 100 µmol m^{-2} s^{-1}, while the dark is 0 µmol m^{-2} s^{-1}. The experimental values shown, are the base fluorescence (F) (blue) and the maximal fluorescence (F_{m}) (black) at the top and the efficiency of photosystem II (Φ_{PSII}) at the bottom. Three replicates for each measurement were done, but only the mean values are shown and also the standard deviation for the F. The data was taken from the original publication [12], therefore all the other meta-information is to be read there. To change the model to the *E. aureum* version, only the Fitted quencher factor corresponding to fastest possible quenching (γ_2) was changed (= 1). However, the conversion of the PPFD to an internal activation rate was done using the following equation: Light $= 0.0004167 \cdot \mathrm{PPFD}^2 + 0.3333 \cdot \mathrm{PPFD} + 862.5$. With these changes the model was simulated using the same PAM protocol and the same results were plotted to the corresponding experimental data (red). This figure is recreated from figure 6 of the original publication of the Matuszynska2016 model [12].

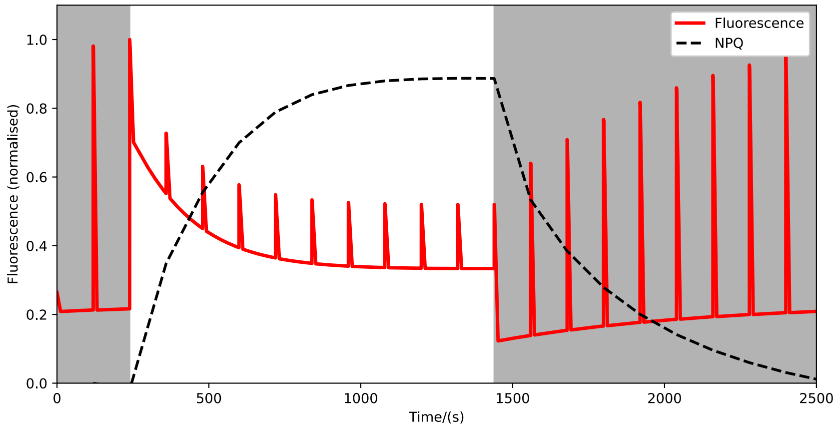

Figure 18: Results of a generic PAM protocol.
The generic Pulse Amplitude Modulation (PAM) protocol starts with a 4 min dark period with a saturating pulse at the 2 min mark. At the end of the dark period, another saturating pulse indicates the start of an actinic light period that goes on for 10 min, with saturating pulses every 2 min. Then another dark period of 18 min starts, again with a saturating pulse at the start and at each 2 min mark. The light intensity used for the actinic light period is $1000\,\mu mol\,m^{-2}\,s^{-1}$, while the dark periods are $40\,\mu mol\,m^{-2}\,s^{-1}$. Each saturating pulse was simulated for 0.8 s at a light intensity of $5000\,\mu mol\,m^{-2}\,s^{-1}$. The simulation is run using the default parameters and initial conditions of the model, except for the reaction rate constant of cyclic electron flow (k_{cyc}) ($= 0$) to match an organism with no cyclic electron flow. The values of light intensity were inputted directly to the Photosynthetic Photon Flux Density (PPFD) parameter of the model. The results shown are the fluorescence (F) which was normalized to the maximum value of that series (red), and the Non-Photochemical Quenching (NPQ) (black), which was calculated by using the F and maximal fluorescence (F_m) (see Eq. 3). This figure is recreated from figure 2 of the original publication of the Saadat2021 model [48].

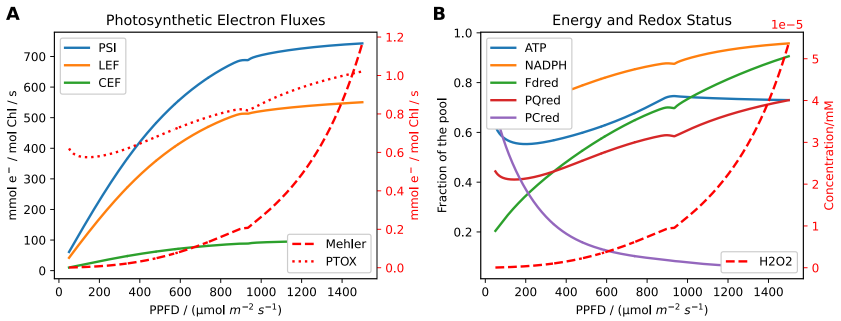

Figure 19: Results of a steady-state scan of PPFD.
The model was simulated to steady-state under different Photosynthetic Photon Flux Density (PPFD) values, ranging from $50\,\mu mol\,m^{-2}\,s^{-1}$ to $1500\,\mu mol\,m^{-2}\,s^{-1}$. The results are separated in two different plots, differentiating between photosynthetic electron fluxes and the energy and redox status. The left side shows the PSI rate (v_{PSI}) (blue), the Linear Electron Flow (LEF) (orange, and calculated by doubling the PSII rate (v_{PSII})), the rate of the cyclic electron flow (v_{cyc}) (green), the mehler reaction lumping the reduction of O_2 instead of Fd (v_{mehler}) (red and dashed), and the oxidation of the PQ pool through cytochrome and PTOX ($v_{PQ_{ox}}$) (red and dotted). On the right side the ratios of Adenosine Triphosphate (ATP) (blue), Nicotinamide Adenine Dinucleotide Phosphate (NADPH) (orange), reduced ferredoxin (Fd_{red}) (green), reduced plastoquinone (PQ_{red}) (red), and reduced plastocyanin (PC_{red}) (purple) to their total pools are shown. Additionally the concentration of hydrogen peroxide (H_2O_2) (red, dashed) is also plotted. The simulation is run using the default parameters and initial conditions of the model, while changing only the Photosynthetic Photon Flux Density (PPFD) to the desired value for each simulation. This figure is recreated from figure 3 of the original publication of the Saadat2021 model [48].

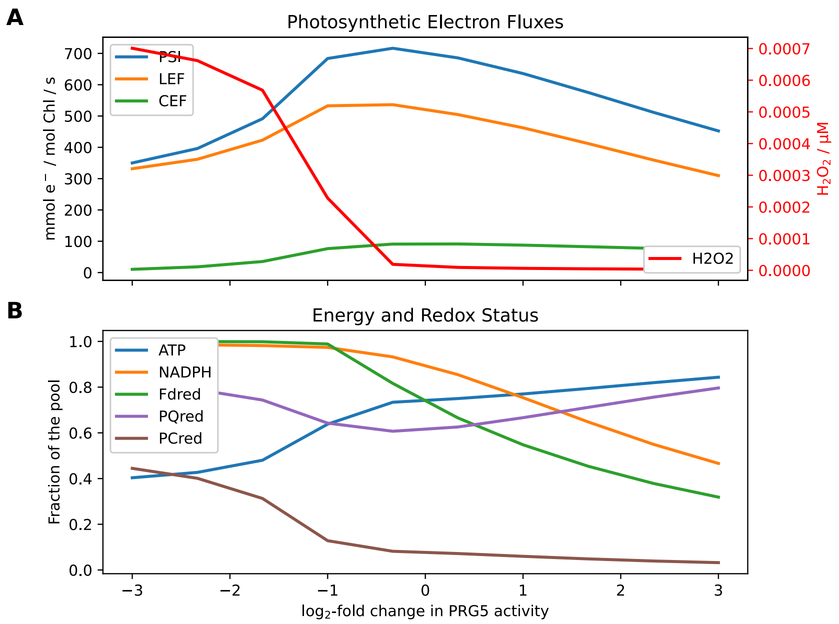

Figure 20: Results of a steady-state scan of aletered CEF.
The model was simulated to steady-state under different reaction rate constant of cyclic electron flow (k_{cyc}) values representing $\log_2$-fold changes ranging from negative three to three. The results are separated in two different plots, differentiating between photosynthetic electron fluxes and the energy and redox status. The top plot shows the PSI rate (v_{PSI}) (blue), the Linear Electron Flow (LEF) (orange, and calculated by doubling the PSII rate (v_{PSII})), the rate of the cyclic electron flow (v_{cyc}) (green), and the concentration of hydrogen peroxide (H_2O_2) (red). In the bottom plot the ratios of Adenosine Triphosphate (ATP) (blue), Nicotinamide Adenine Dinucleotide Phosphate (NADPH) (orange), reduced ferredoxin (Fd_{red}) (green), reduced plastoquinone (PQ_{red}) (red), and reduced plastocyanin (PC_{red}) (purple) to their total pools are shown. The simulation is run at a Photosynthetic Photon Flux Density (PPFD) of $1000\,\mu mol\,m^{-2}\,s^{-1}$, otherwise using the default parameters and initial conditions of the model, while changing only the k_{cyc} to the desired value for each simulation. Due to issues of singular ranges of k_{cyc} not being able to be simulated to steady-state, only a few values could actually be plotted, to bee seen by the jaggedness of the lines. This figure is recreated from figure 3 of the original publication of the Saadat2021 model [48].

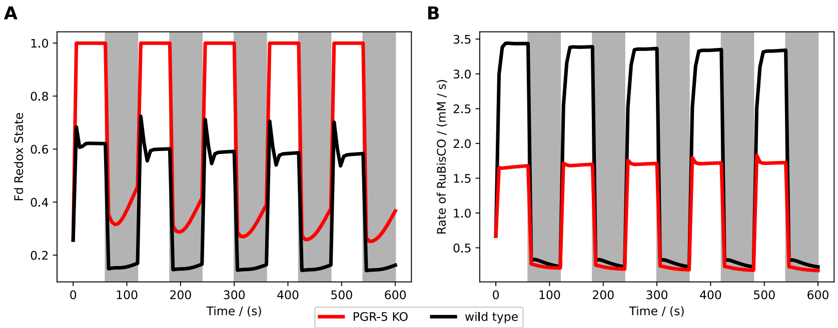

Figure 21: Comparison of results of a wildtype and knockout mutant simulation under varying light intensities.
A simple fluctuating light protocol was used to simulate a wildtype (black) and a knockout mutant (red) of the model. The protocol undergoes a total of 10 periods, each of them lasting 1 min. The light intensities of the periods alternate between light and dark, using a light intensity of 600 µmol m^{-2} s^{-1} and 40 µmol m^{-2} s^{-1}, respectively. The wildtype simulation was run using the default parameters and initial conditions of the model, while the knockout mutant was simulated by setting the reaction rate constant of cyclic electron flow (k_{cyc}) to zero. Each light intensity was inputted into the Photosynthetic Photon Flux Density (PPFD) parameter of the models. The results shown are the ratio of reduced ferredoxin (Fd_{red}) to its total pool on the left, and the RuBisCO carboxylation rate ($v_{RuBisCO|carb}$) on the right. This figure is recreated from figure 5 of the original publication of the Saadat2021 model [48].

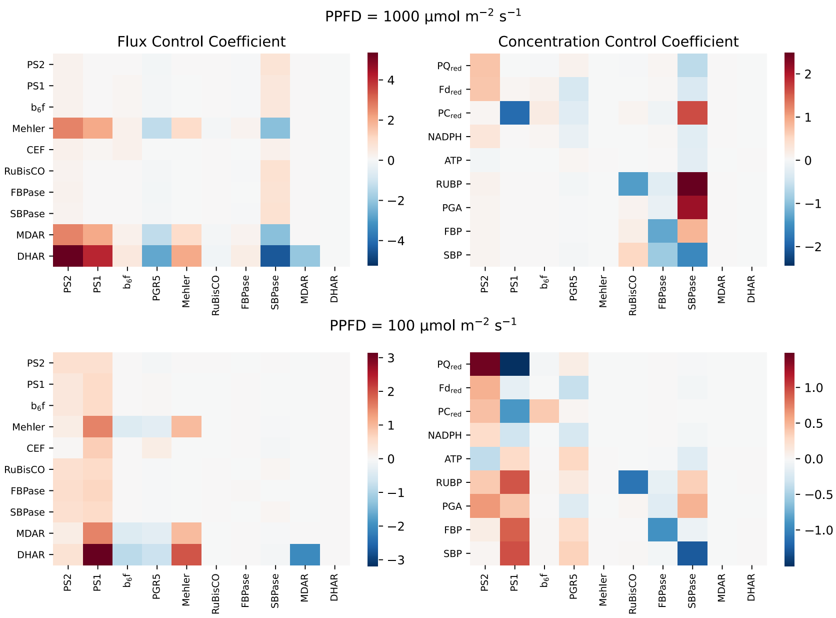

Figure 22: MCA of key aspects of the model under two different light intensities.
A Metabolic Control Analysis (MCA) was done to the model under two different light intensities, 1000 µmol m^{-2} s^{-1} (top) and 1000 µmol m^{-2} s^{-1} (bottom). The results of the fluxes can be found on the left, while on the right are the variables. The parameters that were used for the MCA are the control coefficients of the fluxes with the same names. The following were used, given from left to right on the x-axis of each heatmap: total PSII reaction centers ($PSII^{tot}$), total PSI (PSI_{tot}), rate constant of the cytochrome b6f complex reaction (k_{Cytb6f}), reaction rate constant of cyclic electron flow (k_{cyc}), estimated rate constant for summarized hydrogen peroxide production (k_{Mehler}), catalytic rate constant of RuBisCO for carboxylation ($k_{RuBisCO}$), catalytic rate constant of FBPase (k_{FBPase}), catalytic rate constant of SBPase (k_{SBPase}), turnover rate of monodehydroascorbate reductase (k_{MDAR}), and turnover rate of dehydroascorbate reductase (k_{DHAR}). These parameters were displaced by $\pm 1\%$. The simulations were otherwise done using the default parameters and initial conditions of the model, while changing only the Photosynthetic Photon Flux Density (PPFD) to the desired value for each simulation. This figure is recreated from figure 6 of the original publication of the Saadat2021 model [48].

not be completed at all, as the specific range of the k_{cyc} parameter where these oscillations are found is not well documented.

3.2 Model Demonstrations

3.2.1 Daylight Simulation

The light intensity during the chosen day period follows an approximate bell curve, common for these sorts of graphs. As sunlight is not a constant source, but may change dynamically due to weather conditions, the light intensity shows simple peaks and valleys during the entire day. This allows to test the capabilities of the models to simulate complex and dynamic light protocols, which is a common issue for many models. In this case, all but the Li2021 model show results to varying degree (see Fig. 23).

The Bellasio2019 model shows a clear response to the changing light conditions, both the $v_{\mathrm{RuBisCO|carb}}$ and the ATP and NADPH ratio rising and falling with the light intensity during the day. However, at approximately 10:30, both reach a plateau, that goes on until approximately 16:00, where the light intensity begins to fall. There the $v_{\mathrm{RuBisCO|carb}}$ slowly begins to fall to 0 until the end of the simulation, while the ATP and NADPH ratio slowly rises until 18:00 and then falls following the pattern of the $v_{\mathrm{RuBisCO|carb}}$ line. As this model does not have a quantity representing F, it cannot be plotted.

The Fuente2024 model only has a representative quantity for F, which follows the light intensity pattern very closely. However, in the moments of lower light, the F values are more sensitive to the light fluctuations, as for example the peak at 08:00 causes the F to rise to the same values as during the highest light intensities. Additionally, the F values show a small rise when the day begins to end, which does not follow the light intensity pattern. However, in the higher light intensities, the F values do follow the light intensity pattern very closely.

The Matuszynska2016 model, also only shows a representative quantity for F, which also follows the light intensity pattern very closely. Just like with the Fuente2024 model, the F values follow the light pattern closest during the higher light intensities, while during the lower light intensities, they are more sensitive to the light fluctuations. Interestingly, this model also shows a small rise in F values when the day begins to end.

The Saadat2021 model shows all three representative quantities, $v_{\mathrm{RuBisCO|carb}}$, ATP and NADPH ratio, and F. The $v_{\mathrm{RuBisCO|carb}}$ values show a clear correlative response to the light intensity, rising and falling with the light intensity during the day. Comparatively to the Bellasio2019 model, these values also reach a plateau at 10:30 and slowly fall after 16:00. Contrary to the other models though, both the ATP and NADPH ratio, and the F values show an anticorrelative response to the light intensities. Especially the F which drops significantly during the higher light intensities, while rising during the lower light intensities. It also reaches a plateau at 10:30, but then slowly rises at 16:00, to reach approximately the same starting value as of the start of the day. The ATP and NADPH ratio also shows a similar pattern, but with less significant changes.

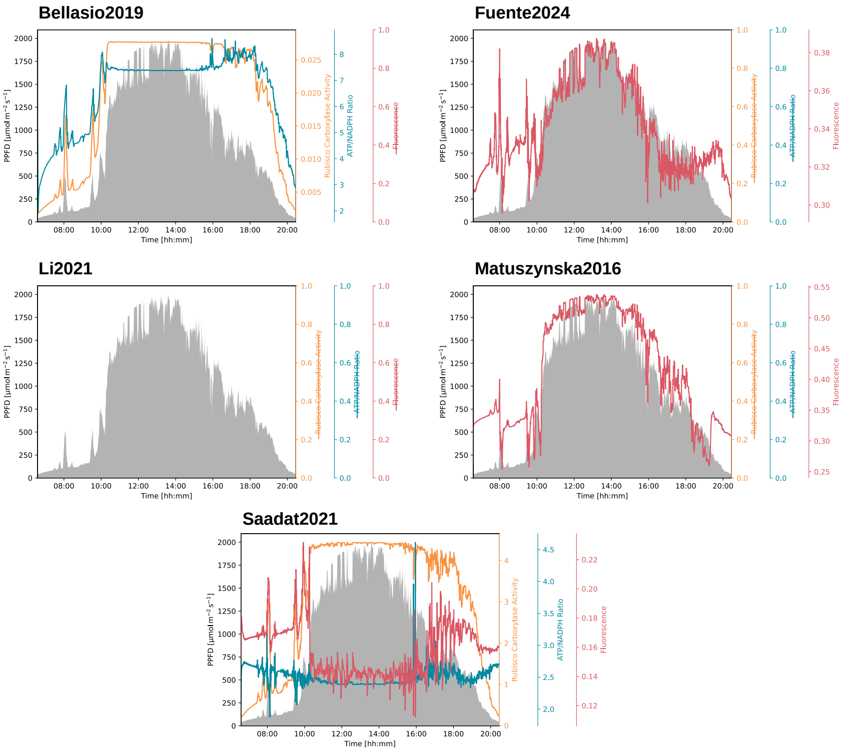

Figure 23: Combined Daylight Simulation demonstrations of all models.
Sample simulation of a day cycle using real Photosynthetic Photon Flux Density (PPFD) data from Kansas, USA on June 19, 2023. The data was obtained from the National Ecological Observatory Network (NEON) data portal [49] and is used to create a protocol for the light intensity PPFD over the course of the day, in a minute interval. The data used is filtered to only show a PPFD that equals or is higher than $40\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$. This threshold is chosen as it has shown to allow most models to still simulate the photosynthetic machinery, while still being a decent representation of the actual daylight conditions. The simulation is run using the default parameters and initial conditions of each model, and the RuBisCO carboxylation rate ($v_{\mathrm{RuBisCO|carb}}$), Adenosine Triphosphate (ATP) and Nicotinamide Adenine Dinucleotide Phosphate (NADPH) ratio, and fluorescence (F) results is plotted over the course of the day, if possible. The results do not represent actual plant behavior, but show the capabilities of the model to simulate complex and more realistic light protocols.

3.2.2 FvCB Add-on

Only the Bellasio2019 and Saadat2021 models have a successful demonstration of the FvCB add-on, as the other models do not have the required quantities' (see Fig. 24). The Fuente2024, Li2021 and Matuszynska2016 models all do not include a representation for $v_{\text{RuBisCO|carb}}$ and CO_2. Without these, a FvCB style assimilation could not be calculated.

The Bellasio2019 on the other hand, shows a very distinct correlation between the simulation results and the min-W FvCB model, with general parameters used [16]. However, once the simulation is done at higher C_{i} values, both the $v_{\text{RuBisCO|carb}}$ and A values drop below the FvCB model. On top of that, they are small divots in the simulation results, which may be resulted in an unsuccessful steady-state analysis, and therefore an automatic rescue to a quasi-steady-state, 1800 s, was performed.

The Saadat2021 model results have approximately the same shape of curve than the FvCB model, but the ascent is much steeper. This can be seen, by the A of the Saadat2021 model beginning lower but ending higher than the FvCB A. Both of the $v_{\text{RuBisCO|carb}}$ start at the same point, namely $0\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$, but the Saadat2021 model also reaches a higher value.

3.2.3 Standard PAM Simulation

All models show a successful demonstration of the standard PAM simulation (see Fig. 25), except the Bellasio2019 model, which does not have a quantity representing F nor NPQ. Additionally, the Li2021 model only includes a quantity for NPQ, which shows a typical curve of NPQ for the standard protocol. However, at the start of the protocol, an uncommon small peak can be seen, with a slow ascent and descent. Due to the fact, that the model does not contain a quantity for F, it is harder to see the different points of time when a saturating pulse was used. Still, due to the continuous simulation of the NPQ, the curve is much smoother than the other models, and small spikes can be seen that can be attributed to time points of saturating pulses.

The Fuente2024, Matuszynska2016, and Saadat2021 models all follow the typical pattern for F a PAM protocol. The F values starting relatively high after the dark adaptation period, with the F_{m} at saturating pulses being the highest in the first dark period. Then the base F values stay stagnant during the actinic light period, where they are higher than during the prior dark period for the Fuente2024 and Matuszynska2016 models, but lower for the Saadat2021 model. In the same light period, the F_{m} values quickly drop to a lower and stable value in all three models. Then during the second dark period for all three models, the base F values drop even lower than in the first dark period, but slowly rising back following the simulation time. The F_{m} values all gradually rise during the second dark period, but do not reach the same values as during the first period.

The NPQ curves of all three models also follow a typical pattern, with them start at near 0 in the first dark period, then quickly rising to a stable value in a curve during the actinic light period. Only the Saadat2021 model, shows a NPQ curve in the actinic light period that reaches over its stagnating point at the start. However, then all three models show a curved drop during the second dark period, reaching a stable value that is higher than at the starting point. As the Matuszynska2016 and Saadat2021 models do not have a quantity for NPQ, it had to be calculated using F and F_{m} (see Equation 3), meaning the amount of points for the NPQ

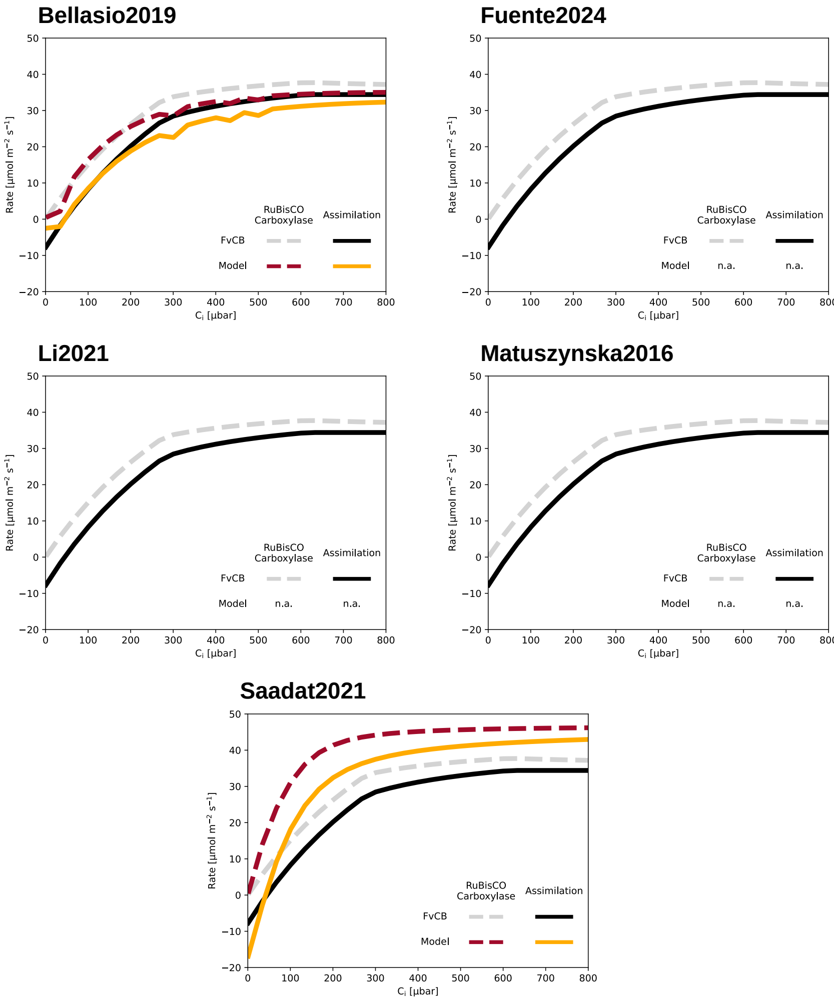

Figure 24: Combined FvCB Addon demonstrations of all models.
Comparison of modelled carbon assimilation (A) and RuBisCO carboxylation rate ($v_{\mathrm{RuBisCO|carb}}$) against the Farquhar, von Caemmerer, and Berry (FvCB) model. The FvCB model is calculated using the min-W approach as described by Lochoki and McGrath (2025) [16]. To be able to simulate A, there are two mandatory quantities that need to be present in the model: carbon dioxide (CO_2) concentration and $v_{\mathrm{RuBisCO|carb}}$. If one of these parameters is missing, the FvCB model will still be shown, but no comparison with the model will be possible. Other parameters that are required to calculate the FvCB model will be added as parameters with default values if they are not present in the model. The simulation is then run until steady-state, or quasi-steady-state if not otherwise possible, for different intercellular CO_2 concentration (C_{i}) partial pressure. The carbon assimilation shown does not represent actual values but rather a theoretical curve to compare the kinetic model to the popular FvCB model.

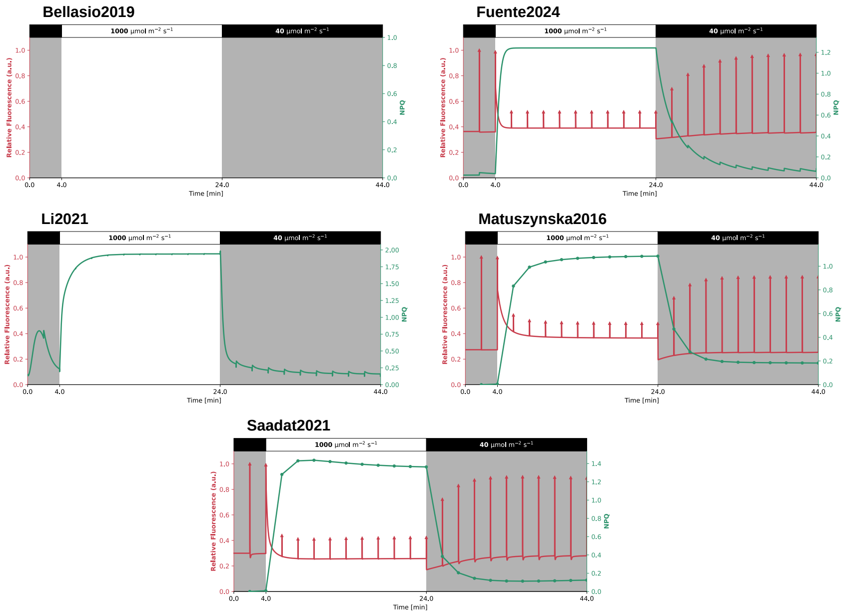

Figure 25: Combined PAM Simulation demonstrations of all models.
Sample simulation of a common Pulse Amplitude Modulation (PAM) protocol to show fluctuations of fluorescence (F) and Non-Photochemical Quenching (NPQ) using saturating pulses. The simulation protocol is as follows: A dark adaptation period that simulates for 30 minutes at a dark light intensity ($40\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$), then the actual protocol starts. The protocol consists of 22 periods with each being 2 minutes of length. That period consists of a specific light intensity of the respective type of period and ends with a saturating pulse with a length of 0.8 s and a light intensity of $3000\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$. First, two dark periods with light intensity of $40\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$, followed by ten light periods with light intensity of $1000\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$, then ten dark periods again. The simulation is run using the default parameters and initial conditions of each model.

curves is the same as the number of F_m present in the simulation. This creates a more jagged curve, while the Fuente2024 model, which has a quantity for NPQ, shows a smoother curve.

3.2.4 MCA of Photosynthesis

No models could complete the entire MCA heatmaps, neither the variables nor the fluxes (see Fig. 26). On top of that, the Li2021 shows completely empty heatmaps, even though it has some of the required quantities to perform the MCA. This is due to the model not being able to achieve steady-state for the parameters scanned, which is required for a MCA analysis on control coefficients.

The Bellasio2019 model shows results for the control coefficients of PSII and RuBisCO for RuBP, CO_2, ATP, and NADPH for the variables, and $v_{\mathrm{RuBisCO|carb}}$, and v_{ATPSynth} for the fluxes. Both control coefficients do not have much control on the variables, except for the RuBisCO coefficient on RuBP, which shows a strong negative control. On the fluxes side, both coefficients do not show much control on both $v_{\mathrm{RuBisCO|carb}}$ and v_{ATPSynth}.

The Fuente2024 model only shows results for the control coefficients of PSII and PSI for $\mathrm{PQ_{ox}}$, ATP, v_{PSI}, and v_{PSII}. The control coefficient of PSII does not have much on either the variables or the fluxes mentioned prior. The control coefficient of PSI shows a strong positive control on $\mathrm{PQ_{ox}}$ and v_{PSI}, but also a smaller positive control on ATP and v_{PSII}.

The coefficients that are included in the Matuszynska2016 model are representative of PSII,

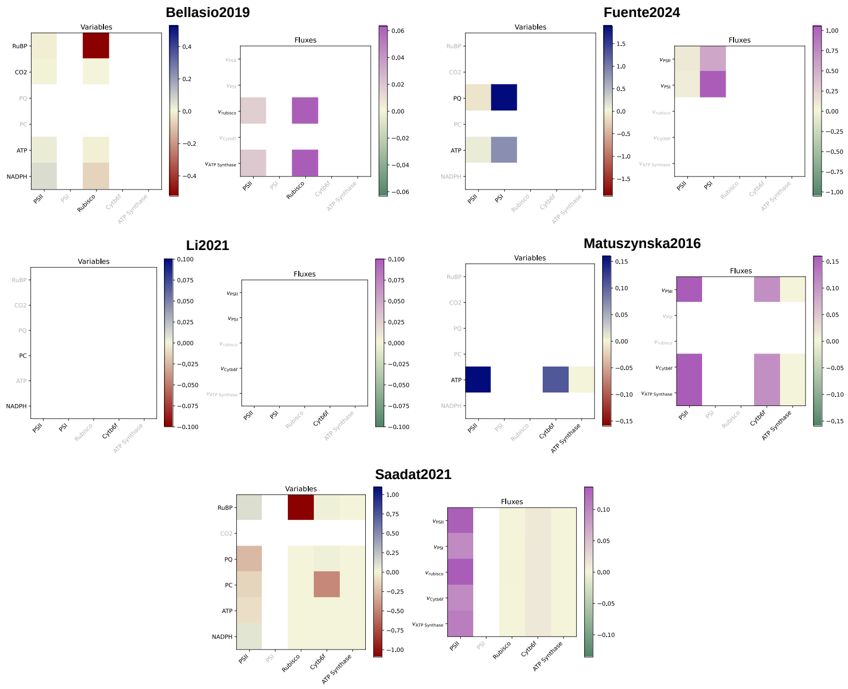

Figure 26: Combined MCA of Photosynthesis demonstrations of all models.
A sample Metabolic Control Analysis (MCA) of typical photosynthesis variables and fluxes. A control coefficient analysis is to be performed, therefore each parameter represents a single coefficient of the photosynthesis rate. The rates chosen should represent RuBisCO carboxylation rate ($v_{\mathrm{RuBisCO|carb}}$), PSII rate ($v_{\mathrm{PSII}}$), PSI rate ($v_{\mathrm{PSI}}$), Cytb$_6$f rate ($v_{\mathrm{b6f}}$) and ATP synthase rate (v_{ATPSynth}). The variables chosen should represent carbon dioxide (CO_2) concentration, Ribulose-1,5-bisphosphate (RuBP), oxidised plastoquinone (PQ_{ox}), oxidised plastocyanin (PC_{ox}), Adenosine Triphosphate (ATP), and Nicotinamide Adenine Dinucleotide Phosphate (NADPH). For each parameter to be scanned, the model is simulated to steady-state, with a displacement of $\pm 0.01\%$ of each respective parameter. The control coefficients are then calculated for each variable and flux by the following formula: $C_p^x = \frac{x_{\mathrm{upper}} - x_{\mathrm{lower}}}{2 \cdot \mathrm{disp} \cdot p}$, where C_p^x is the control coefficient of parameter p on variable or flux x, and disp is the displacement value. x_{upper} and x_{lower} are the steady-state result of x at either $+$disp and $-$disp respectively. It has to be noted that the MCA results can be very dependent on the other values of the parameters in the model, therefore the results shown here are only representative of the default parameter set of the model.

Cytb$_6$f and ATP synthase. The only viable variable in this model is ATP, which is positively controlled by all three coefficients, but most strongly by the coefficient of PSII. The fluxes that are included in this model are v_{PSII}, v_{b6f}, and v_{ATPSynth}. All the coefficients have a positive control on all the fluxes and show a consistent pattern respective to each coefficient, with the coefficient of PSII also having the strongest control on all three fluxes.

The Saadat2021 model includes the most control coefficients of the photosynthesis MCA out of all the models. It includes the coefficients of PSII, RuBisCO, Cytb$_6$f, and ATP synthase. Nearly all coefficients have only a small control on the variables and fluxes. Except the coefficient of Cytb$_6$f, which has a low negative control on PC_{ox} and the coefficient of RuBisCO which has a stronger negative control, but on RuBP. On the fluxes side, only the coefficient of PSII shows a significant positive control on all the fluxes. The other coefficients do not show any control.

3.2.5 Fitting of NPQ

Of all models, only the Bellasio2019 could not produce a fit (see Fig. 27), due to it not having a quantity for F nor NPQ. Additionally, as the Li2021 model does not have quantity for F, only the NPQ curve could be plotted. However, the fitting could still occur, which produced a curve that followed the experimental data in lower light ($90\,\mu mol\,m^{-2}\,s^{-1}$) and dark light ($40\,\mu mol\,m^{-2}\,s^{-1}$) very closely. However, in high light ($903\,\mu mol\,m^{-2}\,s^{-1}$), the curve first overfits and then underfits the data, showing an NPQ curve that shows a peak before getting to a stable value.

The three other models, all show both a curve of F and NPQ, but have varying results. The Fuente2024 model shows a very close fit to the experimental data on the NPQ side, but not on the F side, where both the base F and F_{m} values are repeatedly higher. The model could best fit the NPQ at high light and then first overfits but then underfits during lower light. At dark light, the NPQ consistently overfit. For this model, the parameters that were fitted were maximal extent of NPQ ($\mathrm{NPQ}_{\mathrm{max}}$) and NPQ mechanism deactivation (k_4), by 80.44 % and −19.56 % respectively.

The Matuszynska2016 model shows a very similar fit to NPQ as the Fuente2024 model, but is strikingly better at fitting the F values. At high light, the F and F_{m} seem to be spot on, while following a very similar curve during lower light. There the points do not fit perfectly, but show a much nearer fit. In the last dark period, the base F values are very close to the experimental data, while the fitted F_{m} do not show the same rise from the prior period. For this model, the parameters fitted were Fitted quencher factor corresponding to base quenching not associated with protonation or zeaxanthin (γ_0), Fitted quencher factor corresponding to fast quenching due to protonation (γ_1), Fitted quencher factor corresponding to fastest possible quenching (γ_2), Fitted quencher factor corresponding to slow quenching of Zx present despite lack of protonation (γ_3), and half-saturation constant (relative concentration of Zx) for quenching (K_{ZSat}), by 22.12 %, 573.61 %, 259.39 %, 1035.28 %, and 4288.27 % respectively.

The Saadat2021 model shows a good fit for the F and F_{m} values, however the NPQ is largely underfit during the high and lower light period. The dark period is also overfit, showing a much higher NPQ than the experimental data. For this model, the parameters fitted were half-saturation pH for de-epoxidase activity, highest activity at pH 5.8 (K_{pHSat}), γ_0, and pKa of PsbS activation, kept the same as for VDA (K_{pHSatLHC}), with −14.27 %, −99.99 %, and −16.89 % respectively.

3.3 Website

When arriving on the **GreenSloth** website, the user is first greeted with the Logo and the motto of the project, "Photosynthesis Models at your Pace" (see Fig. 28). At the top of the page is a navigation bar, where the user can click to access different pages of the site. These are "Home", "Models", "GreenSloth" with a GitHub logo, "Compare", "How to Use", "About Us", and "Impressum". The "Home" page is the landing page of the website, that continues with an arrow pointing down to find out more. This brings the user to a section that explains the motivation of the project briefly, and then to another section that explains how to use the site. This section is also accessed by the "How To Use" navigation link in the top navigation bar and is separated into the three main aspects that **GreenSloth** tries to provide: "Search",

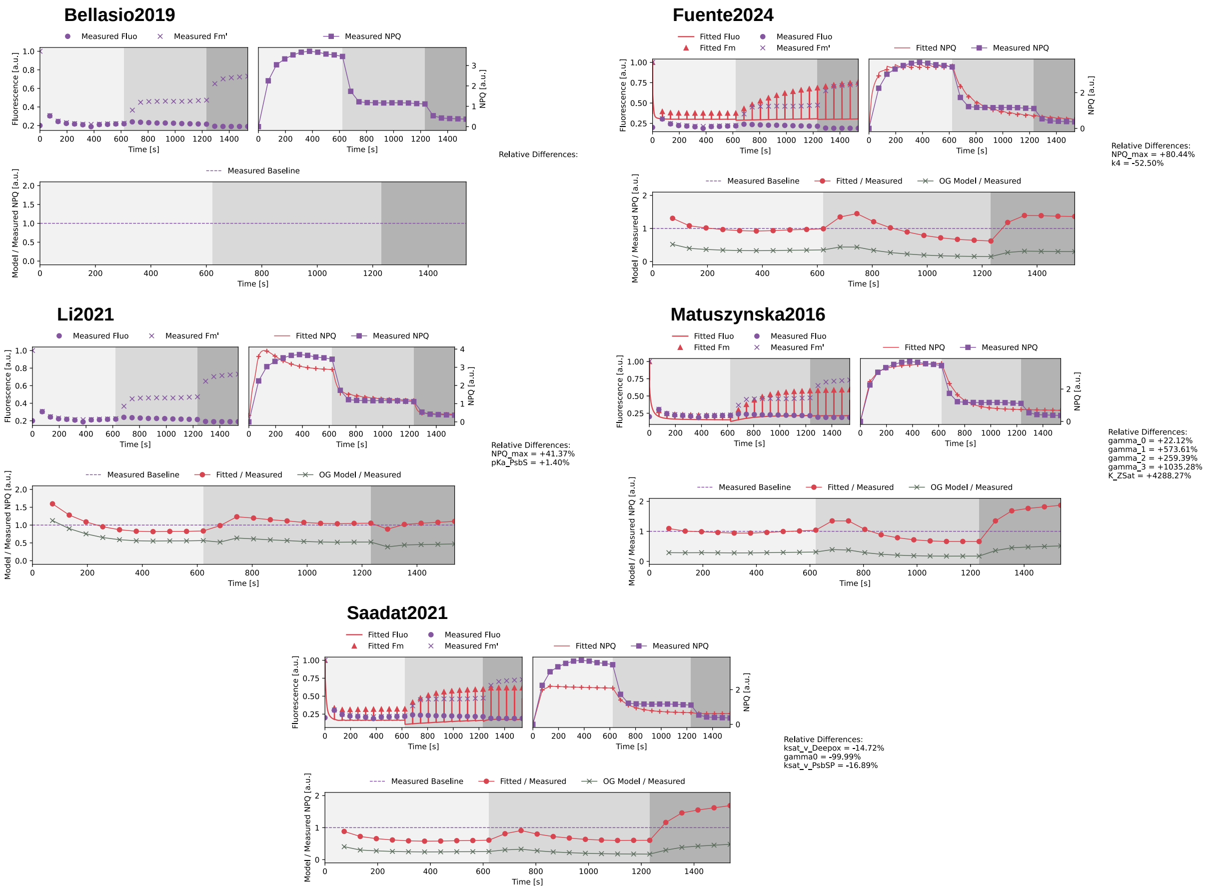

Figure 27: Combined Fitting of NPQ demonstrations of all models.
Sample fitting to experimental Non-Photochemical Quenching (NPQ) data. The NPQ data used is taken from experimental work published in von Bismarck (2022) [53] and was acquired using Maxi Imaging-PAM (Walz, Germany) using Col-0 *Arabidopsis thaliana* (*A. thaliana*) plants. It is assumed that the experiment follows the default PAM protocol of the machine, as no other experimental protocol has been given. Therefore, the protocol of each simulation follows the data given, where the length of one saturating pulse is set to 720 µs at a light intensity of 5000 $\mu mol\, m^{-2}\, s^{-1}$. The light protocol consists of a dark adaptation period of 30 minutes to acclimate the simulation conditions. Then the actual protocol starts with a longer phase of high actinic light (903 $\mu mol\, m^{-2}\, s^{-1}$) for approximately 10 minutes, followed by a lower actinic light of (90 $\mu mol\, m^{-2}\, s^{-1}$) for 10 minutes, and then 5 minutes of a dark period. During each phase, saturating pulses are given approximately every 60 seconds. As the experimental data also provides exact time points for each pulse, these were taken as reference for the protocol and not the general time intervals. In the experimental work, the dark period consists of actual darkness, whereas in the simulation a low light intensity of 40 $\mu mol\, m^{-2}\, s^{-1}$ is used to avoid numerical issues. The fitting is performed using the `lmfit` package in Python with the leastsquare method. On top of that, a standard scaling towards the experimental data is done, to keep the fitting results in the same order of magnitude. To help the fitting converge, weights are applied to the data points, which are defined as the reciprocal of the standard deviation. These settings set are not to be taken as set in stone, as fitting is a highly experimental process and differing settings might be required depending on the model and data used. These settings are a basic starting point for fitting data to a model. The hardest and most impactful decision while fitting is the choice of parameters to fit. There are many ways to find which parameters may be most impactful to fit, such as sensitivity analysis or metabolic control analysis. However, either way experimenting with different parameter sets is always required to find the best fitting practice, which differs for each model and also data to fit to.

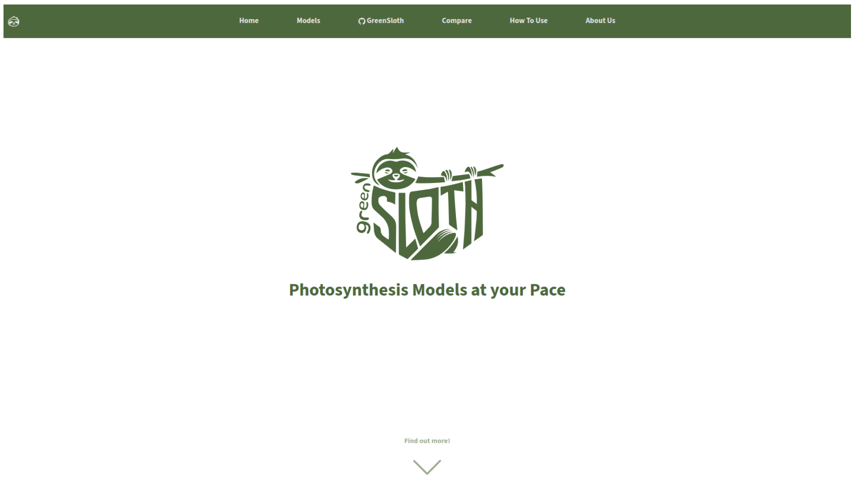

Figure 28: Screenshot of the landing page of the GreenSloth website.
A screenshot was done of the landing page of the **GreenSloth** website, in a full screen mode. The top includes a navigation bar with links to the different pages of the website, and the main section includes the logo and motto of the project.

"Summary", and "Comparison" (see Fig. 28). The "Search" part explains how the website's structure allows users to easily find models of photosynthesis, without the need of doing an intensive literature search. The "Summary" part explains how each model was prepared for **GreenSloth**, including the detailed presentation of the model, the validation of the model, and the demonstrations performed. The "Compare" part explains how a direct comparison between models is made possible on the website. The user can however go one step further down, to a schematic overview of the photosynthetic machinery, where the user can click on different components that directly brings them to the "Models" page with the clicked machinery as a tag pre-selected.

The "Models" page includes a search bar at the top, where the user can search for models solely by their name. Additionally, next to the bar is a button that opens the tag selection box. This allows the user to list all the models that have a specific tag, which can be related to the photosynthetic machinery, the demonstrations that can be performed with the model, or more (see Fig. 29). The models are presented row by row, with their name and schemes for the easiest recognition. The user can click on the name of a model and be brought to that model's page.

After choosing a model, the user is brought onto a page dedicated to that specific model (see Fig. 32). This page includes a sidebar on the left, which enables an easier navigation through the different sections of the page, which are "Summary", "ODE System", "Derived Quantities", "Parameters", "Derived Parameters", "Rates", "Figures", and "Demonstrations". At the top of the page, the name of the model is shown, along with the respective DOI as a link. Below that are two buttons, one to directly bring the user to the GitHub repository of that specific model, with a last update date, and another button that brings the user to the compare page, with this specific model already chosen. Briefly after these buttons, the summary section starts, with the scheme of the model on the left and a brief overview on what the model entails and how the implementation was done on the right. After that section, all the information of the model is shown in a table format, with the respective mathematical equations shown in a LaTeXformat.

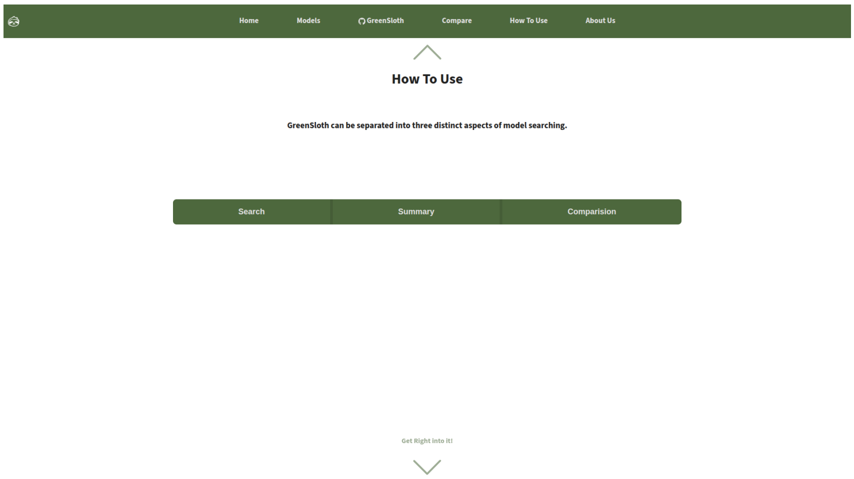

Figure 29: Screenshot of the How To Use page of the GreenSloth website.
A screenshot was done of the How To Use page of the **GreenSloth** website, in a full screen mode. The top includes a navigation bar with links to the different pages of the website, and the main section includes three buttons, "Search", "Summary", and "Compare". They all independently show a section of text explaining the respective aspect of the website. These three aspects represent the three main aspect of model searching, understanding, and comparing that **GreenSloth** tries to provide.

Figure 30: Screenshot of the Machinery page of the GreenSloth website.
A screenshot was done of the page showing the photosynthesis machinery of the **GreenSloth** website, in a full screen mode. The top includes a navigation bar with links to the different pages of the website, and the main section includes a schematic overview of the photosynthetic machinery, where the user can click on different components that directly brings them to the "Models" page with the clicked machinery as a tag pre-selected. On the left is the full schematic, while on the right is the representation, when the user hovers over one part of the machinery, in this case the "PSII" part. The hovering causes the all other parts to lose their color.

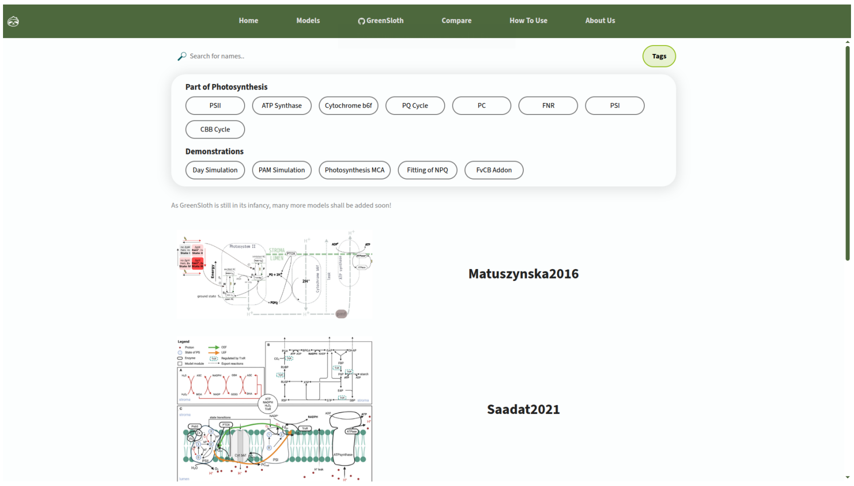

Figure 31: Screenshot of the Models page of the GreenSloth website.
A screenshot was done of the Models page of the **GreenSloth** website, in a full screen mode. The top includes a navigation bar with links to the different pages of the website, and the main section includes a list of models that are available in the database. The list can be filtered by tags that are selected in the tag selection box.

At the end, the figures and demonstrations sections showcase the recreations of the publication figures and the demonstrations that were performed, with a brief description. Each of the figures and demonstrations are in a collapsable box, to avoid overwhelming the user with too much information at once. If the user whishes to see details compared between the different models, they can click on the "Compare" in the sidebar or the top of the model page, which brings them to the compare page with the model already chosen.

Arrived at the compare page, the user can choose two models from a dropdown menu and compare them side by side in five different categories. The first is the "Variables" category, that shows a Venn diagram of all the variables included in the models (see Fig. 33). This Venn diagram is supposed to highlight in which variables the models overlap, but due to design and space issues, a figurative diagram is shown in the middle, with the actual variables listed to the left, right, or under it. The variables on the left are unique to the model on the left, and the same goes for the variables on the right, of course to the right sided chosen model. The variables underneath are all the variables, both models have in common. These are clickable, and when clicked, brings the user to the next category, the "Simulation".

In this category, the user can choose an overlapping variable between both models, and see a simple simulation over time (see Fig. 34). The protocol used always starts with a period of $50\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ light intensity for 10 seconds, followed by $50\,\mathrm{s}$ of light. This category also includes a slider, that lets the user change the light intensity of the light phase, giving a dynamic and instant way to showcase a simple simulation of the models. It has to be noted, that these simulations are not done live on the website, but have been pre-simulated for each of the models, and only the results are shown on the website. If the user, whishes to get more information on the models themselves, they can click on the "Information" tab, which brings them to a side by side numeric table view of all the meta-information of the models, such as the number of variables, parameters, and more.

The penultimate category is that of the "Demonstrations", where the user is able to choose

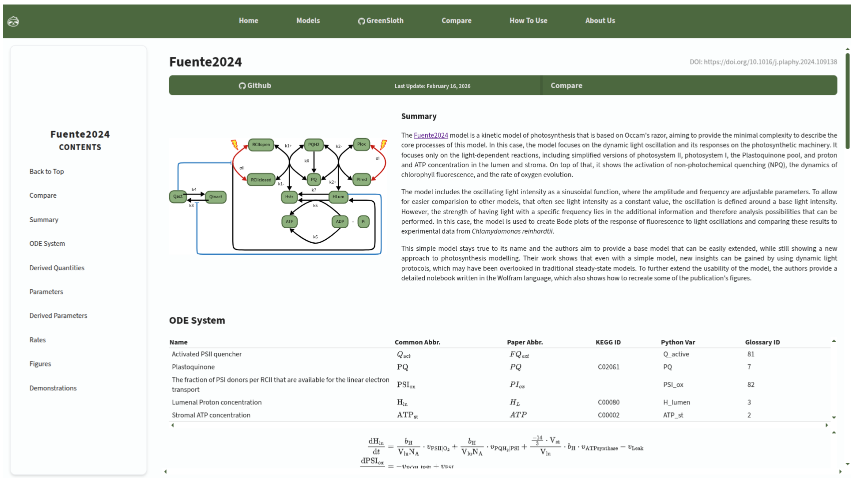

Figure 32: Screenshot of a model page of the GreenSloth website.
A screenshot was done of the Fuente2024 model page of the **GreenSloth** website, in a full screen mode. The top includes a navigation bar with links to the different pages of the website, and the main section includes a sidebar to navigate that specific page on the left and each section, one after another on the right. These sections are: "Summary", "ODE System", "Derived Quantities", "Parameters", "Derived Parameters", "Rates", "Figures", and "Demonstrations".

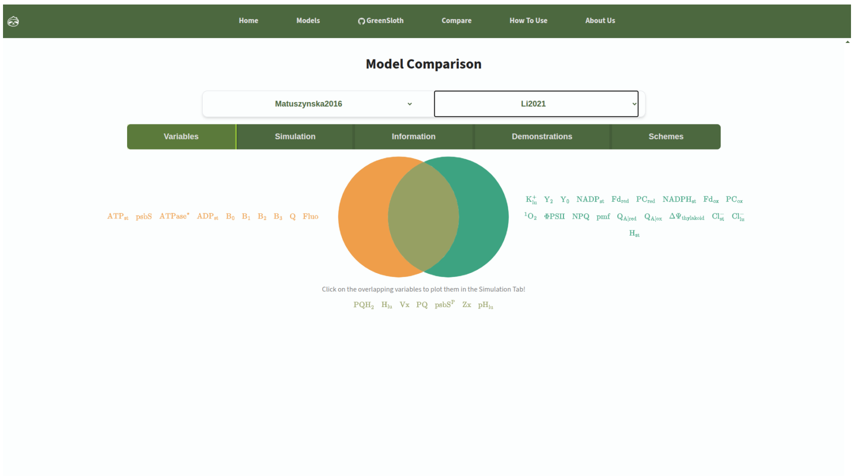

Figure 33: Screenshot of the compare page of the GreenSloth website, with the variables category selected.
A screenshot was done of the compare page of the **GreenSloth** website, in a full screen mode. The top includes a navigation bar with links to the different pages of the website, and the main section includes two select boxes to choose the models to compare, and five different buttons that represent different categories to compare. The "Variables" category is selected, which shows a Venn diagram of all the variables included in the models. The variables on the left are unique to the model on the left, and the same goes for the variables on the right, of course to the right sided chosen model. The variables underneath are all the variables, both models have in common.

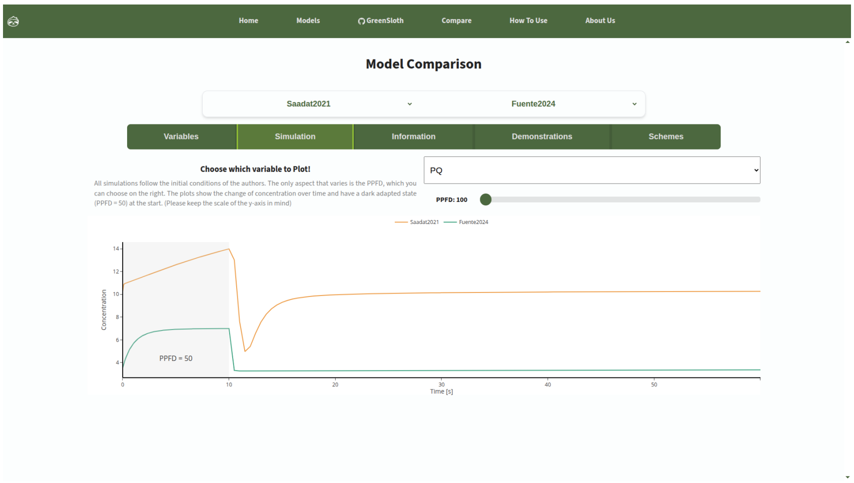

Figure 34: Screenshot of the compare page of the GreenSloth website, with the simulation category selected.
A screenshot was done of the compare page of the GreenSloth website, in a full screen mode. The top includes a navigation bar with links to the different pages of the website, and the main section includes two select boxes to choose the models to compare, and five different buttons that represent different categories to compare. The "Simulation" category is selected, which shows a simple simulation over time of an overlapping variable between both models. Here, the results of oxidised plastoquinone (PQ_{ox}) from the simulation of the Saadat2021 and the Fuente2024 at a Photosynthetic Photon Flux Density (PPFD) of 100 $\mu mol\, m^{-2}\, s^{-1}$ are shown.

from the demonstrations performed on the models, and see the results side by side (see Fig. 35). From the representing figure at the top, to the brief description underneath it. Then, the last category, "Schemes", is where both models' schemes are shown side by side, to easily compare the structure of the models (see Fig. 36).

The last two pages, "About Us" and "Impressum", include either information about the project and the team behind it, or legal information about the project. This, wraps up the entire GreenSloth Website, and can be found here: `https://greensloth.rwth-aachen.de/`.

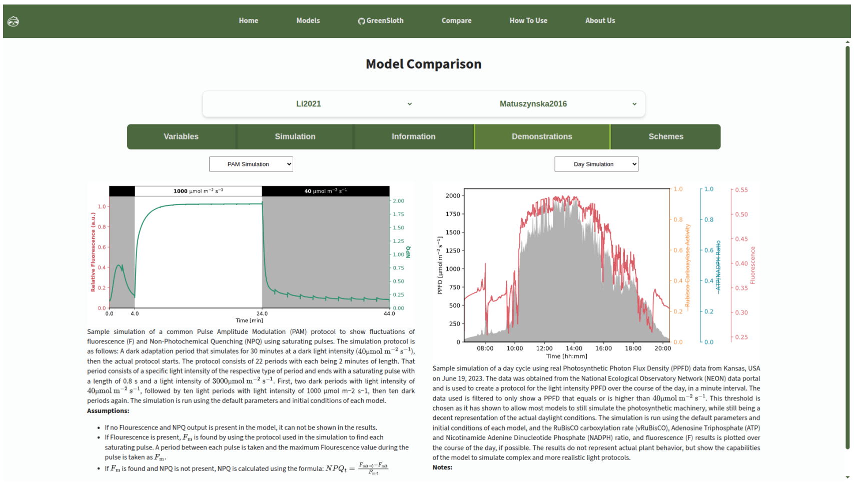

Figure 35: Screenshot of the compare page of the GreenSloth website, with the demonstrations category selected.
A screenshot was done of the compare page of the **GreenSloth** website, in a full screen mode. The top includes a navigation bar with links to the different pages of the website, and the main section includes two select boxes to choose the models to compare, and five different buttons that represent different categories to compare. The "Demonstrations" category is selected, which shows two other select boxes, which contain the five possible demonstrations that were performed on each model. The Li2021 model on the left shows the PAM simulation, while the Matuszynska2016 model on the right shows the Day Simulation. The respective figures of the demonstrations are shown at the top, with a brief description of the demonstration underneath.

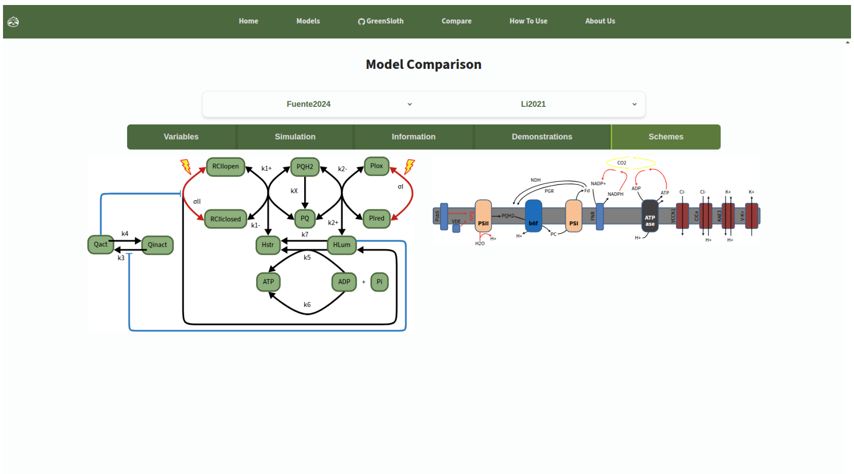

Figure 36: Screenshot of the compare page of the GreenSloth website, with the schemes category selected.
A screenshot was done of the compare page of the **GreenSloth** website, in a full screen mode. The top includes a navigation bar with links to the different pages of the website, and the main section includes two select boxes to choose the models to compare, and five different buttons that represent different categories to compare. The "Schemes" category is selected, which shows both models' schemes side by side. On the left, the Fuente2024 model is shown, while on the right, the Li2021 model is shown.

4 Discussion

4.1 Model Validations

The reimplementation of the five models has worked to varying degrees. However, an overall trend was noticed during the reimplementation process: the documentation of a model is crucial for the success of the reimplementation. The complexity, or the number of quantities in the model, does not seem to be a factor in the reimplementation's success. This can be seen in the Saadat2021 model, which is by far the largest model (see Table 1), but, with the Matuszynska2016 model, is the most successful reimplementation. On the other hand, the Li2021 model, which did not include any information in the publication itself, had not only the most difficult reimplementation process but also the most issues in the results. This shows the need for transparent documentation of the model creation process to enable its use beyond publication. The issues with the model reimplementations are discussed in more detail below.

4.1.1 Bellasio2019

The biggest issue with the reimplementation of the Bellasio2019 model is the C_a, proven primarily by the lacking reproduction of the fifth figure of the publication (see Fig. 6). This figure shows the model's response to the transition of C_a. In both the low-to-high and high-to-low transitions, none of the results match those reported in the publication. Additionally, as the results only change slightly after the transition, it is clear that the model is not responding to the C_a changes as it should. Strange as the quasi-steady-state scans of both the rates and concentrations, match the publication pretty well in the regard of C_a (see Fig. 3 and Fig. 4, both on the right). However, because the x-axis values in both figures match those in the publication, there may be an issue with how the sixth figure was generated. During implementation, it was difficult to follow the model documentation, as there were discrepancies between the text and the equations in the provided Microsoft Excel Workbook. Because of this, and the only minor inconsistencies in the other figures, the model's implementation is classified as a partial success. The math behind the model, especially concerning the C_a dynamics, need to be reevaluted and perhaps corrected to match the implementation used in the figures of the publication.

4.1.2 Fuente2024

Overall, the basic mechanisms of the Fuente2024 model are reproduced, as many results match those reported in the publication. However, a distinct issue is evident at higher frequencies, resulting not only in results shifted away from the publication but also in different dynamics. Further, the results of the F could not be reproduced at all (see Fig. 11), which made it impossible even to attempt to reproduce three of the figures in the publication. The F of the reimplementation shows little to no reaction to the oscillations of the light. In contrast, the publication shows a clear curve.

Great care was taken to ensure the reimplementation of the model was as close as possible to the publication, drawing heavily on the provided Wolfram Notebook. However, differences in the software used during implementation may have caused the issues. This is only an assumption, as the exact details of the issue could not be determined. Still, the reimplementation is classified as a partial success, as most of the results closely match the publication. The issues with the F and the high frequencies need to be investigated further to determine whether they are caused by a software issue or by the model itself.

4.1.3 Li2021

Most of the results of the Li2021 model are reproduced well, however, only for the lower light intensity (see Fig. 13). The higher the light intensity, the more the results deviate from the publication (see Fig. 14, top row). The trend of each result is similar, but does not exactly match the publication. Because the default value of the PPFD is $100\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$, the issues in the higher light intensity most likely stem from an equation in the model that is based on the PPFD. Additionally, no information about the model is given in the publication itself; only changes made to a preexisting model and a link to a GitHub repository that includes a Python script supposedly implementing the model.

This script is not well documented, written, or formatted. It was the only source of information for the reimplementation of the model, making it easy to make mistakes in the code. Additionally, running the script with the default parameters, except for PPFD being set to $500\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$, does not produce the results shown in the publication. These results, however, are very similar to those of the reimplementation; therefore, the script provided is either incomplete or not the actual implementation used for the publication. Because of this, the reimplementation of the model is considered a success, as it reproduced most of the publication's results, albeit with some issues. However, as no other information about the model is provided, it is impossible to investigate the issues in any other way. Additionally, some figures in the publication are not well-explained, making it difficult to reproduce them. The authors should therefore be contacted to ask for more information on the model and its implementation.

4.1.4 Matuszynska2016

The reimplementation of the Matuszynska2016 model was overall successful, as all but one figure could be reproduced very well. This outlier was a phase-contrast projection of lumen pH versus Q at three different light intensities (see Fig. 16, bottom). As these inconsistencies are largely in the y-axis direction, it is fair to assume the lumen pH is the issue, which is reinforced, as the time series of the lumen pH also shows some, but small, inconsistencies (see Fig. 16, top). The exact cause of these errors remains unknown, despite careful investigation of the model. It has to be added that the lumenal pH is calculated through the lumenal proton (H^+) concentration, blowing up a rather small number. Therefore, a small error in concentration can lead to a large error in pH. Still, the reimplementation is considered a success, as the model largely reproduces not only the figures in the publication but also the experimental data the authors used to validate it. The issues with the lumenal pH need to be investigated further, to determine whether they are caused by a software issue or by an issue with the model itself.

4.1.5 Saadat2021

Overall, the reproduction of the Saadat2021 publication was successful, with most figures reproduced well. The only issue is with specific ranges of the k_{cyc} that prevent the model from reaching a steady state. A feature that is presented in the publication, but could not be reproduced well in the reimplementation (see Fig. 19). If this issue is addressed, the model would be able to reproduce all the figures in the publication.

4.2 Model Demonstrations

4.2.1 Daylight Simulations

In the daylight simulations, the models are simulated using actual daylight data to see how they perform in a more realistic "environment". The plots showcase three simulation results: the $v_{\mathrm{RuBisCO|carb}}$, the ATP to NADPH ratio, and the F. The five models vary greatly in their results, with some showing a clear response to daylight changes, while others exhibit more consistent behavior. Some even have no result at all (see Fig. 23).

The $v_{\mathrm{RuBisCO|carb}}$ is only shown by the Bellasio2019 and Saadat2021 models. Both models show a proportional correlation between the $v_{\mathrm{RuBisCO|carb}}$ and the daylight changes, with the Bellasio2019 model showing a more pronounced response. On top of that, they both reach a plateau before the highest light intensity is reached, both at approximately the same time. This means that $v_{\mathrm{RuBisCO|carb}}$ is influenced by PPFD in both models, but only up to a certain level. Additionally, the representation of $v_{\mathrm{RuBisCO|carb}}$ in both models is likely very similar, as they exhibit a similar response.

Only the Saadat2021 model shows a clear response in the ATP to NADPH ratio, with a slight negative correlation with daylight changes; however, the plot remains mostly constant. This can be interpreted as meaning that the rates influencing ATP and NADPH are either not influenced by changes in light, or they are influenced in a way that cancels each other out.

The F results for the Fuente2024 and Matuszynska2016 models show a clear, very similar response to daylight changes, with a positive correlation. Additionally, both models show the curve closely follows daylight changes at higher light intensities, while showing a plateau at lower light intensities. This means that in both models, the F has a minimal base level that is rarely crossed, no matter the light intensity, but easily increases when the light intensity is high enough. This interpretation is supported by the fact that the Fuente2024 model calculates the F with a base level implemented, and the Matuszynska2016 model uses a function based on two states of PSII, both of which are dependent on each other. The only other model showing a response in the F is the Saadat2021 model, which shows a completely different response, with a negative correlation to the daylight changes. While this reaction is counterintuitive, it actually follows data from diurnal experiments [56]. It was shown that F actually falls after a certain light intensity, reaches a lower plateau, and rises again once the light intensity is lower. A very similar curve to the one shown by the Saadat2021 model. This could be explained by the fact that the NPQ mechanism is activated at higher light intensities, leading to a decrease in F. This would mean that the Saadat2021 model better describes the NPQ phenomenon than the other models, even though all implement a NPQ mechanism.

The Li2021 model shows no response to changes in daylight in any of the three results. This is not because the simulation could not be done, but because that model does not contain any of the quantities plotted. This shows a clear issue with this demonstration, as it does not apply to all models. A possible solution to this issue would be to add additional quantities to be plotted, or at least present a visual result to demonstrate that the model is responding to daylight changes, even if the specific quantities plotted are not present in the model.

4.2.2 FvCB Add-on

Only the Bellasio2019 and Saadat2021 models have quantities representative of CO_2 and $v_{\mathrm{RuBisCO|carb}}$, thereby allowing the FvCB to be added in (see Fig. 24). The Bellasio2019 model shows a very similar trend in both curves to the FvCB curves, actually matching at lower C_{i} pressures. In this range, the FvCB curve with the default parameters has the W_{c} as the limiting factor, which means the Bellasio2019 model can capture this limitation pretty well. However, as the C_{i} pressure increases, the limiting factor switches to W_{j} and then W_{p}, which is not shown in the Bellasio2019 curves. Still, the Bellasio2019 model has a good relationship with the FvCB model, even if the curves include some jaggedness at some points. These happen because the model has difficulty simulating a steady state. Adding the FvCB to the Bellasio2019 model was easy, since it already includes its simplest form of calculating the A (see Equation 1, top).

The Saadat2021 model has its own interpretation of $v_{\mathrm{RuBisCO|carb}}$, which was taken to calculate the A with the default parameters of the FvCB model. The resulting curves are close to those of the FvCB model but slightly higher. As both the $v_{\mathrm{RuBisCO|carb}}$ and the A curves are higher, the reason for the shift can be directly attributed to the $v_{\mathrm{RuBisCO|carb}}$. Both the $v_{\mathrm{RuBisCO|carb}}$ and the W_{c} of the Saadat2021 and FvCB models use Michaelis-Menten kinetics, yet the parameters differ. Additionally, the $v_{\mathrm{RuBisCO|carb}}$ of the Saadat2021 model is much more complex and includes many more quantities than the W_{c} of the FvCB model. This means that the $v_{\mathrm{RuBisCO|carb}}$ of the Saadat2021 model is vastly different than the W_{c} of the FvCB model. However, as the relationship of both curves shows similar trends to the FvCB curves, it can be said that the addition of the FvCB to the Saadat2021 model was successful, even though a more detailed investigation into the equations and the ways to bridge the two models should be done.

The FvCB model could not be added to the other three models because they do not include the necessary quantities. The Matuszynska2016 and Fuente2024 models do not include $v_{\mathrm{RuBisCO|carb}}$ or CO_2 related quantities. In contrast, the Li2021 model includes a very simplified term for the CBB cycle but also lacks a quantity related to CO_2. The fact that these quantities are missing makes it hard to incorporate the simple FvCB model, yet it does not degrade the models at all. They all have a different scope of photosynthesis, and these three models focus more on reactions not related to the CBB cycle.

4.2.3 Standard PAM Simulation

Only the Bellasio2019 model could not be used for the standard PAM simulation (see Fig. 25), as it does not include a quantity for F or NPQ. All other models, however, produced very similar outputs. The Fuente2024, Matuszynska2016, and Saadat2021 models all show a very similar reaction for F, with the same number of F_{m} and similar responses to light changes. The Matuszynska2016 and Saadat2021 models use these F to calculate the NPQ (see Equation 3), resulting in jagged curves for both. However, the Fuente2024 model provides a dynamic description of NPQ, allowing it to show a smooth curve that also shows that NPQ reacts to the strong saturating pulse, a feature that cannot be shown in the other two models. While the Li2021 model does not have a quantity for F, it does have a term for the NPQ, which also showcases a similar curve as the other three models, with the added benefit of showcasing the reaction of NPQ to the saturating pulses.

All possible simulations of the standard PAM protocol show a commonly observed trend in experiments [12, 57], even the Li2021 model, which had difficulty simulating the NPQ, as shown in the publication. These new results of the NPQ, match up much more than the NPQ seen at a light intensity of $500\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$ (see Fig. 12), which can be interpreted that the red parts of the figures in the publication are actually run at a light intensity of $1000\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$. However, to be certain, a deeper investigation into the model and the publication is needed.

The results of this demonstration, as with the others, cannot be quantified or directly compared. As all models are parametrized differently and have different scopes, it is not expected that their results will match. However, as the F of the Fuente2024, Matuszynska2016, and Saadat2021 models are very similar, it can be said that these three models have a similar description of this quantity.

4.2.4 MCA of Photosynthesis

Many of the models show an incomplete heatmap for the MCA of photosynthesis (see Fig. 26), as they do not include all the quantities chosen to be integral for the depiction of photosynthesis. In fact, even the model showing the most responses, the Saadat2021 model, is missing two quantities. The CO_2 and a parameter representing the coefficient of PSI. The former is considered lacking in this regard, as it is included in the model as a parameter rather than a variable. The latter could not be assigned a value because no definitive proportional parameter was found for the rate of PSI in the model. The heatmaps presented by this model show little response. On the variables side, only the parameter representing RuBisCO shows a control on RuBP.

In contrast, all the rest show little control over the variables. On the fluxes side, the parameter for PSII shows some control on all the fluxes. This suggests that the chosen representatives of the photosynthetic machinery do not have much control over one another, except for PSII.

The other models have very little representation in the heatmaps because they do not include many of the chosen quantities. Therefore, commenting on the system's control is difficult. On top of that, the Li2021 model shows no response in the heatmaps; however, since some labels are not grayed out, it is not because there were no quantities. As an MCA of control coefficients is done by simulating to steady-state, these heatmaps directly show that the Li2021 model does not simulate to steady-state.

While this MCA demonstration is not a good way to assess the control of the system, it does directly enable comparison of the availability of the photosynthesis machinery across the different models. With the grayed-out labels and empty squares, the user can, with one glance, see which parts are included in the model and which are not. In this case, the Saadat2021 model covers most of photosynthesis; therefore, if a user is looking for a more complete model, they choose that one. On the other hand, if a user is looking for a more focused model, for example, for PSII and RuBisCO, they can choose the Bellasio2019 model. A bonus is that the user can also see which model actually simulates to steady-state, as the Li2021 model does not, which can be an important factor for some users.

4.2.5 Fitting of NPQ

As the Bellassio2019 model does not have a quantity for F nor NPQ, it is the only model that cannot be used for the fitting (see Fig. 27). All other models were fitted to the experimental data; however, to varying degrees of success. The best fit is with the Li2021 model, while the Matuzynska2016 model is also a strong contender. The former, however, does not have a quantity for F, while the F result of the Matuzynska2016 fit shows a very similar pattern to the data. The worst fit is done by the Saadat2021 model, even though the F fits better here than with the Fuente2024 model. However, as all models show a better fit to the data than the normal parameterization, it can be said that all these models can be fitted to that data.

The problem with this demonstration is that it is very difficult to find the best way to fit these models. A fitting routine is strongly dependent on the chosen parameters and whether they have bounds. Additionally, the fitting routine itself can be based on different algorithms, leading to different results. Therefore, this demonstration cannot be used to directly compare the models, as the fitting process cannot be standardized across models. However, it can give an idea of what to do with a model and create the starting blocks for a fitting process, which can be very useful for users who are not familiar with fitting models to data.

4.3 Website

As the internet is now an integral part of most societies worldwide, everyone knows how to use a website [42]. It keeps it simple and accessible for everyone, as it only requires an active internet connection. On top of that, it is a quick way to share information worldwide, as direct sharing only requires a link. However, it is vital to give a website a good design tailored to the target audience. This ensures the best way to convey the information it intends to convey [43]. The advantages of a website are clear: accessibility, ease of use, and quick sharing. Advantages that support the goal of **GreenSloth**.

The website created for **GreenSloth** is designed to be as simple as possible, with a clear structure and easy navigation (see Fig. 28). It follows a modern, common style to convey its message while separating key aspects of the work into distinct parts using models. The lists of implemented models and the ability to search for models by custom tags (see Fig. 31) can help the user narrow their focus to the model they wish to look at. This would alleviate the need for an extensive literature search to find a model. After selecting a model, the user can look at all the details of the model, all on one page (see Fig. 32). This includes the model equations, parameters, results, and a link to the code. This allows the user to quickly get an overview of the model and decide whether it is relevant to their work. As the information is directly sourced from the GitHub repository documentation, it is always up to date with the latest code changes. Creating a direct bridge between the code and the website.

From each model page, the user can also navigate to the comparison page, where they can compare all the models on **GreenSloth** (see Fig. 33). There, they choose the models and the aspects they want to compare. By choosing between Variables, Simulation Examples, Model Information, Demonstrations, and Schemes, they can get a good overview of the differences and similarities between the models. These comparisons are ways people naturally compare models. However, now they are all in one place and can be done systematically. These five ways are suggestions and have significant potential for expansion, especially as more models are added to **GreenSloth**. Many ideas embedded in the ecosystem are heavily inspired by the models already included. A good example of how a comparison can be expanded on is to change the simulation examples from static, pre-calculated examples to a more interactive, dynamic, server-side simulation. This would allow the user not only to compare the models' results but also to test their own parameters and ideas. This idea is already being worked on, separately from **GreenSloth**, but shows the potential to be added to the ecosystem [58].

Overall, the **GreenSloth** website is designed to be easy to understand and accessible to everyone, while also showcasing the models in their entirety. Because the documentation for each model is done in the same format, it is very clear when information is missing. This fact can be used to force the authors of models to provide all the necessary information, while helping them in the process through the structures of the **GreenSloth** ecosystem. The website is not meant to replace a publication, but instead support it, by providing a dedicated platform for the models, where they are held to a specific standard, not in a sense of their value to the community, but in a sense of transparency, availability, and reusability. In turn, helping the model and its publication to follow the FAIR and CURE. The website is a key way to support the sharing of models of photosynthesis, which may otherwise be lost in the onslaught of publications.

4.4 Conclusion

The **GreenSloth** ecosystem is a solution for a wide variety of problems in the field of photosynthesis modelling. It provides support in creating model documentation to make the model's details as transparent as possible. It also provides ideas for demonstrations to make comparisons between models easier and reduce the user's effort. Finally, it provides an accessible, easy-to-use website to share and display models in a compact format. The website is also set up to compare each model side by side, providing another level of help for choosing models.

Five models of photosynthesis have been chosen to show the power of **GreenSloth**. They first had to be reimplemented, which was validated by the recreation of the figures in the publications. Then the complete documentation was completed to the **GreenSloth** standard, including a summary, the model components, recreated figures, and demonstrations. These demonstrations are common but unofficial ways to compare models, methods that modellers already use to present their models. These demonstrations are standardized to allow at least a basic level of comparison between models and to reduce the effort required or even inspire users of the models. The models are implemented and shared openly on GitHub in a reproducible and clear way, along with the code for the demonstrations and figure recreation, and their documentation. Finally, all the information about these models is also shared openly on the **GreenSloth** website, including additional ways to compare the models, such as sample simulations over time. This allows the user to access these five models of photosynthesis more interactively without detracting from the publication itself. The **GreenSloth** ecosystem must not be seen as a replacement for the publication system or as a way to bypass it; it is merely a way to help model authors curate and share their work with the community.

The reimplementation of the five models was not entirely successful, as documentation is often misrepresented or missing. However, the overall trend is that the better the documentation, the more successful the reimplementation is. It does not matter how complex the model is; if documentation is taken care of, it can be reimplemented. This shows the importance of the **GreenSloth** ecosystem, which provides a clear, easy-to-follow guide for documenting a model, with tools and features that support the author. The proposed demonstrations keep comparisons at a very basic level; however, this is necessary, as even with a focus on kinetic models of photosynthesis, the scope of models can vary dramatically. Keeping the demonstrations basic and more proof-of-concept-like allows them to encapsulate as many models as possible and show key differences between them. Still, the demonstrations shown at the current state need to be revised and worked on to allow even the basics of modelling to understand and use them. At the moment, some do not show the extreme cases well, such as the Li2021 model in the MCA demonstration (see Fig. 26), and others are too dependent on the author's expertise in the system, such as the fitting demonstration. These demonstrations are not perfect, but as more models are added to the **GreenSloth** ecosystem, they will be expanded and improved. Additionally, more will be added as new inspiration is found from other models.

The website has been built with ease of access and use in mind for a wide range of users. The design is simple and modern, allowing users already familiar with the internet to feel comfortable with the website. However, the website has been designed only for use on a computer, without considering that a large part of internet users are on mobile devices [59]. This is a clear issue, as it limits the website's accessibility and, by extension, the models'. Additionally, as not many models are included on the website, the list is manageable; therefore, the search function has not been expanded. However, as more and more models are added, the search for models can quickly become as confusing as it is with publications; therefore, a more advanced search function is needed. The tag system is a good start; however, at its current state, it is too minimal and lacks sufficient information to be useful. More categories need to be considered, for example, how the models were simulated or which data were used to validate them. The side-by-side comparison page allows the user to compare two models directly on a specific aspect; however, it can be expanded further.

The **GreenSloth** ecosystem is a project that incorporates many branches, all connected by a central idea. Making the creation, presentation, and sharing of models of photosynthesis easier and more accessible. The current state of the ecosystem presented in this thesis is only a starting block, like a sapling. This project has significant potential to grow and expand, with more models added, more demonstrations created, and the website improved. As a tree needs a crow to remove pests, fungi to refresh the soil, or a squirrel to plant its seeds, so does **GreenSloth** need the help of the community to grow and thrive. A web developer to continue adapting the website, a web designer to make the website more appealing, a modeller to add more models, and a software developer to help with code and documentation. This additional help will not only let **GreenSloth** grow, but will cement the community around it. How the ecosystem is built, which branches grow, which fall off, the main bark of **GreenSloth** will keep being:

Photosynthesis Models at your Pace

5 How to Contribute

If a user wishes to contribute a model to the **GreenSloth** ecosystem, they can follow the steps outlined in this section. The process is designed to be straightforward and minimize effort for the user, while ensuring the model is well-documented and easily shareable with the community. The steps follow the same separation of power as **GreenSloth**: 1) Model Creation, 2) Model Presentation, and 3) Model Sharing

5.1 Model Creation

The actual writing of the model cannot be assisted by **GreenSloth**, as it is too complex a task. However, once a model is written in Python, using the `mxlpy` package, **GreenSloth**, can take over many tasks. The biggest is a bridge between the Python code and the actual documentation. To start, the user downloads the **GreenSlothUtils** package (`https://github.com/ElouenCorvest/GreenSlothUtils`), and installs it as an editable package into their Python environment. This allows the user to use all the utilities of the **GreenSlothUtils** package, including a step-by-step guide to creating a well-formatted model repository. The details of this process can be found in the **GreenSlothUtils** documentation and in the materials and methods section (see Section 2.1) of this thesis. With the information and documentation complete, the user can then recreate the publication's figures in the already-generated template Jupyter Notebook. In the README file, these figures shall then be imported into the "Figures" section and shortly described. Due to rights and permissions, the user may not be able to share the original figure caption from the publication. The captions for these figures must describe not only what is visible, but also how the figures were obtained. The scientific context of the figures is not needed, as they are part of the publication. With the model now well packaged, validated, and documented, the user can proceed to the next step.

5.2 Model Presentation

The model presentation is the recreation of the demonstrations later used to compare the model. As these demonstrations are standardized, the user does not need to worry about much code. They can use the already-generated Jupyter Notebook template and fill in the missing names of the quantities needed for each demonstration, all collected at the start of the notebook. If the current model is missing any required quantities, the user shall instead provide `None`. In this case, each demonstration will decide what happens with the missing quantity. For example, if the model does not have a quantity for NPQ, the demonstrations of the PAM protocol and the fitting will both calculate it from the F. Sadly, in some cases, the demonstrations have specific rules or requirements. For example, the FvCB add-on needs the $v_{\mathrm{RuBisCO|carb}}$ to be in a unit of $\mathrm{\mu mol\,m^{-2}\,s^{-1}}$. If that is not the case in the current model, the user needs to either create a new quantity with the correct unit or let the demonstration handle it if the unit fits. These extra requirements are listed in the documentation of the **GreenSlothUtils** package, where all the required functions are located. With the demonstrations now filled in, the user can add notes to the README file, if applicable. The README file will automatically import the figures generated by the demonstrations if the user followed the previous model-creation process. At this point, the model is well packaged and documented, and is now prepared to be shared with the community.

5.3 Model Sharing

Once the model has reached the required state of being well packaged, validated, demonstrated, and documented, it can now be uploaded to the site. Before that, though, the user has to check whether their model overlaps with models already on the site. To do that, the main **GreenSloth** repository contains two main glossaries in the overarching models directory. These two files contain all the rates and variables used in the models already in the ecosystem. The user has to go through these lists and see if any match their model. This is a tedious process; however, if there is a match, the user only has to include the respective Glossary ID in the correct column. Actually doing this during model creation is recommended. However, it had to be mentioned as the final obligation before the model could be shared. Once the user has done this, they can then create a pull request of their branch with the new model. The pull request will then be reviewed by the maintainers of **GreenSloth** and accepted if all of the previous steps have been followed. If the pull request is accepted, the user does not need to do anything else, as the maintainers will handle the rest.

5.4 After Sharing

Since the user-uploaded model follows the required format, it can now be added to the website and made available to the community for use. To keep maintainers' workload as low as possible, this step has been automated as much as possible. Once the model directory is in the main **GreenSloth** repository, the maintainers switch to the website branch (`https://github.com/ElouenCorvest/GreenSloth/tree/website`), and add the new model name inside a small JSON file. This file predicts which models are extracted from the main repository and also serves as the storage for the tags. These tags also need to be added manually by the maintainer. Once the JSON file is updated, the maintainer only needs to duplicate one of the other model HTML files, rename it to the new model name, and that is it. The maintainer should then check on the website's dev environment that the new model is displayed correctly. If everything is done correctly, the rest of the work is done automatically. The maintainer can then build and deploy the website, making the new model available on the live website.

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